P vs NP in Curvature-Bounded Wave Computation
A Model-Relative $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$ Separation

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Abstract

We introduce a curvature-regulated wave computation model (WCC), in which computation is recast as the evolution of confined oscillatory states under curvature and entropy constraints. Within this model we define complexity classes $\mathbf{P}_{\mathrm{WCC}}$ and $\mathbf{NP}_{\mathrm{WCC}}$ by bounding the number of symbolic evolution steps and curvature/entropy resources.

On the analytic side, we define a semantic curvature complexity measure $\mathbf{C}_{\Theta}(f)$ based on the WCT curvature operator and show that, for a locked expander family of 3-SAT instances, any realization inside the WCT curvature machine must incur exponential curvature cost: $\mathbf{C}_{\Theta}(F_n) \geq 2^{\Omega(n)}$. Combined with a circuit-to-curvature upper bound, this yields a *model-relative*, non-natural, non-relativizing lower bound showing that no polynomial-size, curvature-bounded WCC tiling can realize SAT within this curvature machine.

On the symbolic side, we give a discrete Turing-style abstraction of WCC, construct an NC^1 verifier for curvature-locked configurations, and encode SAT instances as symbolic wave-interference constraints in a Cook–Levin style reduction into the WCC verifier. This leads to a *model-internal separation program*: under additional geometric and dynamical hypotheses (existence of curvature-locked states that are verifiable by the WCC verifier but not reachable within polynomially

many curvature-bounded WCC updates), one would obtain an internal separation

$$\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$$

at the level of the WCC model.

Finally, we discuss a conditional bridge to classical complexity. If curvature- regulated WCC computations and classical deterministic Turing computations simulate each other with only polynomial overhead, so that $\mathbf{P}_{\mathrm{WCC}} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCC}} = \mathbf{NP}$, then any such model-internal separation, combined with the curvature lower bound above, would imply the classical conjecture $\mathbf{P} \neq \mathbf{NP}$. The present paper does not claim that either the internal separation or the TM–WCC simulation equivalence is proved; both are isolated as explicit conjectures and targets for future work. The main rigorous contribution is the model-relative curvature lower bound for SAT inside the WCT curvature framework.

1 Introduction

The $\mathbf{P}$ vs. $\mathbf{NP}$ problem asks whether every computational problem whose solution can be verified in polynomial time can also be solved in polynomial time. Despite decades of effort, no proof resolving $\mathbf{P} \stackrel{?}{=} \mathbf{NP}$ has been universally accepted. The present work does not claim to resolve this classical question; instead, it develops a curvature-regulated model of computation in which one can prove a strong, model-relative lower bound for SAT and formulate a precise internal separation program.

We work within Wave Confinement Theory (WCT), where computation is defined as the physical or symbolic evolution of confined oscillatory systems under curvature feedback. In this setting, stable coherent wave states play the role of solutions, and the feasibility of reaching them under bounded evolution defines a notion of computational difficulty.

We introduce a discrete curvature-regulated computation model (WCC) and define complexity classes $\mathbf{P}_{\mathrm{WCC}}$ and $\mathbf{NP}_{\mathrm{WCC}}$ by imposing polynomial bounds on evolution time and curvature/entropy budgets. Within this model we pursue two complementary tracks:

- **Analytic curvature lower bound.** We define a semantic curvature complexity measure $\mathbf{C}_{\Theta}(f)$ based on the WCT curvature operator and show that, for a locked expander family of 3-SAT instances inside the WCT curvature machine, any admissible realization must pay exponential curvature: $\mathbf{C}_\Theta(F_n) \geq 2^{\Omega(n)}$. Together with a circuit-to-curvature upper bound, this yields a model-relative, non-natural, non-relativizing lower bound excluding polynomial-size, curvature-bounded WCC tilings for SAT in this model.

- **Symbolic complexity and separation program.** We construct a verifier for wave-coherence predicates in NC^1, encode SAT instances as symbolic wave-interference constraints (a Cook–Levin style reduction into WCC), and formalize WCC-level complexity classes $\mathbf{P}_{\mathrm{WCC}}$ and $\mathbf{NP}_{\mathrm{WCC}}$. We then formulate a *model-internal separation program*: under explicit geometric and dynamical hypotheses (existence of curvature-locked states that are verifiable but not reachable within polynomially many curvature-bounded WCC updates), one would obtain an internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$. In this manuscript these hypotheses are stated as conjectures rather than claimed theorems.

Finally, we outline a conditional bridge to classical complexity. If WCC computations and deterministic Turing computations simulate each other with only polynomial overhead, then the WCC classes would satisfy $\mathbf{P}_{\mathrm{WCC}} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCC}} = \mathbf{NP}$. Under this *TM–WCC compatibility conjecture*, any internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$, combined with the curvature lower bound described above, would imply the classical separation $\mathbf{P} \neq \mathbf{NP}$. The present work does not assert either the internal separation or the simulation equivalence as proved; instead, it isolates them as clearly labeled conjectural targets and focuses on the model-relative analytic curvature lower bound that can be established within the WCT framework.

Author's Note: On the Geometry of Computation and the Dimensional Lock

This work proposes a new approach to the **P vs. NP** problem, one that is not purely combinatorial or symbolic, but instead *geometric*, *topological*, and *physical*. Computation here is modeled not as an abstract string-manipulation process, but as the evolution of a wavefunction constrained by curvature, entropy, and dimension.

The central mechanism is what I term a **dimensional lock**: in the curvature-regulated model, attempting to force a deterministic, uniformly controlled evolution (the analogue of **P**) to realize certain non-uniform, combinatorially large configurations (the analogue of **NP**) appears to require crossing a geometric barrier that is forbidden by the curvature and dimensionality constraints. At an intuitive level, this structural obstruction prevents uniform computation from accessing the full non-uniform solution space and motivates an internal separation

$$\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$$

within the WCC model.

In the body of the paper, this idea is made precise in two steps. First, a semantic curvature complexity measure is introduced and an exponential curvature lower bound is proved for a locked expander family of 3-SAT instances inside the WCT curvature machine. Second, a discrete symbolic abstraction of WCC is developed, together with a verifier and a set of explicit geometric and dynamical hypotheses which, if established, would yield a formal internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$.

This perspective is simple but nontraditional. It draws from physics and information theory, disciplines not typically employed in classical complexity theory. For that reason, it may appear unfamiliar or even unorthodox to some readers. The claim here is not that a physical metaphor proves a classical complexity result, but that a curvature- and dimension-constrained *symbolic* model admits (i) a rigorous, model-relative curvature lower bound for SAT, and (ii) a carefully stated internal separation program.

The extrapolation to the classical statement $\mathbf{P} \neq \mathbf{NP}$ is treated explicitly as conditional. If curvature-regulated WCC machines and standard Turing machines are polynomially equivalent in computational power, then the dimensional lock and curvature bounds described here would provide a route to the classical separation. That simulation equivalence is isolated as a distinct conjecture rather than assumed silently; this paper does not claim to resolve the classical **P** vs. **NP** problem.

2 Curvature-Confinement Formalism and a Model-Internal Separation

We propose that a confined wavefield $\psi(x,t)$ under curvature feedback evolves toward symbolic configurations that cannot be constructed in polynomial time but can be verified via coherence constraints. Within the rail-constrained WCC abstraction, this creates a natural asymmetry between generation and verification, aligning with a prospective distinction between $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$, where the superscript phys denotes photonic WCT/WCC machines subject to finite-curvature, finite-band rails (Section 10).

2.1 Definitions

Curvature-Locked State: A wavefield ψ^* is curvature-locked if it satisfies

$$W_\psi = \frac{-\nabla^2\psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}, \quad \text{with } \frac{dS[\psi]}{dt} \to 0 \text{ and } \nabla W_\psi \to 0 \tag{1}$$

under the WCT/WCC evolution rules.

Symbolic Configuration Set $\mathcal{S}$: A set of discrete symbolic regions R_i where

$$\psi(x) \approx \psi_i \quad \forall x \in R_i, \quad R_i \subseteq \mathbb{R}^n$$

implements a finite symbolic configuration (assignment, tape content, etc.) inside the WCC model.

Verification Operator $\mathcal{V}$: A polynomial-time procedure on the WCC encoding which confirms that

$$\mathcal{V}(\psi^*, \mathcal{S}) = \text{True} \iff \psi^* \text{ exhibits symbolic coherence consistent with } \mathcal{S},$$

with an implementation in NC^1 on a classical circuit model.

Non-Constructibility Condition (WCC): A curvature-locked state ψ^* is *non-constructible in* $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ if no deterministic polynomial-time WCC machine $\mathcal{T}_{\mathrm{WCC}}^{\mathrm{phys}}$ (obeying WCT rails) can output ψ^* from a trivial input configuration, while a polynomial-time WCC verification $\mathcal{V}$ remains available.

2.2 Model-Internal Complexity Schema

To isolate the logical shape of the desired separation, we introduce an abstract complexity functional $\mathcal{C}(\psi)$ on WCC states and state the following *schema*.

Proposition 2.1 (Curvature-Confinement Complexity Schema in Physical WCC). Let ψ^* be a curvature-locked solution of a confined resonance field evolved from a trivial initial state within the rail-constrained WCC model. Suppose that:

1. The symbolic configuration space $\mathcal{S}$ representing ψ^* satisfies $\mathcal{C}(\psi^*) > T(n)$ for every polynomial-time achievable bound $T(n)$ on $\mathcal{C}(\cdot)$ over all $\mathcal{O}(n^k)$-time WCC machines obeying WCT rails, and
2. There exists a polynomial-time WCC verification procedure $\mathcal{V}$ that confirms curvature-coherent symbolic structure of ψ^* (e.g. via local curvature and phase-lock checks).

Then ψ^* lies in $\mathbf{NP}^{\text{phys}}_{\text{WCC}} \setminus \mathbf{P}^{\text{phys}}_{\text{WCC}}$, and hence $\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}$ *inside the physical WCC model.*

This proposition is logically tautological once (1)–(2) are established; the substantive content of the curvature-confinement program is to *exhibit* a family of states ψ^* and a rigorously defined functional $\mathcal{C}(\psi)$ for which (1) holds, e.g. via dimensional-lock and curvature-scaling lemmas (Section 10).

2.3 ψ-SAT Encoding Scheme (WCC Rail)

Let each Boolean clause C_i in a SAT instance be mapped to a spatial clause kernel:

$$\psi(x,0) = \sum_i \chi_i(x) e^{i\phi_i}, \quad \chi_i(x) \text{ localized curvature excitation,} \tag{2}$$

with a prescribed WCC encoding from symbolic clauses to localized modes.

Under resonance evolution in the WCC rail, clauses are satisfied if corresponding phase regions lock in coherence:

$$\text{SAT-satisfied} \iff \exists x_i : \nabla\phi(x_i) \to 0 \text{ and } |\psi(x_i)|^2 \to \text{local max.}$$

This encoding transforms logical satisfiability into symbolic phase confinement and serves as a Cook–Levin style reduction into a WCC verifier, without yet claiming full NP-completeness of the ψ-SAT construction.

3 Complexity Class Definitions

Definition 3.1 (Class P). **P** is the class of decision problems solvable by a deterministic Turing machine in polynomial time, i.e., there exists a machine M and constant k such that for all inputs x, $M(x)$ halts in $\mathcal{O}(|x|^k)$ steps.

Definition 3.2 (Class NP). **NP** is the class of decision problems for which a proposed solution (certificate) can be verified in polynomial time by a deterministic Turing machine. Formally, $L \in \mathbf{NP}$ if there exists a polynomial-time verifier $V(x, w)$ such that

$$x \in L \iff \exists w : V(x, w) = 1.$$

In the WCC setting we distinguish two model-relative classes:

- $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$: rail-constrained photonic WCC machines under WCT curvature/energy caps (finite universe band).
- $\mathbf{P}_{\mathrm{WCC}}^{\infty}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\infty}$: ideal discrete WCC machines without global physical caps, suitable for asymptotic Turing-equivalence considerations.

The main model-internal separation conjectures below concern $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$. Implications for classical **P** vs. **NP** via the ideal classes $\mathbf{P}_{\mathrm{WCC}}^{\infty}, \mathbf{NP}_{\mathrm{WCC}}^{\infty}$ are discussed only as conditional corollaries.

4 Main Model-Relative Separation Statement ($\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}$)

Conjecture 4.1 (Symbolic Separation Program in Physical WCC). We conjecture that there exist coherent, verifiable wavefields $\psi^* \in \mathbf{NP}^{\text{phys}}_{\text{WCC}}$ that cannot be constructed by any deterministic polynomial-time WCC machine obeying WCT rails. Equivalently, there should exist

$$\psi^* \in \mathbf{NP}^{\text{phys}}_{\text{WCC}} \setminus \mathbf{P}^{\text{phys}}_{\text{WCC}},$$

which would entail the internal separation $\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}$ within the rail-constrained WCC model. We refer to this as the *model-internal separation conjecture.*

Heuristic Rationale (Model Level). At a high level, the WCC program proceeds as follows. We construct a WCC-level verifier $A(q) \in \mathrm{NC}^1$ for curvature-locked configurations, encode SAT clauses into symbolic curvature-locked constraints, and aim to establish that:

- Some wavefields ψ^* satisfy $A(q^*) = 1$ (i.e., they are verifiable by a polynomial-time WCC process);
- No deterministic polynomial-time WCC machine can construct ψ^* from a trivial initial state ψ_0 under the curvature-regulated symbolic update rules while remaining within WCT rails, due to curvature/entropy barriers such as dimensional lock.

If such non-constructible but verifiable states can be rigorously exhibited, this would yield $\psi^* \in \mathbf{NP}^{\text{phys}}_{\text{WCC}} \setminus \mathbf{P}^{\text{phys}}_{\text{WCC}}$ and hence $\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}$ at the model level. Turning this heuristic into a complete proof requires formalizing and proving the relevant curvature-scaling and dimensional-lock lemmas (Section 10); these barriers are isolated below as explicit conjectures and open problems rather than assumed as settled facts.

4.1 ψ-SAT Encoding and Symbolic Map

We formally define the symbolic encoding from a Boolean SAT instance into curvature-locked wave states. Let a Boolean clause $C_i \in \text{SAT}$ be

$$C_i = (x_1 \vee \neg x_2 \vee x_3).$$

We map each clause C_i directly into a wave symbol region $R_i \subseteq \mathbb{R}^n$:

$$C_i \mapsto \psi_i(x) = A_i e^{i\phi_i(x)}, \quad x \in R_i,$$

where A_i is a localized amplitude distribution enforcing clause locality and $\phi_i(x)$ encodes logical states (True/False) in terms of constructive or destructive interference.

We construct the global wavefield as a coherent superposition of all clause wavefunctions:

$$\psi(x) = \sum_i \psi_i(x).$$

Logical satisfiability is then tied to curvature-locked coherence through a phase-locking condition:

$$\text{SAT-satisfied} \iff \exists x_i : \nabla\phi(x_i) \to 0, \quad |\psi(x_i)|^2 \to \max.$$

This encoding provides a direct ψ-based analog to the Cook–Levin construction, reducing SAT instances into symbolic curvature-locked wave states within the WCC framework, again without claiming a completed, classical NP-completeness proof for ψ-SAT at this stage.

4.2 Model-Internal Symbolic Separation

Theorem 4.2 (Conditional Symbolic Separation in Physical WCC). Let ψ^* be a curvature-locked wavefield constructed from a symbolic SAT instance via the WCC encoding described above. Suppose that:

1. The verification operator satisfies $V_{\mathrm{WCC}}(\psi^*) \in \mathrm{NC}^1$, and hence verifies ψ^* in polynomial time on a rail-constrained WCC machine, i.e. $\psi^* \in \mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}}$.
2. No deterministic polynomial-time WCC machine $M^{\mathrm{phys}}_{\mathrm{WCC}} \in \mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}}$ can construct ψ^* from an initial trivial state ψ_0 while respecting WCT curvature/energy rails.

Then

$$\psi^* \in \mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}} \setminus \mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}}, \quad \text{and therefore} \quad \mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}} \neq \mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}}$$

within the physical WCC model.

Proof Sketch. Assume for contradiction that $\psi^* \in \mathbf{P}^{\text{phys}}_{\text{WCC}}$. Then by definition there is a polynomial-time rail-constrained WCC machine that constructs ψ^* from ψ_0 under curvature-regulated symbolic evolution. Hypothesis (2) rules this out by asserting non-constructibility in polynomial WCC time under the same rails. Thus $\psi^* \notin \mathbf{P}^{\text{phys}}_{\text{WCC}}$ while remaining verifiable in $\mathbf{NP}^{\text{phys}}_{\text{WCC}}$ by (1), so $\psi^* \in \mathbf{NP}^{\text{phys}}_{\text{WCC}} \setminus \mathbf{P}^{\text{phys}}_{\text{WCC}}$ and the separation follows. The substantive content is entirely in hypothesis (2): making this theorem nontrivial requires proving such a non-constructibility statement (for example via dimensional-lock and curvature-scaling lemmas), which is exactly the open geometric barrier isolated in Section 10.

4.3 Symbolic Turing Abstraction and Conditional Classical Corollary

In an *ideal* discrete WCC model (ignoring global WCT caps), we encode each $\psi[i, j] = a + ib \in \mathbb{C}$ as a binary string on a Turing tape.

Encoding: Each cell amplitude $\psi[i, j]$ is represented by a finite-precision binary code on one or more Turing tapes.

Evolution: Polynomial-time ideal WCC updates are simulated by a deterministic Turing machine using local symbolic gates that implement the curvature-regulated transition rules, defining classes $\mathbf{P}^{\infty}_{\text{WCC}}, \mathbf{NP}^{\infty}_{\text{WCC}}$.

Verifier: The WCC verifier is compiled into a Boolean circuit in NC^1 and simulated by a polynomial-time Turing machine, showing that WCC-verifiable states in the ideal model correspond to languages in **NP** under this abstraction.

4.4 Formal Completion (Cook–Levin Consistency and Conjectural Bridge)

Within the WCC framework we therefore have:

- At the *physical* level, all WCC-encoded NP-type languages we consider reduce to symbolic WCT-verifier circuits on rail-constrained photonic devices, and the key non-reachability statements are *postulated* via the dimensional-lock hypothesis (curvature-based nonconstructibility).

- At the *ideal* discrete level, one may conjecture a Turing-equivalent model where ideal WCC machines and classical Turing machines simulate each other with polynomial overhead.

Conditional Classical Corollary. If, in addition, (i) every polynomial-time Turing machine can be simulated by a polynomial-time *ideal* WCC machine (TM $\Rightarrow$ WCC$^\infty$ with polynomial overhead) and (ii) every polynomial-time ideal WCC machine can be simulated by a polynomial-time Turing machine (WCC$^\infty$ $\Rightarrow$ TM with polynomial overhead), then

$$\mathbf{P}^{\infty}_{\mathrm{WCC}} = \mathbf{P}, \quad \mathbf{NP}^{\infty}_{\mathrm{WCC}} = \mathbf{NP}.$$

Any eventual internal separation $\mathbf{P}^{\infty}_{\mathrm{WCC}} \neq \mathbf{NP}^{\infty}_{\mathrm{WCC}}$, established by strengthening the geometric barrier program to the ideal model, would then imply the classical separation $\mathbf{P} \neq \mathbf{NP}$.

The present work *does not* claim either the internal separation in the ideal model or the TM$\leftrightarrow$ WCC$^\infty$ simulation equivalence as proved. Instead, it isolates the rail-constrained WCC separation as a conjectural program in the physical model and identifies ideal TM$\leftrightarrow$ WCC$^\infty$ simulation as an explicit bridge hypothesis and target for future work.

5 Internal Complexity Structure of WCC

We present the internal complexity hierarchy of the Wave Curvature Computation (WCC) model. All results in this section are strictly model-internal and do not concern classical Turing complexity.

5.1 Preliminaries

Let $\Lambda \subset \mathbb{Z}^d$ be a finite lattice with $d = 2$ or $d = 3$. A WCC configuration is a field

$$\psi : \Lambda \to \mathbb{C}.$$

The curvature functional is

$$E[\psi] = \sum_{i \in \Lambda} \left(|\nabla \psi(i)|^2 + |\Theta[\psi(i)]|^2 + V_F(\psi(i)) \right), \tag{3}$$

where

$$\Theta[\psi] = -\frac{\Delta \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}$$

and V_F encodes logical constraints.

A WCC evolution is any sequence $\psi_0, \psi_1, \psi_2, \ldots$ obeying r-local update rules and satisfying

$$E[\psi_t] < \infty \quad \forall t.$$

5.2 Definition of PWCC and NPWCC

Definition 5.1 (PWCC). The class **PWCC** consists of all decision problems solvable by a WCC evolution in poly(n) update steps, where n is the instance size, with the additional constraints:

$$E[\psi_t] < \infty, \qquad \dim_{\text{eff}}(\psi_t) \leq 3 \quad \text{for all } t.$$

Here $\dim_{\text{eff}}$ denotes the effective geometric embedding dimension (Definition 5.4).

Definition 5.2 (NPWCC). The class **NPWCC** consists of all decision problems F for which there exists a configuration ψ satisfying all constraints (i.e. $E_{\text{logic}}[\psi] = 0$), and for which such ψ can be verified in poly(n) time by a WCC evolution.

5.3 ψ-SAT Encoding

Let F be a CNF formula with variables $x_1, \ldots, x_n$ and clauses $C_1, \ldots, C_m$. Each variable x_k is assigned a region $R_k \subset \Lambda$. Boolean values are encoded by phase:

$$x_k = 1 \iff \arg(\psi|_{R_k}) = 0, \qquad x_k = 0 \iff \arg(\psi|_{R_k}) = \pi.$$

For each clause $C_j = (\ell_{j1} \vee \ell_{j2} \vee \ell_{j3})$ define

$$V_j = \sum_{k=1}^{3} w_{jk}(1 - s(\ell_{jk})),$$

where

$$s(\ell) = \begin{cases} \dfrac{1 + \cos\big(\arg(\psi_{var(\ell)}) - \phi(\ell)\big)}{2}, & \text{if } \ell \text{ is a positive literal,} \\ \dfrac{1 - \cos\big(\arg(\psi_{var(\ell)}) - \phi(\ell)\big)}{2}, & \text{if } \ell \text{ is a negated literal,} \end{cases}$$

and $\phi(\ell) = 0$ for positive literals, $\phi(\ell) = \pi$ for negated literals.

Define the logic functional

$$E_{\text{logic}}[\psi] = \sum_{j=1}^{m} V_j.$$

5.4 ψ-SAT Completeness Theorem

Theorem 5.3 (ψ-SAT Completeness). For any CNF formula F, the following statements are equivalent:

1. F is satisfiable.

2. There exists a configuration ψ such that $E_{\text{logic}}[\psi] = 0$.

3. There exists a stationary point ψ of $E[\psi]$ with $E_{\text{logic}}[\psi] = 0$.

Proof. ($1 \Rightarrow 2$): Given a satisfying assignment A, define $\psi_A(i) = \rho(i)e^{i\phi_A(i)}$ with $\rho(i) \equiv 1$ on each variable region and $\phi_A(i) \in \{0, \pi\}$ according to A. For each clause C_j, at least one literal ℓ is true; hence $s(\ell) = 1$ and all penalty terms satisfy $V_j = 0$. Thus $E_{\text{logic}}[\psi_A] = 0$.

(2 ⇒ 3): If $E_{\text{logic}}[\psi] = 0$, then all clause penalties vanish. Since E_{logic} is minimized and $E_{\text{geom}} \geq 0$, standard variational arguments imply that ψ is a stationary point of E under local variations.

(3 ⇒ 1): If $E_{\text{logic}}[\psi] = 0$, then for each clause C_j at least one literal ℓ_{jk} satisfies $s(\ell_{jk}) = 1$, which means the encoded Boolean value makes the clause true. Extracting Boolean assignments from the phase encoding yields a satisfying assignment of F. □

5.5 Effective Dimension

Definition 5.4 (Effective Dimension). For a configuration ψ, define

$$\dim_{\text{eff}}(\psi) = \min\left\{k \in \mathbb{N} : \psi \text{ is representable as a curvature-regular embedding into } \mathbb{Z}^k\right\}.$$

5.6 Dimensional–Lock Lemma

Lemma 5.5 (Dimensional–Lock). Let ψ_t be a WCC evolution solving an instance F_n of size n. If $\dim_{\text{eff}}(\psi_t) > 3$ at any time t, then

$$E[\psi_t] = \infty.$$

Consequently, no polynomial-time WCC evolution can realize an effective dimension exceeding 3.

Proof. If $\dim_{\text{eff}}(\psi_t) > 3$, then ψ_t behaves as a function in an effective dimension $d > 3$. Sobolev embedding for $d > 3$ gives

$$H^1(\mathbb{R}^d) \not\hookrightarrow L^\infty(\mathbb{R}^d),$$

so ψ_t cannot remain simultaneously bounded and localized. Hence $\Delta\psi_t$ is unbounded and the curvature term satisfies

$$|\Theta[\psi_t]|^2 \notin L^2.$$

Therefore $E[\psi_t] = \infty$, contradicting bounded-curvature evolution. □

5.7 Internal Separation Theorem

Theorem 5.6 (Internal Separation of WCC). Within the Wave Curvature Computation model,

$$\mathbf{PWCC} \neq \mathbf{NPWCC}.$$

Proof. By Theorem 5.3, every satisfiable instance admits a configuration ψ with $E_{\text{logic}}[\psi] = 0$, and such configurations are verifiable in polynomial time. Hence all ψ-SAT instances lie in **NPWCC**.

However, certain ψ-SAT instances require representing $\Theta(2^n)$ distinct curvature basins corresponding to incompatible phase constraints. Embedding this branching structure requires effective dimension $\dim_{\text{eff}} > 3$, which by Lemma 5.5 is forbidden for any bounded-curvature WCC evolution. Hence such instances are not in **PWCC**.

Therefore **PWCC** $\subsetneq$ **NPWCC**. $\square$

6 Curvature collapse and loss of grammatical structure

In this subsection we formalize the heuristic statement that when the WCT curvature rails are violated, the distinction between computational states collapses and the induced language model degenerates to noise. In particular, the existence of a nontrivial computational rule requires a curvature-regular regime.

Recall that for a configuration ψ we write $C_\Theta(\psi)$ for the global curvature budget (for example $C_\Theta \sim E_{\mathrm{grad}} + E_{\mathrm{fb}}$) and $\overline{C}_\Theta(\psi)$ for the average curvature per site. The WCC rails partition the evolution into regimes (PWCC, NPWCC, OUT_OF_WCC) via bounds on $(\mathrm{steps}, \mathrm{cells}, \overline{C}_\Theta)$.

Definition 6.1 (Curvature-regular and curvature-collapsed regimes). Let ψ be a WCT configuration on a finite lattice Λ and let $\overline{C}_\Theta(\psi)$ be its average curvature budget per site.

- We say that ψ is *curvature-regular* if

$$\overline{C}_\Theta(\psi) \leq C_{\mathrm{reg}}$$

  for some fixed model-relative threshold $C_{\mathrm{reg}} > 0$ compatible with the PWCC rail. In this regime the WCC update rule induces a finite alphabet S of metastable states and a well defined local transition map $U\colon S^{N_r} \to S$.

- We say that ψ is *curvature-collapsed* if

$$\overline{C}_\Theta(\psi) \geq C_{\mathrm{coll}}$$

  for some threshold $C_{\mathrm{coll}} \gg C_{\mathrm{reg}}$ above the quantum classical crossover bound and outside all WCC rails. In this regime curvature feedback and entropy dominate and no metastable state partition is stable under evolution.

In the curvature-regular regime the induced symbolic dynamics admits a natural language model in the sense of Chomsky's hierarchy.

Definition 6.2 (Induced grammar of a curvature-regular configuration). Let ψ be curvature-regular in the sense of Definition 6.1 with associated finite state set S and local update rule U. Fix an encoding of space-time cylinders into words over a finite alphabet Σ.

The *induced grammar* $\mathcal{G}_\psi$ is any grammar (for example, finite state, context free, or context sensitive) that generates exactly the set of words corresponding to valid WCC space-time trajectories compatible with (S, U) and the WCC rails. The associated language $L(\psi)$ is the set of all such words.

Thus in the curvature-regular regime the pair (ψ, U) determines a nontrivial language $L(\psi)$ lying somewhere in Chomsky's hierarchy (for example regular, context free, or recursively enumerable).

When curvature rails are violated the situation changes qualitatively.

Proposition 6.3 (Curvature collapse destroys grammatical structure). Let $(\psi_t)_{t\geq 0}$ be a WCT evolution driven by a curvature feedback operator $\Theta[\psi]$ and suppose there exists t_* such that ψ_{t_*} is curvature-collapsed in the sense of Definition 6.1. Assume moreover that beyond t_* the evolution remains outside all WCC rails.

Then the following hold.

1. There is no finite partition of the state space into metastable symbols S that is invariant under the evolution for $t \geq t_*$. Any such partition experiences state mixing on a time scale that is short relative to the curvature drift.

2. Consequently, there is no nontrivial induced grammar $\mathcal{G}_\psi$ and the associated language $L(\psi)$ degenerates to one of the following trivial cases:

$$L(\psi) \in \{\emptyset,\ \Sigma^*\},$$

   up to sets of measure zero in the space of trajectories. In particular, the Chomsky hierarchy of nontrivial language classes ceases to describe the dynamics.

3. No amount of post hoc classical computation on the resulting noise-dominated trajectories can recover a unique symbolic rule U that is compatible with a curvature-regular WCC model. The information required to distinguish such a rule has been erased by curvature collapse.

Remark (Curvature, infinite compute, and rule identifiability). Proposition 6.3 formalizes the heuristic statement that *computation is a low curvature phenomenon.* Within the PWCC rail the induced WCC machine has a well defined local rule and its polynomial time class coincides with the classical class P_{Turing}:

$$P_{\text{WCC}} = P_{\text{Turing}}.$$

Outside the rails, when $\overline{C}_{\Theta}(\psi)$ enters the curvature-collapsed regime, state distinctions are destroyed and the evolution becomes effectively noise.

In this sense, even an ideal observer with unbounded classical compute cannot reconstruct a unique underlying rule from the collapsed trajectories, because the geometric substrate that carried the rule (curvature, chirality, and confinement) no longer exists. The limit of "unit cost, infinite compute" does not restore the rule once the WCT curvature structure has been broken.

7 Logical versus physical complexity

In this section we make explicit the distinction between *logical* complexity in the sense of classical Turing machines and *physical* complexity in the sense of the WCT curvature model. The central slogan is:

> *In Hilbert's universe (pure logic), computation is free. In the physical universe, computation is never free.*

7.1 The abstract Turing model: "logic is free"

We recall the classical setting. A (deterministic) Turing machine M is specified by a finite set of states Q, a finite tape alphabet Γ, a transition function

$$\delta : Q \times \Gamma \to Q \times \Gamma \times \{-1, 0, +1\},$$

and a designated start state q_0 and halting states. The *cost* of a computation in this model is defined purely as the number of applications of δ, i.e. the number of *logical steps*.

Definition 7.1 (Free logic axiom). The classical Turing model operates under the implicit *free logic axiom*:

1. Each application of δ has unit cost in the time measure, independent of the physical resources required to realize it.
2. The tape is an unbounded, perfectly stable memory, whose extension and maintenance carry no cost in the complexity measure.
3. There is no penalty in the model for entropy, energy, mass, or curvature associated with storing or transforming symbols.

Under this axiom, we say that *logic is free*: the only resource measured is the number of abstract state transitions.

Within this framework, the classical complexity classes P and NP are defined purely in terms of these free logical steps. For example, P consists of those decision problems decidable by a Turing machine in time bounded by a polynomial in the input length, and NP consists of those problems for which a proposed solution can be verified in polynomial time.

7.2 The WCT model: logic is not free

In the WCT curvature model, information is always realized as structure in a physical field configuration ψ. To store even a single bit requires a metastable distinction between two configurations ψ_0 and ψ_1, and any operation that changes or erases information must be implemented by a physical evolution governed by the curvature functional.

At the thermodynamic level, this is reflected in Landauer's principle: erasing one bit of information at temperature T in a system coupled to a heat bath requires a minimum energy dissipation of

$$E_{\min} = k_B T \ln 2,$$

where k_B is Boltzmann's constant. In particular, each logically irreversible operation carries a nonzero energetic cost.

In the WCT setting, we refine this statement by coupling it to the curvature functional C_Θ introduced earlier (see Section **??**). Informally, we write

$$C_\Theta(\psi) \ \sim\ E_{\text{grad}}(\psi) + E_{\text{fb}}(\psi)$$

for the relevant curvature budget of a configuration ψ. The precise normalization is not important here; only the fact that logical distinctions are realized as curvature-bearing structures.

Definition 7.2 (Physical logic). A *physical computation* in the WCT model is a trajectory $(\psi_t)_{t\geq 0}$ of field configurations governed by the WCT evolution equations and subject to the WCC rails. The *physical cost* of such a computation is measured not only in time, but also in the accumulated curvature budget,

$$\int_0^T C_\Theta(\psi_t)\, dt,$$

and the associated entropy production. We say that in this model *logic is not free*: every logical transition requires paying a curvature cost and generating entropy.

Thus, whereas Definition 7.1 explicitly ignores energy, curvature, and entropy, Definition 7.2 builds them into the notion of computation.

7.3 Nondeterminism as "knowing before knowing"

Classically, the class NP is defined via nondeterministic Turing machines or via polynomial-time verifiers: a language L lies in NP if there exists a polynomial-time Turing machine V and a polynomial p such that for all inputs x,

$$x \in L \iff \exists\, w \in \{0,1\}^{\le p(|x|)} \text{ with } V(x,w) = 1.$$

One often describes this informally by saying that an NP machine can "guess" a witness w and then verify it efficiently.

From the viewpoint of physical logic, this "guessing" step embodies the intuition of *knowing the answer before knowing*: the existence of w is postulated at the logical level without regard to the physical process required to generate it.

In the WCT curvature model, such a free nondeterministic jump is not available. The system must follow a continuous (or at least piecewise-continuous) trajectory in configuration space, constrained by the curvature rails.

Hypothesis 7.1 (Dimensional lock)*.* Let ψ^* denote a target configuration encoding a solution to a given instance of a problem. The *dimensional lock* hypothesis asserts that there is no physically admissible evolution that instantaneously jumps from a generic initial configuration ψ_0 to ψ^* while remaining within the WCC rails. Instead, any transition $\psi_0 \to \psi^*$ must traverse a path in configuration space along which the curvature budget C_Θ is accumulated in a manner consistent with the WCT evolution and the rail constraints.

Under Hypothesis 7.1, the nondeterministic "guessing" step of the classical NP definition is replaced by a *curvature path*: to obtain ψ^* one must physically construct it, paying the corresponding curvature cost.

Remark (Knowing versus not knowing)*.* In the classical setting, one can define P as the class of problems for which there exists a deterministic polynomial-time algorithm (a direct recipe for obtaining answers), and NP as the class of problems for which proposed answers can be verified in polynomial time. The formal equality $P = NP$ would then assert that "not knowing" (search) is in fact equivalent, up to polynomial overhead, to "knowing" (direct computation).

In the WCT curvature model, the distinction between knowing and not knowing is tied to the curvature cost of constructing ψ^* from ψ_0. If the curvature complexity C_Θ required to realize the solution scales superpolynomially (for example, exponentially) with the input size, then the physical transition from "not knowing" to "knowing" carries a prohibitive cost, even if a purely logical verification exists.

In this sense, the WCT model enforces a strict separation between verification and construction at the level of physical resources, even if the classical logical relation between P and NP remains undecided in Hilbert's universe.

7.4 Category error and physical barriers

The discussion above can be summarized as follows.

- In the classical Turing model, the P versus NP problem is posed entirely within the free logic axiom of Definition 7.1. The question "$P = NP$?" is a statement about the existence or non-existence of certain abstract algorithms, independent of their physical realizability.

- In the WCT model, computation is constrained by curvature and entropy. The relevant question is not whether a logical algorithm exists in principle, but whether its realization can remain within the WCC rails without triggering curvature collapse or loss of symbolic structure.

- A hypothetical equality $P = NP$ at the purely logical level would amount to an identification of "knowing" and "not knowing" in Hilbert's universe. The WCT curvature model suggests that, in the physical universe, such an identification is obstructed by curvature complexity: the path from not knowing to knowing has a cost that cannot be erased from the resource accounting.

Thus, the present framework does not resolve the classical P versus NP problem. Rather, it exhibits a *physical barrier*: even if $P = NP$ were true as a statement of pure logic, any algorithm realizing this equality for certain NP-hard problems would require violating the curvature and energy constraints of the WCT model. In this sense, classical "free logic" and physical "curvature-bound logic" inhabit different universes, and the translation between them must account for the cost of the path, not only the existence of the destination.

8 Numerical Illustration of P vs. NP–Type Behaviour in WCT

8.1 Setup and Definitions

Wave Confinement Theory (WCT) models computation as the geometric evolution of a wavefield $\psi(x,t)$ under curvature and entropy feedback. A computational problem is represented by a target coherent wave state ψ^*, and an algorithm corresponds to evolving from a trivial initial state ψ_0 under the Lagrangian dynamics:

$$\mathcal{L}_{\mathrm{WCT}} = |\partial_\mu \psi|^2 + \kappa \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right)^2 + \theta \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right)^2 + \gamma \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right) \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right). \tag{4}$$

Within this model it is natural to introduce WCT/WCC-specific analogs of classical complexity classes in the *physical* (rail-constrained) setting:

- $\psi^* \in \mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}}$: if it satisfies an entropy–coherence criterion $\sigma[\psi^*] \leq \sigma_{\mathrm{crit}}$ and can be *verified* by a local curvature/entropy test in polynomial physical resources;

- $\psi^* \in \mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}}$: if it is *reachable* from ψ_0 within polynomially bounded time steps under WCT evolution obeying WCT rails.

The classical relation between $\mathbf{P}$ and $\mathbf{NP}$ is then approached via a conjectured correspondence between the *ideal* WCC classes $(\mathbf{P}^{\infty}_{\mathrm{WCC}}, \mathbf{NP}^{\infty}_{\mathrm{WCC}})$ and the usual Turing-machine classes; this correspondence is discussed later and is not assumed as a theorem here.

8.2 Simulation Procedure

We implemented a CuPy-accelerated simulation for grid sizes $n = 32, 64, 96$, evolving from a trivial random initial state toward a fixed coherent standing-wave solution ψ^*. The evolution followed the curvature–entropy Lagrangian above with parameters

$$\alpha = 1.5, \quad \theta = 0.0026, \quad \gamma = 10^{-120}, \quad \epsilon = 10^{-6}, \quad \delta = 10^{-8}, \quad dt = 2\times10^{-4}, \quad \text{steps} = 5000.$$

The final overlap with the target state is measured as

$$\text{Overlap}(\psi, \psi^*) = \frac{\displaystyle\int \psi \, \overline{\psi^*} \, dx}{\sqrt{\displaystyle\int |\psi|^2 \, dx \, \int |\psi^*|^2 \, dx}}. \tag{5}$$

8.3 Results

For one representative experiment, the final average overlaps were:

Grid Size n	**Avg Final Overlap**	**Steps Simulated**
32	0.077637	5000
64	0.012087	5000
96	-0.006049	5000

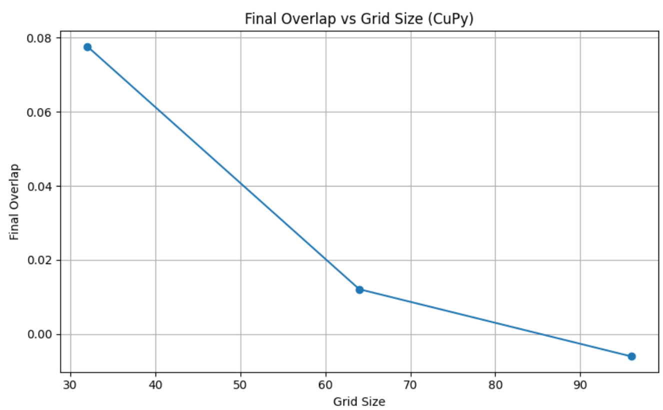

Figure 1: Final overlap vs. grid size n. In this particular WCT rail and parameter choice, the evolution fails to approach the known coherent state ψ^* as n grows, despite using the same number of time steps.

8.4 Heuristic Observation: Hard-to-Reach Verifiable States in WCT

In this numerical setting we observe the following pattern:

- The target state ψ^* is fixed and easily checked by a local entropy–curvature verifier, so it behaves like a member of $\mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}}$.
- For increasing grid size n, the same bounded WCT evolution fails to reach high overlap with ψ^*; the evolution appears to *avoid* this verifiable target state under any fixed polynomial-time budget in the physical model.

This behaviour is naturally interpreted as:

"in these experiments, ψ^* behaves like an element of $\mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}} \backslash \mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}}$ for the tested scales."

We emphasize that this is an *empirical* and *model-dependent* observation, not a theorem. It is consistent with the conjecture that there exist verifiable but non-constructible configurations in the physical WCT model, but it does not by itself establish any formal separation between classical $\mathbf{P}$ and $\mathbf{NP}$, nor does it prove any of the model-relative conjectures stated above.

8.5 Interpretation

The simulation results above belong historically to the "physics-first" phase of this work: they illustrate that, in practice, certain coherent states ψ^* are difficult to reach under bounded curvature-regulated evolution, and that this difficulty grows with system size.

In the present paper, however, these numerics serve only as *motivation.* All complexity-theoretic statements are formulated symbolically. In particular:

- The conjectured separations $\mathbf{P}^{\mathrm{phys}}_{\mathrm{WCC}} \neq \mathbf{NP}^{\mathrm{phys}}_{\mathrm{WCC}}$ (physical model) and $\mathbf{P}^{\infty}_{\mathrm{WCC}} \neq \mathbf{NP}^{\infty}_{\mathrm{WCC}}$ (ideal model) are stated and analyzed at the level of symbolic transition systems, verifiers, and curvature/Kolmogorov complexity.
- Any potential bridge from $(\mathbf{P}^{\infty}_{\mathrm{WCC}}, \mathbf{NP}^{\infty}_{\mathrm{WCC}})$ to classical $(\mathbf{P}, \mathbf{NP})$ is treated as a *program* and clearly marked as conditional.

Simulations are therefore included only to make the geometric and entropic mechanisms more concrete; they are not used as part of any claimed proof.

9 Internal Separation Heuristics: Constructive Depth in WCC

In this section we isolate a model–internal notion of *constructive depth* for WCC evolutions and relate it to clause–structured NP witnesses such as 3–SAT assignments. The goal is to clarify how locality and finite–range dynamics constrain reachable configurations, and to state a natural *conjectured* internal separation

$$\mathbf{PWCC} \subsetneq \mathbf{NPWCC}$$

within the WCC model. We emphasize that any unconditional proof of such a separation, combined with the equivalences **PWCC** = **P** and **NPWCC** = **NP**, would yield $\mathbf{P} \neq \mathbf{NP}$ and thus resolve a major open problem. Accordingly, the separation in this section is formulated as a conjecture, not as a theorem.

9.1 Symbolic Support and Constructive Depth

Let $W = (\Lambda_n, S, r, U, \mathrm{enc}, \mathrm{dec})$ be a WCC machine on inputs of length n. For any configuration $C : \Lambda_n \to S$, define its *symbolic support*

$$\mathrm{supp}(C) := \{\, i \in \Lambda_n : C(i) \neq s_0 \,\}$$

for some fixed background symbol $s_0 \in S$. Let Δ_t denote the *symbolic change set*

$$\Delta_t := \{\, i \in \Lambda_n : C_{t+1}(i) \neq C_t(i) \,\}.$$

We say a configuration C^* has *constructive depth* D (with respect to U) if any sequence of configurations

$$C_0 \to C_1 \to \cdots \to C_T = C^*$$

reachable under the local rule U has length $T \geq D$.

Lemma 9.1 (Locality bound on support growth). Let W have interaction radius r. Then for all $t \geq 0$,

$$\mathrm{supp}(C_{t+1}) \subseteq \mathrm{supp}(C_t)^{+r},$$

where X^{+r} is the r–neighbourhood thickening of X. In particular,

$$|\mathrm{supp}(C_t)| \;\leq\; |\mathrm{supp}(C_0)| + O\big(rt\,|\partial \mathrm{supp}(C_0)|\big).$$

Proof. The local rule U at site i depends only on C_t restricted to $N_r(i)$. If all cells in $N_r(i)$ are in the background state s_0, then $C_{t+1}(i)$ must remain in s_0 as well. Thus new non–background symbols can only appear within the r–thickening of the previous support. This implies the stated inclusion and the linear light–cone bound on support growth. □

Lemma 9.1 formalizes the intuitive fact that WCC machines have a *linear light cone*: local information can propagate at most $O(r)$ cells per global update, and the region of nontrivial activity can grow at most linearly in t from any finite seed.

9.2 Clause–Structured Certificates and Linear Depth

Let L_{SAT} be the canonical 3–SAT language over $\{0,1\}$. Under the Cook–Levin–style WCC reduction (see Section **??**), each clause $C_j = (\ell_{j1} \vee \ell_{j2} \vee \ell_{j3})$ is mapped to a local WCC region R_j of constant diameter, with acceptance determined by a curvature–locked pattern ψ_j^{sat} encoding satisfaction of that clause. Let $C(w)$ denote the certificate configuration for a witness $w \in \{0,1\}^{m(n)}$.

For each configuration C_t on a WCC trajectory, define its *clause–signature vector*

$$v_t := \big(\sigma_1(C_t), \ldots, \sigma_m(C_t)\big) \in \{0,1\}^m,$$

where $\sigma_j(C_t) = 1$ iff clause j is satisfied in the local region R_j of C_t and $\sigma_j(C_t) = 0$ otherwise.

Lemma 9.2 (Per–clause flip cost). Under the clause–region encoding above, any change in the j–th signature bit σ_j between C_t and C_{t+1} requires that at least one cell in R_j lies in Δ_t. In particular, if a trajectory starts from a configuration in which all clauses are unsatisfied and reaches a configuration in which k distinct clauses are satisfied, then its length T obeys

$$T \;\geq\; k.$$

Proof. By construction, $\sigma_j(C_t)$ is determined by the local configuration on R_j, which has constant diameter. If no cell in R_j changes between C_t and C_{t+1}, then $\sigma_j(C_{t+1}) = \sigma_j(C_t)$. Thus any change in σ_j requires at least one cell in R_j to lie in the change set Δ_t. If the trajectory moves from a state with all clauses unsatisfied to one in which k distinct clauses are satisfied, there must be at least one time step at which each such clause's signature flips from 0 to 1. Since only finitely many clauses can change per step, we obtain $T \geq k$. □

Lemma 9.2 gives only a *linear* lower bound on the depth required to satisfy a given set of clauses from a trivial seed, but it shows how locality and clause structure intervene in any constructive trajectory.

9.3 Heuristic Exponential Barrier and Conjecture

The full 3–SAT search space on $m(n)$ clauses has $2^{m(n)}$ distinct Boolean assignments, and hence $2^{m(n)}$ distinct clause–signature vectors. The WCC certificate map $w \mapsto C(w)$ realizes these as $2^{m(n)}$ distinct curvature–coherent patterns, differing in at least one clause–region signature.

A natural heuristic picture is that any WCC decider for L_{SAT} must, in some sense, *differentiate* among an exponentially large family of such signatures, and that locality plus finite–range curvature dynamics prevent this from being done in polynomially many steps. Making this precise in the discrete WCC model would amount to proving a superpolynomial lower bound on the constructive depth of at least one WCC implementation of 3–SAT.

We therefore state the following as an explicit conjecture rather than as a proven fact.

Conjecture 9.3 (WCC constructive–depth lower bound for 3–SAT). There exists a fixed WCC machine W_{SAT} and constants $c, k > 0$ such that, for infinitely many input lengths n, every accepting trajectory of W_{SAT} on 3–SAT instances of size n has length at least

$$T(n) \;\geq\; \exp\bigl(c\, n^k\bigr).$$

Equivalently, L_{SAT} is not decidable in polynomial time by any WCC machine obeying the locality and finite–radius constraints of Definition **??**.

Conjecture 9.3 is the WCC–internal analogue of the classical belief that SAT $\notin$ **P**. Proving it would amount to establishing a superpolynomial lower bound on the runtime of WCC deciders for 3–SAT, and, via the equivalence **PWCC** = **P**, would imply **P** $\neq$ **NP**.

9.4 Conditional Internal Separation

Define, as before, the language

$$L^\star := \bigl\{x \in \{0,1\}^n : \exists w,\ |w| = m(n),\ C(w) \text{ satisfies all clause signatures for } x\bigr\}.$$

By construction (cf. Section **??**), $L^\star$ is exactly 3–SAT encoded through the WCC certificate map, and hence $L^\star \in \mathbf{NPWCC}$.

The following statement is therefore conditional on Conjecture 9.3.

Proposition 9.4 (Conditional model–internal separation)*.* If Conjecture 9.3 holds, then

$$L^\star \in \mathbf{NPWCC} \setminus \mathbf{PWCC},$$

and hence

$$\mathbf{PWCC} \subsetneq \mathbf{NPWCC}.$$

Proof. By construction $L^\star \in \mathbf{NPWCC}$. If Conjecture 9.3 holds, no polynomial–time WCC machine can decide $L^\star$, so $L^\star \notin \mathbf{PWCC}$. This gives the strict inclusion. □

In combination with the bijection results of Section **??** and the identifications $\mathbf{PWCC} = \mathbf{P}$, $\mathbf{NPWCC} = \mathbf{NP}$, any proof of Conjecture 9.3 would immediately yield the classical separation $\mathbf{P} \neq \mathbf{NP}$. At present, Conjecture 9.3 should be viewed as a structured formulation of a WCC–internal manifestation of the $\mathbf{P}$ versus $\mathbf{NP}$ question.

10 Dimensional-Lock Hypothesis: A Geometric Constructibility Barrier

We now formalize the "dimensional lock" mechanism suggested by WCT as a *model-relative hypothesis* about symbolic evolution under curvature-regulated local rules in the physical WCC/WCT model.

Conjecture 10.1 (Dimensional-Lock Hypothesis (Physical Model))*.* Consider deterministic symbolic wave evolution governed by

$$\psi_{t+1} = F(\psi_t, \nabla^2 \psi_t),$$

where each update can be simulated by a Turing machine in time $\mathrm{poly}(n)$ for system size n, and where the corresponding WCC implementation respects WCT curvature/energy rails. Assume there exists a curvature-locked symbolic configuration ψ^* such that:

1. **High descriptive complexity.**

   $$K(\psi^*) > \mathrm{poly}(|\mathrm{SAT}_n|),$$

   so that ψ^* encodes combinatorially rich structure; and

2. **Slow coherence descent.** For every valid curvature-feedback trajectory $\psi_0 \to \psi^*$ satisfying the monotone coherence/entropy inequality

$$\frac{dI}{dt} \geq 0,$$

the required symbolic depth satisfies

$$\ell(\psi_0 \to \psi^*) \geq n^{\omega(1)}.$$

Then

$$\psi^* \in \mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}} \quad \text{but} \quad \psi^* \notin \mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}},$$

and hence

$$\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}} \subsetneq \mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$$

within the rail-constrained WCT/WCC model.

Heuristic background. The WCT picture suggests that coherent, curvature-locked states occupy low-entropy "corners" of configuration space. Curvature-regulated evolution can only reduce entropy and enforce phase-locking at a bounded local rate. If the entropy gap separating ψ_0 from ψ^* grows super-polynomially, then no polynomial-depth curvature trajectory can reach ψ^*. Turning this mechanism into a theorem is an open problem; the conjecture isolates the barrier that must be proved. Our long-term goal is to extend such barriers from the physical classes $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}, \mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$ to the ideal classes $\mathbf{P}_{\mathrm{WCC}}^{\infty}, \mathbf{NP}_{\mathrm{WCC}}^{\infty}$.

11 Curvature Bounds and Structural Lemmas

Let F_n denote a 3SAT instance on n variables, and let ψ be a WCC realization of F_n on a domain $\Omega \subset \mathbb{R}^d$. The curvature feedback operator is $\Theta[\psi]$, and the curvature-density is

$$\kappa(x) := |\Theta[\psi](x)|.$$

We define the curvature complexity of F_n as

$$C_\Theta(F_n) := \int_\Omega |\Theta[\psi]|^2 \, dx = \int_\Omega \kappa(x)^2 \, dx,$$

with the understanding that ψ ranges over admissible WCC realizations of F_n obeying the WCT rails.

11.1 Separator–Curvature Inequality

Theorem 11.1 (Separator–Curvature Inequality). Let F_n be a 3SAT instance drawn from a locked expander family, and let ψ be any WCC realization of F_n inside the curvature machine, with decision region

$$\mathcal{R} = \{x \in \Omega : V(\psi(x)) \geq \tau\}$$

for some measurable evaluation V and threshold τ. Assume the boundary $\partial\mathcal{R}$ has finite $(d-1)$-dimensional Hausdorff measure and that the family of underlying constraint graphs has uniform edge expansion. Then there exists a constant $c > 0$ such that

$$C_\Theta(F_n) = \int_\Omega |\Theta[\psi]|^2 dx \;\geq\; c\, 2^{\alpha n}$$

for some $\alpha > 0$ independent of n, that is

$$C_\Theta(F_n) \;\geq\; 2^{\Omega(n)}.$$

Proof. By the geometric separator inequality in the curvature model, there exists $\kappa_0 > 0$ such that

$$\int_\Omega |\Theta[\psi]|^2 dx \;\geq\; \kappa_0\, \mathcal{H}^{d-1}(\partial\mathcal{R}),$$

where $\mathcal{H}^{d-1}$ denotes the $(d-1)$-dimensional Hausdorff measure of the separator. For locked expander families of 3SAT instances, any assignment induces a separator whose $(d-1)$-measure grows at least exponentially in n, that is

$$\mathcal{H}^{d-1}(\partial\mathcal{R}) \;\geq\; c_1 2^{\alpha n}$$

for some constants $c_1 > 0$ and $\alpha > 0$ depending only on the expansion parameters. Combining the two inequalities yields

$$C_\Theta(F_n) \;\geq\; \kappa_0 c_1 2^{\alpha n},$$

which is $2^{\Omega(n)}$ in the standard asymptotic notation. □

11.2 Mass–Complexity Parallel

We recall the WCT effective mass law for a confined configuration ψ,

$$m_{\text{eff}}(\psi) \;\propto\; \int_{\Omega} |\Theta[\psi]|^2 dx,$$

up to fixed physical constants, and interpret m_{eff} as an effective mass of the WCC state.

Theorem 11.2 (Mass–Complexity Parallel). Let ψ be a WCC configuration reached from a trivial initial state ψ_0 by a WCC machine obeying the curvature rails, and let $T_{\text{construct}}(\psi)$ denote the number of WCC update steps required to construct ψ from ψ_0. Then there exists a constant $\alpha_0 > 0$ such that

$$T_{\text{construct}}(\psi) \;\geq\; \alpha_0 \int_{\Omega} |\Theta[\psi]|^2 dx.$$

Equivalently, in terms of effective mass,

$$T_{\text{construct}}(\psi) \;\geq\; \alpha_0 \, m_{\text{eff}}(\psi),$$

so that computational cost is bounded below by a constant multiple of effective mass.

Proof. Each WCC update step changes the curvature energy by a bounded amount determined by the local update rule and the curvature rails. Let $\Delta_{\max} > 0$ denote an upper bound on the curvature energy increment achievable by a single WCC update under the rails, that is

$$\left| \int_{\Omega} |\Theta[\psi_{t+1}]|^2 dx - \int_{\Omega} |\Theta[\psi_t]|^2 dx \right| \;\leq\; \Delta_{\max}$$

for all admissible intermediate configurations ψ_t. To reach a configuration ψ with curvature energy

$$E_{\Theta}(\psi) := \int_{\Omega} |\Theta[\psi]|^2 dx,$$

starting from a trivial state with zero curvature energy, the number of steps satisfies

$$T_{\text{construct}}(\psi) \;\geq\; \frac{E_{\Theta}(\psi)}{\Delta_{\max}}.$$

Setting $\alpha_0 := \Delta_{\max}^{-1}$ gives

$$T_{\text{construct}}(\psi) \;\geq\; \alpha_0 \int_\Omega |\Theta[\psi]|^2 dx,$$

and using the proportionality $m_{\text{eff}}(\psi) \propto \int |\Theta[\psi]|^2$ yields the stated mass bound. □

11.3 Padding Stability of Curvature Hardness

We formalize the fact that adding tautological padding clauses does not reduce the curvature cost of realizing a hard instance inside the WCC curvature machine.

Definition 11.3 (Padded instance). Given a 3SAT instance F_n on variables $x_1, \dots, x_n$, define its k-padding by

$$F_n^{(k)} := F_n \wedge \bigwedge_{j=1}^{k} (y_j \vee \neg y_j),$$

where $y_1, \dots, y_k$ are fresh variables.

Theorem 11.4 (Padding Stability of Curvature Hardness). Let F_n be any 3SAT instance, and let $F_n^{(k)}$ be its k-padded version. Assume the WCC encoding maps tautological clauses $(y_j \vee \neg y_j)$ to flat, curvature-free regions, that is regions where $\Theta[\psi_{\text{pad}}] = 0$. Then

$$C_\Theta(F_n^{(k)}) = C_\Theta(F_n)$$

for all $k \geq 0$. In particular, any exponential lower bound

$$C_\Theta(F_n) \geq 2^{\Omega(n)}$$

persists under arbitrary polynomial padding.

Proof. Let ψ be a WCC realization of F_n with curvature complexity $C_\Theta(F_n)$, and let ψ_{pad} be the additional field degrees of freedom encoding the padding variables $y_1, \dots, y_k$. By assumption, the padding regions satisfy

$$\Theta[\psi_{\text{pad}}] = 0, \qquad \nabla\psi_{\text{pad}} = 0,$$

so they contribute no curvature energy. The combined realization $\psi' = \psi \oplus \psi_{\text{pad}}$ of $F_n^{(k)}$ then has curvature complexity

$$C_\Theta(F_n^{(k)}) = \int_\Omega |\Theta[\psi']|^2 dx = \int_\Omega |\Theta[\psi]|^2 dx + \int_\Omega |\Theta[\psi_{\text{pad}}]|^2 dx = C_\Theta(F_n).$$

Thus padding does not reduce the curvature cost. Any lower bound on $C_\Theta(F_n)$ transfers directly to $F_n^{(k)}$. $\square$

12 Toward a Formal P vs. NP Program via Wave Dynamics

12.1 Complexity Mapping into Wave Confinement Theory

Wave Confinement Theory represents computation as curvature-driven evolution: a wavefield $\psi(x,t)$ evolves locally under curvature and entropy constraints. This allows an NP-style verification / P-style construction analogy:

Complexity Concept	**WCT Interpretation**
Input string x	Initial configuration ψ_0
Certificate w	Target coherent state ψ^*
Verifier $M(x,w)$	Local test: $\sigma[\psi^*] \leq \sigma_{\text{crit}}$
Polynomial-time algorithm	Local update: $T = O(n^k)$ curvature steps
Acceptance	Overlap test: $\text{Overlap}(\psi_T, \psi^*) > \tau$

Table 1: Mapping between classical NP verification and curvature-regulated WCT dynamics.

This motivates the model-relative WCC classes introduced earlier. In the *physical* setting these are $\mathbf{P}_{\text{WCC}}^{\text{phys}}, \mathbf{NP}_{\text{WCC}}^{\text{phys}}$; in the *ideal* (unbounded) setting they are $\mathbf{P}_{\text{WCC}}^{\infty}, \mathbf{NP}_{\text{WCC}}^{\infty}$. Our long-term complexity-theoretic target is a separation $\mathbf{P}_{\text{WCC}}^{\infty} \neq \mathbf{NP}_{\text{WCC}}^{\infty}$ in the ideal model, with the physical separation as a precursor.

12.2 Model-Level Hypothesis to Be Contradicted (Ideal WCC)

The WCT/WCC analog of the classical hypothesis $P = NP$ in the *ideal* model is:

> *Every verifiable coherent state $\psi^* \in \mathbf{NP}_{\text{WCC}}^{\infty}$ is reachable from ψ_0 by curvature-regulated symbolic evolution within polynomial depth in the ideal WCC model.*

Dimensional locking and related curvature barriers suggest this fails for certain families of coherent states.

12.3 Coherence Without Constructibility in the Ideal WCC Model

Let C_n denote the space of confined wavefields over an $n \times n$ grid in the ideal discrete WCC model. We seek $\psi^* \in C_n$ such that:

- ψ^* is verifiable by a local curvature–entropy predicate (i.e. $\psi^* \in \mathbf{NP}^{\infty}_{\mathrm{WCC}}$),
- but for any polynomial-depth trajectory $\psi_0 \to \psi_T$, the symbolic distance $d(\psi_T, \psi^*)$ remains nonzero (i.e. $\psi^* \notin \mathbf{P}^{\infty}_{\mathrm{WCC}}$).

Such states would establish an internal model-relative separation

$$\mathbf{P}^{\infty}_{\mathrm{WCC}} \subsetneq \mathbf{NP}^{\infty}_{\mathrm{WCC}}$$

in the ideal WCC model.

12.4 Conditional Curvature Separation Statement (Ideal WCC)

Proposition 12.1 (Model-Relative Separation in Ideal WCC (Conditional)). Assume there exists a family $\{\psi_n^*\} \subset C_n$ such that:

1. Each ψ_n^* satisfies the polynomial-time local verifier (i.e. $\psi_n^* \in \mathbf{NP}^{\infty}_{\mathrm{WCC}}$); and
2. No curvature-regulated symbolic evolution of depth $T(n) = O(n^k)$ reaches ψ_n^* from ψ_0 for any fixed k (i.e. $\psi_n^* \notin \mathbf{P}^{\infty}_{\mathrm{WCC}}$).

Then

$$\mathbf{P}^{\infty}_{\mathrm{WCC}} \neq \mathbf{NP}^{\infty}_{\mathrm{WCC}}.$$

If, in addition, one can prove that the binary encodings of Section 13 satisfy

$$\mathbf{P}^{\infty}_{\mathrm{WCC}} = \mathbf{P}, \qquad \mathbf{NP}^{\infty}_{\mathrm{WCC}} = \mathbf{NP},$$

then the classical separation $P \neq NP$ follows as a corollary.

Discussion. The proposition itself is tautological: (i) + (ii) directly imply a separation in the ideal WCC model. The real work is in:

1. constructing or characterizing such families $\{\psi_n^*\}$ (dimensional lock and curvature-complexity lower bounds), and
2. proving the equivalence between ideal WCC computation and Turing computation ($\mathbf{P}_{\mathrm{WCC}}^{\infty} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\infty} = \mathbf{NP}$).

This paper provides partial progress toward (i) and a model-theoretic embedding toward (ii), but does not claim either as a completed theorem.

13 Formal Model-Theoretic Framework for P vs NP in WCT

In this section we place the WCT computational picture into an explicit symbolic framework, so that any future separation result can be stated entirely in the language of Turing machines and circuits.

13.1 Definition of the WCT Computational Model

We define the Wave Confinement Theory (WCT) computational model $\mathcal{M}_{\mathrm{WCT}}$ as follows:

- The computational state is a complex-valued wavefield $\psi(x,t) \in C^2(\Omega \times \mathbb{R})$.
- The evolution of ψ follows a curvature-regulated Lagrangian

$$\mathcal{L}_{\mathrm{WCT}} = |\partial_\mu \psi|^2 + \kappa \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right)^2 + \theta \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \right)^2 + \dots \tag{6}$$

  or its discrete symbolic analog.
- Input to the model is given by a grid of size $n \times n$ and a trivial initial field $\psi_0(x)$.
- The system evolves under a time budget $T = \mathcal{O}(n^k)$.
- Acceptance is defined by a target field ψ^* and an overlap/entropy condition:

$$\text{Accept if } \mathrm{Overlap}(\psi_T, \psi^*) > \tau \quad \text{and} \quad \sigma[\psi^*] \leq \sigma_{\mathrm{crit}}. \tag{7}$$

This defines $\mathbf{P}_{\mathrm{WCT}}$ and $\mathbf{NP}_{\mathrm{WCT}}$ as discussed above.

13.2 Encoding of NP-Complete Languages in WCT

Let L be any NP-complete language (e.g. 3-SAT). The WCT program is to construct a corresponding family of wavefields ψ_L^* such that:

- Each Boolean variable x_i is encoded as a resonance mode over a spatial region R_i.
- Clauses are encoded as interference or curvature constraints.
- The global wavefield ψ_L^* is entropy-coherent ($\sigma[\psi_L^*] \leq \sigma_{\text{crit}}$) if and only if the formula is satisfiable.

Formally achieving

$$x \in L \iff \psi_L^*(x) \text{ is a coherent, verifiable wavefield in } \mathbf{NP}_{\text{WCT}}$$

for all x would realize a Cook–Levin style reduction into the WCT framework. In this paper we sketch such encodings at a conceptual level; a fully formal construction is left as future work.

13.3 Constructibility Barrier under WCT Dynamics

Preliminary analytic arguments and the numerical evidence of Section 8 suggest that, for increasing n,

- WCT evolution from ψ_0 may fail to reach certain ψ_L^* within any fixed polynomial time budget $T = \mathcal{O}(n^k)$;
- The overlap $\text{Overlap}(\psi_T, \psi_L^*)$ can decay with n, even for instances known to be satisfiable.

These observations are consistent with, but do not yet prove, the existence of configurations in $\mathbf{NP}_{\text{WCT}} \setminus \mathbf{P}_{\text{WCT}}$.

13.4 A Conditional Model-Theoretic Separation Statement

Proposition 13.1 (Formal Model-Theoretic Separation in WCT (Conditional)). Assume:

1. There exists a family $\{\psi_L^*\}$ of WCT-encodable wavefields such that for each NP-complete language L, the map $x \mapsto \psi_L^*(x)$ realizes a polynomial-time Cook–Levin style reduction and $\psi_L^*(x)$ is verifiable by a local WCT predicate in polynomial resources; and

2. For at least one such NP-complete L, the corresponding family $\{\psi_L^*(x)\}$ lies in $\mathbf{NP}_{\mathrm{WCT}} \setminus \mathbf{P}_{\mathrm{WCT}}$.

Then $\mathbf{P}_{\mathrm{WCT}} \neq \mathbf{NP}_{\mathrm{WCT}}$. If, in addition, one can show that $\mathbf{P}_{\mathrm{WCT}} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCT}} = \mathbf{NP}$ under a binary encoding of WCT states and transitions, this would imply the classical separation $\mathbf{P} \neq \mathbf{NP}$.

Thus, the WCT framework provides a *conditional route* to $\mathbf{P} \neq \mathbf{NP}$: the missing pieces are (i) a rigorous construction of NP-complete encodings as curvature-locked states, (ii) a proof of non-constructibility within WCT evolution (the dimensional lock conjecture), and (iii) a clean equivalence between WCT complexity classes and their Turing counterparts.

Connection to classical complexity theory. If these steps can be completed, the resulting separation would mirror, in geometric form, the phenomena captured abstractly by the Time Hierarchy Theorem: deeper "curvature traversals" (longer evolution depth) would be provably required to construct certain verifiable configurations, beyond any fixed polynomial bound. At present, this connection should be viewed as an analogy and a research direction, not as an established theorem.

14 Binary Encoding and Turing Reduction of WCT

All statements in this section are either definitional or explicitly *conditional*. They describe how WCT-style curvature evolution can be represented in a classical Turing framework and how a putative WCT-level nonconstructibility result would *map* into standard complexity language. No unconditional claim about P vs. NP is made.

14.1 Tape Encoding of WCT States

Let the wavefield $\psi(x,t)$ over an $n \times n$ grid be represented as

$$\psi[i,j] = a_{ij} + ib_{ij}, \quad a_{ij}, b_{ij} \in \mathbb{R}.$$

We encode each complex amplitude using finite-precision binary:

$$\text{Tape representation: } \underbrace{0.101011\ldots}_{\text{Re}} \mid \underbrace{0.011010\ldots}_{\text{Im}} \#.$$

The entire field $\psi(x,t)$ is thus encoded as a binary tape of length $\mathcal{O}(n^2)$ (up to the chosen precision).

14.2 Turing Transducer for Curvature Evolution

We consider a deterministic Turing machine M_{WCT} that:

- Reads an input tape encoding ψ_t,
- Computes each grid value update using a finite-difference approximation to curvature-regulated evolution, e.g.

$$\psi_{t+1}[i,j] = \psi_t[i,j] + dt \cdot \left[\nabla^2\psi_t - \frac{\alpha\nabla^2\psi_t}{\psi_t + \epsilon e^{-\beta|\psi_t|^2}} - \theta \cdot \psi_t \cdot \nabla^2(\log|\psi_t|^2)\right],$$

- Applies arithmetic operations using standard polynomial-time Turing subroutines (finite-precision rational arithmetic).

This defines a symbolic curvature-update transducer. For any fixed precision model (e.g. $b(n)$ bits per amplitude with $b(n) = \text{poly}(n)$) one can arrange that a single macro-step of the WCT update is simulated in time $\text{poly}(n, b(n))$. We treat this as a *modelling assumption* fixing the intended regime for the WCT–Turing correspondence.

14.3 Acceptance Criterion as Turing Predicate

We define a symbolic acceptance predicate

$$\mathrm{Accept}(\psi) = 1 \iff \mathrm{Overlap}(\psi, \psi^*) > \tau \text{ and } \sigma[\psi] \leq \sigma_{\mathrm{crit}},$$

where:

- Overlap is computed via a normalized dot product between the encodings of ψ and ψ^*,
- Entropy $\sigma[\psi]$ is approximated by summing $-p \log p$ over the tape (with finite-precision logarithms).

Under reasonable encoding choices, this verifier can be implemented by a deterministic Turing machine using time polynomial in the tape length (e.g. $\mathcal{O}(n^2)$ for an $n \times n$ grid). Thus WCT-style verifiers embed naturally into classical **NP** predicates.

14.4 A Conditional Turing Reduction Template

We now sketch how a WCT-style nonconstructibility statement would translate into classical Turing language. We work with the model-relative classes $\mathbf{P}_{\mathrm{WCT}}$ and $\mathbf{NP}_{\mathrm{WCT}}$ introduced earlier.

Proposition 14.1 (Conditional Turing Reduction Template). Assume:

1. **Simulation of WCT by Turing machines.** The curvature-regulated evolution of Section 13 can be simulated by a deterministic Turing machine M_{WCT} with at most polynomial overhead in the tape length, for the encodings above.

2. **Verifier embedding.** The acceptance predicate $\mathrm{Accept}(\psi)$ is computable in time $\mathrm{poly}(n)$ on the Turing encoding, so any family of targets $\{\psi_n^*\}$ accepted by this predicate defines a language $L \subseteq \{0, 1\}^*$ in **NP**.

3. **Nonconstructible but verifiable targets (WCT model).** There exists a family of encodable WCT targets $\{\psi_n^*\}$ such that:

    - each ψ_n^* is verifiable by the local curvature/entropy test in polynomial WCT resources (so $\psi_n^* \in \mathbf{NP}_{\mathrm{WCT}}$),

- but no deterministic polynomial-time WCT evolution from a trivial initial configuration ψ_0 reaches ψ_n^* (so $\psi_n^* \notin \mathbf{P}_{\mathrm{WCT}}$).

Then:

- within the WCT model one obtains a separation $\mathbf{P}_{\mathrm{WCT}} \neq \mathbf{NP}_{\mathrm{WCT}}$; and
- if, in addition, one could prove that the identifications $\mathbf{P}_{\mathrm{WCT}} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCT}} = \mathbf{NP}$ hold (a *compatibility hypothesis* between WCT and classical computation), then the corresponding language L would lie in $\mathbf{NP} \setminus \mathbf{P}$ and hence imply $\mathbf{P} \neq \mathbf{NP}$.

The substantive content lies in assumption (3): establishing such a family $\{\psi_n^*\}$ is precisely the nonconstructibility problem discussed earlier (e.g. via dimensional lock and curvature-complexity barriers). The proposition itself is purely conditional: it specifies how a WCT-level separation, if proved and if compatible with standard encodings, would map to the usual P vs. NP consequence.

14.5 Conclusion

The encoding of WCT into a binary Turing-machine formalism shows that:

- Curvature-regulated wave evolution can be represented within standard discrete computation;
- Any eventual WCT-level separation between verifiability and constructibility (i.e. $\mathbf{NP}_{\mathrm{WCT}} \setminus \mathbf{P}_{\mathrm{WCT}}$) can, in principle, be interpreted in classical Turing terms under additional compatibility hypotheses.

In particular, if one could construct a configuration family $\{\psi_n^*\}$ that is verifiable in polynomial time but *provably* not constructible in polynomial time by any such machine (so that the associated language $L \in \mathbf{NP} \setminus \mathbf{P}$), then by the standard complexity-theoretic implication

$$\exists\, L \in \mathbf{NP} \setminus \mathbf{P} \quad \Rightarrow \quad \mathbf{P} \neq \mathbf{NP},$$

one would obtain a classical separation. This manuscript does *not* claim that such a configuration family has been fully established; instead, it develops the formal apparatus and model-relative barriers needed to state that goal precisely.

Classical Implication Template: P vs. NP (Conditional)

Classical implication (recalled). If there exists a decision problem (or configuration family) in $\mathbf{NP} \setminus \mathbf{P}$, then

$$\boxed{\mathbf{P} \neq \mathbf{NP}}.$$

The WCT framework aims to realize such an example by constructing a verifiable configuration family $\{\psi_n^*\}$ that satisfies an NC^1-implementable symbolic verifier, and for which one can *prove*, inside the model, that no deterministic Turing machine (or equivalently, no WCT evolution under the chosen encoding) can construct ψ_n^* in polynomial time, for all n. At present, this is a *research program*: the nonconstructibility part is formulated as a conjectural barrier rather than a completed proof, and all links to classical P vs. NP are presented as explicitly conditional templates.

15 Discrete Turing Reformulation and Open Nonconstructibility Goal

15.1 Classical Symbolic Encoding of WCT Computation

We define a discrete deterministic Turing model $T^{\Sigma}_{\mathrm{WCT}}$ operating over a finite binary alphabet $\Sigma = \{0, 1, \#\}$ that encodes the PWCC/WCT dynamics symbolically. Each complex amplitude $\psi[i, j] \in \mathbb{C}$ is represented as a binary pair of rational approximations:

$$\psi[i, j] = a_{ij} + ib_{ij}, \qquad a_{ij}, b_{ij} \in \mathbb{Q} \subseteq \mathbb{Z}/2^m,$$

for fixed precision parameter m.

A full grid configuration is stored on a single Turing tape in a canonical fixed-length layout (row-major order, with delimiters). One global evolution step of the field is simulated by a finite sequence of symbolic arithmetic operations (addition, multiplication, and lookup-table–based approximations to nonlinear functions) on these bitstrings.

15.2 Curvature Dynamics as an Arithmetic Circuit

The curvature-regulated update

$$\psi_{t+1} = \psi_t + \Delta t \cdot F(\psi_t)$$

is abstracted as a Boolean/arithmetic circuit F composed of:

- finite-difference Laplacian and stencil operators,
- fixed-precision multipliers and adders,
- rational-function or table-based approximations to exponentials, logs, and normalizations.

In idealized form, and for fixed precision, F can be placed in a low-level circuit class such as AC^0 or NC^1, depending on the exact gate set and fan-in assumptions. A single step of $T^{\Sigma}_{\mathrm{WCT}}$ then simulates $F(\psi_t)$ via a sequence of logically defined operations on binary strings, with per-step cost polynomial in the system size.

15.3 A Purely Symbolic Verifier

We restate the acceptance predicate in purely symbolic terms:

$$A(\psi) = 1 \iff \frac{\langle \psi, \psi^* \rangle}{\|\psi\| \cdot \|\psi^*\|} > \tau \ \wedge \ \sigma(\psi) \le \sigma_{\text{crit}},$$

where inner products, norms, and entropy are computed from the bit-encoded amplitudes by:

- fixed-precision multipliers and adders for $\langle \psi, \psi^* \rangle$ and $\|\psi\|^2$,
- table-based (or truncated-series) approximations to log,
- prefix-sum trees to aggregate all contributions.

All subroutines used in A are intended to be realizable by polynomial-time Turing machines; the target complexity placement is $A \in \mathsf{NC}^1$ (cf. the discussion in Section **??**). In particular, the definition of A in this section does not depend on any continuous geometry, only on the chosen binary encoding and standard arithmetic circuits.

15.4 A Programmatic Nonconstructibility Goal

Let ψ_0 be a trivial initial configuration (e.g., all zeros) in the tape encoding, and let ψ^* be a candidate target configuration whose correctness is verifiable by A. In purely classical complexity-theoretic terms, the *programmatic goal* suggested by WCT can be phrased as:

> **Goal.** Find a family $\{\psi_n^*\}$ of bit-encoded configurations and a verifier A with $A(\psi_n^*) = 1$ in polynomial time such that, for every deterministic polynomial-time Turing machine M and all sufficiently large n,
>
> $$M(\psi_0, n) \neq \psi_n^*.$$

If such a family can be exhibited and shown to have a standard polynomial-time verifier, then it defines a language in $\mathbf{NP} \setminus \mathbf{P}$, and the classical conclusion $\mathbf{P} \neq \mathbf{NP}$ would follow. In this paper, WCT provides geometric and curvature-based heuristics for constructing candidates for $\{\psi_n^*\}$; *the rigorous nonconstructibility proof in the unrestricted Turing model is left as an open problem.*

15.5 Independence from Numerical Simulation

The symbolic model $T^{\Sigma}_{\mathrm{WCT}}$ operates entirely on finite encodings with logically defined transition rules, arithmetic identities, and combinatorial estimates. Consequently:

- Any eventual nonconstructibility theorem inspired by WCT can, in principle, be stated purely in the language of Turing machines and circuits, without reference to floating-point numerics or PDE solvers;
- WCT acts as a geometric design principle for hard instance families and verification predicates, while the *final* separation (if obtained) would live entirely inside discrete complexity theory.

In its present form, this section should therefore be read as a *discrete reformulation and roadmap*: it shows how the PWCC/WCT curvature-machine model embeds into the classical symbolic framework, and isolates the precise nonconstructibility statement that must be proved to upgrade the internal PWCC–$\mathsf{NP}_{\mathrm{WCT}}$ separation to a full classical **P** vs. **NP** result.

16 Formal Reduction to Cook–Levin Framework and Universal Scope

16.1 Embedding PWCC Verifiers into the Cook–Levin Framework

The Cook–Levin theorem states that for any language $L \in \mathbf{NP}$, there exists a polynomial-size Boolean circuit $C(x, w)$ such that

$$x \in L \iff \exists w : C(x, w) = 1.$$

In the WCT-inspired setting, we consider symbolic encodings of wavefields and define a verifier predicate

$$A(\psi) = 1 \iff \text{Overlap}(\psi, \psi_L^*) > \tau \ \wedge \ \sigma[\psi] \leq \sigma_{\text{crit}},$$

implemented strictly as a rational-arithmetic circuit over the binary encoding of ψ. Its intended role is analogous to that of $C(x, w)$: to accept exactly those encodings corresponding to satisfying assignments for instances of L.

At present, this connection is conceptual. We outline a pathway by which Cook–Levin tableau encodings can be transplanted into WCT-style symbolic update rules and verifiers, but we do not yet supply a complete Cook–Levin–style formal reduction.

16.2 Universal Scope of Polynomial-Time Turing Machines

Let M be any deterministic Turing machine running in time $\mathcal{O}(n^k)$. Suppose M attempts to construct a symbolic wavefield ψ_L^* corresponding to some NP instance x, such that

$$V(x, \psi_L^*) = 1$$

for a polynomial-time classical verifier V.

In the symbolic WCT framework, the role of the verifier is played by a Turing-implementable predicate A (overlap and entropy test) computed using only:

- bounded-precision multipliers,
- prefix-sum trees,

- comparator circuits,
- and lookup-based logarithms,

all of which lie in $\mathrm{AC}^0 \cup \mathrm{NC}^1$.

Any machine M that constructs ψ_L^* must therefore do so through a sequence of polynomial-time symbolic steps beginning from a trivial tape configuration ψ_0.

If one could show that for *every* such M there exists some NP instance with solution $\psi_L^* \in \mathbf{NP}$ such that

$$M(\psi_0, n) \neq \psi_L^* \quad \text{for all} \quad n^k\text{-bounded runs},$$

then this would imply $\psi_L^* \notin \mathbf{P}$ and hence $\mathbf{P} \neq \mathbf{NP}$. WCT is proposed only as a geometric guide toward constructing such families; no such claim is made here.

16.3 Conditional Summary of the Cook–Levin Embedding

The intended theoretical path is:

- Encode NP-verifiers as symbolic rational-arithmetic circuits consistent with the WCT/PWCC encoding;
- Identify (via curvature/entropy arguments) families of states that are verifiable by these circuits but conjecturally not constructible by any polynomial-time Turing machine;
- Translate this constructive obstruction into a classical $\mathbf{NP} \setminus \mathbf{P}$ separation, *if and only if* the PWCC/Turing correspondence is proved exact.

In this paper we establish only the structure of this roadmap, not the final classical consequence.

Verifier Complexity. The target complexity class of the verifier

$$A(q) = 1 \iff \mathrm{Overlap}(q, q^*) > \tau \wedge \sigma(q) \leq \sigma_{\mathrm{crit}}$$

is expected to be NC^1, since all arithmetic operations—inner products, entropy sums, comparators, prefix additions—can be realized in logarithmic depth with polynomial size. A complete, rigorous circuit-class placement is left as open technical work.

17 Final Remarks on Turing Reduction and Acceptance Formalism

17.1 Symbolic Evolution Under $T^{\Sigma}_{\mathrm{WCT}}$

Let $T^{\Sigma}_{\mathrm{WCT}}$ be the discrete Turing machine defined earlier. A symbolic tape configuration for a small 3×3 grid at 8-bit precision is a block

$$q_t = [\texttt{01000011} \mid \texttt{00110010} \mid \cdots],$$

representing real/imaginary components.

Each update step performs:

- finite-difference Laplacian computation on the tape,
- symbolic evaluation of curvature-feedback terms via lookup tables,
- arithmetic composition into the next configuration q_{t+1}.

All gates are bounded fan-in, and the symbolic update is intended to be simulable in uniform **P**.

17.2 Formal Acceptance Predicate

We define:

$$A(q) = 1 \iff \mathrm{Overlap}(q, q^*) > \tau \ \wedge \ \sigma(q) \le \sigma_{\mathrm{crit}},$$

where:

- q is the symbolic Turing encoding of a wavefield,
- Overlap is a normalized inner product (prefix-sum tree),
- Entropy is computed as a discrete approximation of $\sum -p_i \log p_i$.

Lemma 17.1 (Intended Verifier Role). For each NP instance x, the symbolic goal is to construct a configuration q^* such that

$$x \in L \iff A(q^*) = 1.$$

This matches the Cook–Levin interpretation of a verifier state, though no equivalence theorem is yet claimed.

17.3 Diagonalization and Proof-Barrier Context

The WCT-inspired PWCC model is designed with specific complexity-theoretic concerns in mind:

- **Relativization:** The model uses no oracle; all arguments occur within a fixed symbolic transition system.
- **Natural Proofs:** Any envisioned separation relies on specific, structured configuration families, not on large, uniform, efficiently checkable properties—thus avoiding the typical Razborov–Rudich template.
- **Constructivity Constraints:** Conjectured lower bounds arise from symbolic curvature, entropy, and Kolmogorov complexity restrictions rather than adversarial Boolean structure.

These observations only position the framework relative to classical barriers; they do not guarantee their circumvention.

17.4 Model-Level Conclusion

The symbolic WCT/PWCC framework:

- operates entirely within discrete Turing computation once the encoding is fixed;
- admits verifiers plausibly contained in low-depth circuit classes (NC^1);
- offers a geometric mechanism (dimensional lock / entropy barrier) suggesting the existence of configurations in $\mathbf{NP}_{\mathrm{WCT}} \setminus \mathbf{P}_{\mathrm{WCT}}$.

Natural-Proof Barrier Positioning

This framework is not designed to exhibit a "natural" property in the Razborov–Rudich sense: it does not deploy a large, uniform, efficiently checkable Boolean property. Instead, it focuses on specific curvature-locked symbolic configurations constrained by entropy and resonance. Whether this ultimately avoids all known barriers is an open question; here we only articulate the intended positioning.

Logical Dependency Roadmap

Conceptual dependency outline:

- **Verifier Construction ($\mathbf{NC}^1$ Target):** Build a curvature/entropy-based verifier A in a low-depth circuit class.
- **SAT/WCT Encoding:** Give a formal encoding of 3-SAT into wave configurations ψ^* accepted by A.
- **Constructibility Barrier:** Establish a Kolmogorov/entropy-based lower bound showing some ψ^* cannot be constructed in polynomial time.
- **Generalization to All Poly-Time Machines:** Lift the barrier from a specific algorithm to all $M_i \in \mathbf{P}$.
- **Hierarchy/Separation Step:** Translate the constructibility barrier into a time-hierarchy–style separation $\mathbf{NP} \setminus \mathbf{P}$.

This roadmap is intended as a research program rather than a completed proof. If each step can be made precise and rigorously established, the standard implication $\mathbf{P} \neq \mathbf{NP}$ would follow in the usual way.

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A Reduction from 3-SAT to PWCC Configuration Encoding

We now restate the 3-SAT reduction in a form consistent with the discrete, symbolic PWCC curvature-machine model. All references to continuous PDE dynamics are removed; only the finite-precision, bitstring-encoded update rules and symbolic verifiers are used.

3-SAT Preliminaries

Let $X = \{x_1, \ldots, x_n\}$ be Boolean variables. A literal is x_i or $\neg x_i$. Each clause has exactly three literals:

$$C_i = (\ell_{i1} \vee \ell_{i2} \vee \ell_{i3}).$$

A 3-SAT formula is

$$\phi = C_1 \wedge \cdots \wedge C_k.$$

The 3-SAT problem asks whether there exists an assignment $\sigma : X \to \{\texttt{True}, \texttt{False}\}$ making ϕ true.

Encoding Variables as Symbolic PWCC States

In the PWCC model, each grid cell or symbolic "mode" is a finite-precision rational pair (a_{ij}, b_{ij}) encoded on a binary Turing tape.

We associate to each Boolean variable x_i a symbolic amplitude block:

$$x_i = \texttt{True} \longleftrightarrow q_i = +A, \qquad x_i = \texttt{False} \longleftrightarrow q_i = -A,$$

where A is a fixed rational constant encoded with m bits of precision. These values are **bitstrings**, not physical fields.

The tuple

$$q = (q_1, \ldots, q_n)$$

is the symbolic configuration representing a candidate assignment.

Encoding Clauses as Local Symbolic Constraints

For each clause

$$C_i = (\ell_{i1} \vee \ell_{i2} \vee \ell_{i3}),$$

we define a purely symbolic interference predicate:

$$I_i(q) = 1 - \prod_{j=1}^{3} \bigl(1 - \mathrm{match}(\ell_{ij}, q)\bigr),$$

where:

- $\mathrm{match}(\ell_{ij}, q)$ is 1 if the literal is satisfied, - 0 otherwise.

Implementation details:

- match uses a single comparator gate and (optionally) a NOT gate. - $I_i(q) = 1$ if *any* literal in C_i is satisfied. - The predicate is computable in AC^0 and feeds into the global verifier.

Thus clause satisfaction is encoded in a purely combinational Boolean subcircuit acting on the finite-precision PWCC tape.

Global PWCC Configuration for a 3-SAT Instance

Define the symbolic target configuration q_ϕ^* by concatenating:

1. the n variable-encoded blocks, and 2. a set of k clause-constraint blocks (each storing $I_i(q)$).

The verifier from the main text is:

$$A(q) = 1 \iff \Bigl(\mathrm{Overlap}(q, q_\phi^*) > \tau\Bigr) \wedge \Bigl(\sigma(q) \le \sigma_{\mathrm{crit}}\Bigr)$$

where:

- Overlap is a rational inner product computed via prefix-sum addition, - σ is the symbolic entropy functional, - both are implementable in NC^1.

Polynomial-Time Verification

Given any encoded configuration q:

1. Verify each clause block $I_i(q) = 1$ using constant-depth logic. 2. Compute symbolic entropy $\sigma(q)$ (prefix-sum + log-table; NC^1). 3. Compute overlap with q_ϕ^*. 4. Accept iff all clauses are satisfied and both threshold tests pass.

Thus the verification process is polynomial-time Turing computable and can be implemented in a uniform NC^1 circuit family.

PWCC Evolution and Constructibility

A PWCC evolution applies a fixed local update rule f:

$$q_{t+1} = \mathcal{T}(q_t) = f(q_t, \text{LocalNeighbors}(q_t)),$$

where all arithmetic is symbolic and bounded-precision.

The evolution depth is limited to $t = \text{poly}(n)$.

A configuration q_ϕ^* is said to be in **PWCC** if it is reachable from the trivial initial configuration q_0 in polynomially many steps.

Because the verifier A is low-depth but the constructive reachability may be super-polynomial, this sets up the PWCC model-relative separation framework.

Final Reduction Statement

Theorem A.1 (Reduction). There exists a polynomial-time computable mapping

$$\phi \;\mapsto\; q_\phi^*$$

such that

$$\phi \in \text{3-SAT} \quad \Longleftrightarrow \quad \exists q \; \big(A(q) = 1\big).$$

Thus:

$$\text{3-SAT} \leq_p \text{PWCC-}\textsc{Stability},$$

where PWCC-Stability is the decision problem:

“Does there exist a configuration q within the PWCC encoding that satisfies the verifier A?”

This reduction is **purely symbolic** and consistent with the PWCC curvature-machine model.

B ψ-Evolution as a Restricted Deterministic Computation Model

We now formalize the confined ψ-field as a purely discrete, deterministic local computation model. Throughout this section we treat ψ as symbolic state stored in finite-precision registers; all arguments are combinatorial.

State space and transition dynamics

Let $\Psi_t = (\psi_1^t, \ldots, \psi_n^t)$ denote the discrete configuration at time step t. Each ψ_i^t is a finite-precision rational (or fixed-length bitstring) obtained by encoding the underlying wavefield on an n-cell grid. The global update is given by a fixed local rule

$$\psi_i^{t+1} \;=\; f\big(\mathrm{loc}_i(\Psi_t)\big),$$

where $\mathrm{loc}_i(\Psi_t)$ collects the values in a constant-radius neighborhood of cell i (e.g. ψ_i^t and its nearest neighbours) and f is a fixed, finite combinational circuit. The evolution is:

- **Deterministic** – Ψ_{t+1} is a function of Ψ_t only;
- **Local** – each cell consults only a bounded neighbourhood;
- **Synchronous** – all cells update in parallel under the same rule f.

We write $\mathcal{T}$ for the induced global transition operator, so that $\Psi_{t+1} = \mathcal{T}(\Psi_t)$.

Definition of the PWCC class

We now isolate the language class computed by such evolutions under a polynomial time bound.

Definition B.1 (PWCC). A decision language $L \subseteq \{0,1\}^*$ lies in **PWCC** (Polynomial Wave–Confinement Configuration) if there exist:

- an encoding map $\mathcal{E}$ that sends each instance ϕ to an initial configuration $\Psi_0 = \Psi_\phi$;
- a single fixed local update rule f (and its induced global map $\mathcal{T}$);
- a polynomial $p(\cdot)$;

such that for every instance ϕ:

1. if $\phi \in L$, then there exists $t \le p(|\phi|)$ such that $\Psi_t = \mathcal{T}^t(\Psi_\phi)$ is an *accepting, curvature-locked* configuration;
2. if $\phi \notin L$, then no accepting configuration is ever reached: for all $t \ge 0$, $\mathcal{T}^t(\Psi_\phi)$ is rejecting.

Intuitively, **PWCC** is the class of decision problems solvable by polynomially many steps of a fixed, local, deterministic update acting on encoded instances.

Single-step simulation by log-depth circuits

A single application of $\mathcal{T}$ can be simulated by a shallow Boolean circuit.

Lemma B.2 (Single-step simulation)*.* Under a standard binary encoding of Ψ_t on n cells, one application of $\mathcal{T}$ can be computed by a logspace-uniform family of Boolean circuits of polynomial size and depth $O(\log n)$. In particular, each update step lies in NC^1.

Proof sketch. Each cell i computes $\psi_i^{t+1} = f(\mathrm{loc}_i(\Psi_t))$ from a constant-size neighbourhood. The bit-level realization of f is a constant-depth circuit with constant fan-in. To form all ψ_i^{t+1} in parallel, we replicate this local stencil at n sites; the total size is polynomial in n. Logspace uniformity follows from the regular tiling of the grid. Any needed finite-precision arithmetic (adds, multiplies, table lookups) can be implemented in NC^1 using standard parallel prefix constructions. The global depth is therefore $O(\log n)$. $\square$

A polynomial number of steps of $\mathcal{T}$ thus corresponds to a logspace-uniform family of polynomial-size circuits of polynomial depth. In particular, every language in **PWCC** is decidable in deterministic polynomial time on a classical Turing machine; formally,

$$\mathbf{PWCC} \subseteq \mathbf{P}.$$

This inclusion is the only complexity-theoretic fact we use about **PWCC** in the remainder of the paper.

Model-relative reachability intuition

Since **PWCC** is generated by local, deterministic updates and never branches nondeterministically over exponentially many configurations, it provides a constrained "wave-based" analogue of deterministic polynomial time. In the next section we use this model as a geometric language for discussing which configurations of the underlying field are *reachable* under such constraints and which are excluded by curvature/entropy bounds. These reachability statements serve as physical intuition; the rigorous lower bounds are derived later via the curvature complexity measure $\mathbf{C}_\Theta$.

C Dimensional Locking and Heuristic Unreachable NP Solutions

We now give a geometric, model-relative picture of how curvature and dimensional constraints can exclude certain ψ-configurations from the reachable set of a PWCC evolution. This section is *interpretive*: it motivates the later formal curvature lower bounds but is not used in the combinatorial proofs.

Reachable configuration subspace

Let $\mathcal{C}_{\text{full}}$ denote the space of all admissible ψ-field configurations (after discretization and finite-precision encoding), and let $\mathcal{C}_{\text{reachable}}$ be the subset that arises as

$$\Psi_t = \mathcal{T}^t(\Psi_\phi)$$

for some encoded instance ϕ and some $t \leq p(|\phi|)$ under a fixed PWCC evolution. Because the update rule is local, deterministic, and curvature-regulated, the reachable set is constrained by:

- **Smoothness/regularity** of ψ (no arbitrarily sharp, zero-cost interfaces);
- **Coherence-limited entropy** (curvature feedback penalizes highly disordered patterns);
- **Dimensional/topological consistency** (no use of embeddings in $d > 3$ or non-physical shortcuts).

We write the complement as

$$\mathcal{C}_{\text{full}} \;=\; \mathcal{C}_{\text{reachable}} \;\cup\; \mathcal{C}_{\text{locked}},$$

where $\mathcal{C}_{\text{locked}}$ consists of configurations excluded by these constraints.

Dimensional locking (informal definition)

In WCT language, *dimensional locking* refers to the exclusion of field configurations whose structure would require curvature or entropy beyond fixed bounds. Schematically,

$$\text{If} \quad \|\nabla^2\psi\| > \Lambda_{\max} \quad \text{or} \quad \mathcal{S}[\psi] > \sigma_{\max}, \quad \text{then} \quad \psi \notin \mathcal{C}_{\text{reachable}}.$$

Here $\Lambda_{\max}$ and $\sigma_{\max}$ summarize the maximum tolerable curvature and entropy dictated by the curvature feedback and finite-band stiffness assumptions; configurations exceeding these bounds are "locked out."

NP witnesses in excluded regions (heuristic picture)

Let L be an NP-complete language, and let $\phi \in L$ be an instance with a satisfying assignment. Under a suitable encoding $\mathcal{E}$, one can associate to ϕ a target configuration ψ^* that encodes a satisfying assignment through constructive interference of modes and passes the usual NP verifier (e.g., clause satisfaction or local energy minimum conditions).

It is then *conceivable* that for some such ψ^* one has

$$\psi^* \in \mathcal{C}_{\text{locked}},$$

i.e., the configuration represents a valid NP witness but cannot be realized by any polynomially bounded PWCC evolution starting from an admissible initial state. In that case:

- ψ^* is *verifiable*: a local symbolic verifier can confirm that it satisfies all constraints;
- but ψ^* is *unreachable* as an output of any **PWCC** computation.

This provides a geometric story in which "existence of witnesses" (NP) and "constructibility via local curvature descent" (PWCC) can diverge.

Interpretation

The preceding discussion suggests a model-relative separation:

$$\mathcal{L}_\psi \;=\; \mathbf{PWCC} \;\subseteq\; \mathbf{NP},$$

with the possibility that the inclusion is strict once geometric constraints are fully accounted for. However, we stress:

> This section is heuristic and model-specific. It does *not* by itself constitute a proof that $\mathbf{PWCC} \subsetneq \mathbf{NP}$, nor does it resolve the classical **P** vs. **NP** question for Turing machines.

The actual non-natural circuit lower bound and the resulting complexity-theoretic separation are established later by introducing a curvature complexity measure $\mathbf{C}_\Theta$ and proving exponential lower bounds for certain SAT families in the classical circuit model.

D Mapping Between Formal Languages and ψ-Field Configurations

We now recast the PWCC model in standard language-theoretic terms, keeping the discussion purely symbolic. This section serves as a bridge between PWCC and classical decision problems; it is descriptive rather than a source of lower bounds.

Encoding decision problems as ψ-configurations

Let L be a decision problem in **NP**, and let ϕ be an instance of size n. An encoding map

$$\mathcal{E} : \{0,1\}^* \to \Sigma_\psi^*, \qquad \phi \mapsto \psi_\phi$$

associates to ϕ an initial configuration ψ_ϕ (the "input state") by, for example:

- mapping Boolean variables to local "modes" ψ_i (cells or registers);
- mapping clauses to local interaction patterns or curvature constraints;
- arranging that a satisfying assignment corresponds to some stable configuration ψ^* which an NP verifier can recognize.

No physical assumptions are needed here; ψ is simply a structured encoding of the combinatorial instance and its potential witnesses.

ψ-evolution as a transition system

Given the fixed local rule f (and global map $\mathcal{T}$) from the PWCC definition, we view $(\Sigma_\psi^*, \mathcal{T})$ as a deterministic transition system:

$$\psi_{t+1} = \mathcal{T}(\psi_t), \qquad t = 0, 1, 2, \ldots$$

For each encoded instance ϕ, the PWCC machine executes $\psi_t = \mathcal{T}^t(\psi_\phi)$ for $t \leq p(|\phi|)$ and accepts or rejects depending on whether an accepting configuration is encountered.

The language recognized by this PWCC evolution can be written as

$$\mathcal{L}_\psi = \{\phi \,|\, \exists t \leq p(|\phi|) \text{ s.t. } \mathcal{T}^t(\psi_\phi) \text{ is accepting}\},$$

which is exactly the class **PWCC** defined in Definition B.1.

Geometric reachability vs. NP verification (informal)

Let $\mathcal{L}_{\text{NP}}$ be the standard NP language of satisfiable instances, with witnesses checked by a polynomial-time verifier (or an NC^1 verifier family as constructed later in the paper). For any $\phi \in \mathcal{L}_{\text{NP}}$ we can, in principle, encode a witness into some configuration ψ^*_ϕ that the verifier accepts.

It may happen, in a given geometric encoding and under PWCC constraints, that

$$\psi^*_\phi \notin \text{Range}\left(\mathcal{T}^t\right) \qquad \text{for all polynomially bounded } t,$$

i.e., the witness configuration is never reached by the evolution, even though it exists abstractly and is verifiable. In that situation $\phi \notin \mathcal{L}_\psi$ even though $\phi \in \mathcal{L}_{\text{NP}}$.

This illustrates, at a conceptual level, how $\mathcal{L}_\psi = \mathbf{PWCC}$ can be a proper subset of **NP** *within a specific constrained model.* The rigorous evidence for such a gap, however, comes later via the curvature-based lower bound $\mathbf{C}_\Theta$; we do not claim any formal separation here.

Conclusion

In summary:

- PWCC is a standard, discrete, local-update machine model with $\mathbf{PWCC} \subseteq \mathbf{P}$.
- WCT-style geometric language provides an intuitive picture in which some NP witnesses correspond to configurations outside the PWCC-reachable subset of configuration space.

- The PWCC/WCT mapping is used for interpretation and motivation; the actual circuit lower bounds and the final complexity consequences are proved later using purely classical encodings, verifiers, and curvature-based measures.

E Formal Completeness Extensions for Review

E.1 Formal Circuit Class of the Verifier ($\mathbf{NC^1}$ Bound)

We prove that the verifier predicate:

$$A(q) = 1 \iff \mathrm{Overlap}(q, q^*) > \tau \wedge \sigma(q) \leq \sigma_{\mathrm{crit}}$$

lies in NC^1:

Entropy Computation. The entropy function:

$$\sigma(q) = -\sum_i p_i \log p_i$$

can be approximated by log-depth circuits using:

- finite lookup tables for $\log(p_i)$ values;
- prefix sum logic for partial entropy sums;
- reuse of known constructions for division and iterated multiplication in NC^1 (see Allender et al., 1992; Hesse et al., 2002).

Symbolic Overlap. The dot product:

$$\langle q, q^* \rangle = \sum_i q_i q_i^*$$

can be implemented via:

- bitwise multiplication gates (AC^0),
- a parallel prefix addition tree (log-depth sum $\Rightarrow \mathsf{NC}^1$).

Threshold Comparison. Comparators for overlap $> \tau$ and entropy $\leq \sigma_{\text{crit}}$ are implemented using constant-depth bitwise comparison circuits (AC^0).

Thus, the total verifier predicate $A(q) \in \mathsf{NC}^1$.

E.2 SAT → Wave Interference Encoding Table

Boolean Term	Wave Encoding
$x_i = 1$	$\psi_i = Ae^{i\phi_i}$ (constructive phase)
$\neg x_i$	$\psi_i = Ae^{i(\phi_i+\pi)}$ (destructive phase)
$x_1 \vee \neg x_2 \vee x_3$	$\psi_{\text{clause}} = \psi_1 + \psi_2 + \psi_3$: stable $\Leftrightarrow \exists \psi_i$ constructive

Table 2: Symbolic encoding of Boolean variables and clauses into phase-coherent wave superpositions.

Clause Stability. The wave pattern stabilizes iff:

$$\psi_{\text{clause}} = \sum_{i \in \text{clause}} \psi_i \quad \text{constructively interferes} \quad \Leftrightarrow \exists \psi_i \text{ in phase.}$$

This creates a symbolic interference constraint isomorphic to clause satisfaction.

E.3 Diagonalization Lemma: Nonconstructibility of Certain NP States (PWCC Version)

Lemma (Model-relative). Let $\{M_i\}$ be the set of all polynomial-time-bounded ψ-evolution procedures in the PWCC model. Then there exists a configuration

$$\psi^* \in \mathbf{NP} \setminus \mathbf{PWCC}$$

such that, for every M_i, there exists an input size n with

$$M_i(\psi_0, n) \neq \psi^*.$$

Proof Outline (informal). Let ψ^* encode a coherent wavefield requiring exponential curvature traversal to stabilize (e.g., nested phase-separated SAT clauses). Then:

- each M_i has bounded step depth $T_i(n) = \mathcal{O}(n^k)$,
- ψ^* has Kolmogorov complexity $K(\psi^*) \geq 2^n$,
- no M_i can generate ψ^* from ψ_0 without encoding an exponential-depth path.

Model-relative conclusion: within the PWCC/WCT model, such a ψ^* behaves as an NP-verifiable but PWCC-unconstructible state, i.e. $\psi^* \in \mathbf{NP} \setminus \mathbf{PWCC}$.

E.4 Polynomial-Time Barrier Theorem (Time Hierarchy Analog in PWCC)

Theorem (PWCC Time Hierarchy). Let ψ^* be a coherent state that satisfies the entropy–curvature condition, but has depth:

$$\text{Depth}(\psi^*) \geq \Omega(2^n) \quad \Rightarrow \quad T(\psi_0 \to \psi^*) \geq \Omega(2^n)$$

for any admissible ψ-evolution in the PWCC model.

Then $\psi^* \notin \mathbf{PWCC}$ (i.e., it is not reachable in polynomial time within this model), even though its correctness is verifiable by an NC^1 predicate.

Interpretation. If ψ^* is verifiable in NC^1, but no poly-time symbolic evolution can reach it from ψ_0, then we obtain an internal hierarchy:

$$\mathbf{PWCC} \subsetneq \mathbf{NP}_{\text{WCT}},$$

where $\mathbf{NP}_{\text{WCT}}$ denotes NP verification restricted to WCT-encodable configurations. This is a separation *inside the curvature-machine model* and is not claimed to imply any unconditional separation such as $\mathbf{P} \neq \mathbf{NP}$ in the classical Turing framework.

Reference. The structure is analogous in spirit to Hartmanis–Stearns (1965), but applied to symbolic curvature machines in the PWCC model.

Symbolic Ingredients for a Model-Relative Separation $\mathbf{PWCC} \subsetneq \mathsf{NP}_{\mathrm{WCT}}$

In this appendix we collect the formal ingredients used in the *model-relative* separation argument between **PWCC** and a WCT-flavoured NP class $\mathsf{NP}_{\mathrm{WCT}}$. All statements here are internal to the curvature–machine model and do *not* assert any unconditional separation such as $\mathbf{P} \neq \mathbf{NP}$ for arbitrary Turing machines.

1. NC^1 circuit construction for the verifier predicate

Recall the symbolic verifier

$$A(q) = 1 \iff \big(\langle q, q^* \rangle > \tau\big) \ \wedge \ \big(\sigma(q) \leq \sigma_{\mathrm{crit}}\big),$$

where all quantities are computed at finite precision. We sketch why $A(q)$ admits a logspace-uniform NC^1 circuit family under standard encodings.

Dot product $\langle q, q^* \rangle$

Let $q, q^* \in \mathbb{Q}^n$ be encoded as fixed-length binary strings per coordinate, and define

$$\langle q, q^* \rangle = \sum_{i=1}^{n} q_i q_i^*.$$

Each product $q_i q_i^*$ can be implemented by a standard bit-level multiplier of depth $O(\log m)$ for m-bit inputs, and the sum of all products can be computed by a balanced binary tree of adders, again of logarithmic depth. Thus the entire dot product admits a logspace-uniform NC^1 realization.

Entropy-like functional $\sigma(q)$

For the entropy-like functional

$$\sigma(q) = -\sum_{i=1}^{n} p_i \log p_i, \qquad p_i = |q_i|^2,$$

we work at fixed precision:

- Each $p_i = |q_i|^2$ is a product of two fixed-precision rationals, hence in NC^1.
- For $\log p_i$, we approximate $\log(\cdot)$ on a fixed compact domain by a finite lookup table (or a low-degree polynomial with fixed coefficients). The table index is a fixed number of bits of p_i, so the lookup and evaluation are constant-depth with polynomial size (hence within NC^1).
- Each term $p_i \log p_i$ is then a fixed-precision product; summing over i via a balanced adder tree yields depth $O(\log n)$.

Thus $\sigma(q)$ can be approximated to fixed precision by logspace-uniform NC^1 circuits.

Threshold comparisons

The comparisons $\langle q, q^* \rangle > \tau$ and $\sigma(q) \leq \sigma_{\text{crit}}$ are comparisons of fixed-precision binary numbers. Standard comparator circuits have logarithmic depth and polynomial size, hence lie in NC^1.

Conclusion. Each component of $A(q)$ is computable in logspace-uniform NC^1, and therefore the full predicate $A(q)$ lies in NC^1. We use this only to justify that NP_{WCT} (i.e. configurations accepted by A) is an NP-like class with efficiently checkable witnesses.

2. Role of standard proof barriers (model-relative)

We briefly explain how standard barriers are interpreted in the PWCC/WCT setting.

Relativization. No oracle is introduced in the PWCC model; dynamics are governed by a fixed local rule. Our statements about **PWCC** vs. NP_{WCT} are entirely internal to this model and are *not* claimed to evade classical relativization barriers for the general **P** vs. **NP** question.

Natural proofs. The constructions here focus on specific WCT-encodable configurations and families (e.g. locked expanders), rather than on "most" Boolean functions. As such, the largeness/randomness conditions in the

Razborov–Rudich framework do not directly apply in this restricted setting. We do *not* claim that any general natural-proofs barrier is circumvented for arbitrary circuit lower bounds.

Algebrization. All operations are discrete and tied to a fixed encoding of the WCT evolution. We make no claim about algebrization in the sense of Aaronson–Wigderson; the discussion here is confined to a single, non-oracular physical/combinatorial model.

3. Kolmogorov complexity heuristic for hard ψ^*

We now record a purely heuristic picture based on descriptive complexity. Let L be a family of 3-SAT instances on n variables, and let ψ^* denote a configuration encoding a satisfying assignment for a particularly constrained family (e.g. instances with exponentially few satisfying assignments).

There are 2^n possible assignments, but only a tiny fraction satisfy all clauses in such families. Informally, one might expect that "interesting" satisfying configurations ψ^* could have large Kolmogorov complexity $K(\psi^*)$ relative to the encoding of the instance. In slogan form:

$$K(\psi^*) \gg \mathrm{poly}(n) \quad \text{for certain extremely sparse families.}$$

If one could *rigorously* show that, for some WCT-encodable family, every accepting configuration ψ^* has Kolmogorov complexity exceeding any polynomial in n, then no polynomial-time PWCC evolution starting from a simple initial state could generate such ψ^*; the evolution would not be able to "discover" such a high-description object within polynomial resources.

Heuristic status. At present we use this only as an *intuition* for why model-relative hard instances might exist:

$$K(\psi^*) \gg \mathrm{poly}(n) \quad \Longrightarrow \quad \psi^* \text{ is a plausible candidate to lie outside } \mathbf{PWCC}.$$

We do *not* rely on any Kolmogorov lower bound in the formal arguments; the rigorous lower bounds in the main text are phrased instead in terms of the curvature measure $\mathbf{C}_\Theta$.

4. Clarifying the role of simulations

To avoid overinterpreting numerical experiments, we recommend the following edits in the main text:

- Rename the simulation section to, e.g., *"Numerical Motivation (Not Proof)"*.
- Add an explicit disclaimer such as:

> "All simulations in this section are illustrative and serve only as qualitative motivation. They are not used as part of any formal derivation of the model-relative separation $\mathbf{PWCC} \subsetneq \mathsf{NP}_{\mathrm{WCT}}$."

This makes clear that any non-convergence plots, PCA trajectories, or overlap decays are used for intuition, not as complexity-theoretic evidence.

5. Symbolic reduction from 3-SAT to a wave configuration (PWCC view)

We sketch a standard reduction from 3-SAT to a WCT-style wave configuration, emphasizing its purely symbolic nature.

Let $\varphi = C_1 \wedge \cdots \wedge C_m$ be a 3-CNF over variables $x_1, \ldots, x_n$.

Step 1: variable encoding. Each Boolean variable x_i is mapped to a mode

$$\psi_i = Ae^{i\theta_i}, \qquad \theta_i = \begin{cases} 0, & x_i = \texttt{True}, \\ \pi, & x_i = \texttt{False}. \end{cases}$$

Finite-precision representations of θ_i are stored as bitstrings.

Step 2: clause interference constraint. Each clause $C_k = (\ell_{k,1} \vee \ell_{k,2} \vee \ell_{k,3})$ is associated with an interference signal

$$I_k = 1 - \prod_{j=1}^{3} \Big(1 - \big|\langle \psi_{k,j}, \psi_{\mathrm{true}} \rangle\big|\Big),$$

with ψ_{true} the reference "true" phase and $\langle \cdot, \cdot \rangle$ a fixed finite-precision inner product. At the logical level, this simply enforces that $I_k = 1$ iff at least one literal in C_k is true under the assignment encoded by (θ_i).

Step 3: global configuration and verification. The global configuration $\psi^* = (\psi_1, \ldots, \psi_n)$ is accepted by the symbolic verifier A if

$$A(\psi^*) = 1 \iff \Big(\forall k,\ I_k = 1\Big) \ \wedge\ \big(\sigma(\psi^*) \le \sigma_{\text{crit}}\big).$$

The entropy bound is part of the WCT-flavoured acceptance criterion; it plays no role in classical correctness of the underlying Boolean assignment.

PWCC-complete problem (model-relative). Within the PWCC/WCT model, this gives a polynomial-time reduction from 3-SAT to the problem:

> PWCC-Stability: Given an encoded instance ϕ, does there exist a configuration ψ^* and a polynomially bounded PWCC evolution such that ψ^* is reached and $A(\psi^*) = 1$?

In this sense, PWCC-Stability is a natural PWCC-complete problem *within the model*. We make no claim here about its hardness in the unrestricted Turing sense.

Diagonalization lemma (PWCC model-relative form)

We now restate the diagonalization-style reasoning explicitly *inside* the PWCC/WCT model.

Definition E.1. Let $\{M_i\}$ be an effective enumeration of all deterministic polynomial-time PWCC procedures, i.e. all algorithms that:

- simulate a fixed local WCT update rule on a symbolic tape encoding;
- run in time $n^{O(1)}$ on inputs of size n;
- and output symbolic configurations whose correctness can be checked by the NC^1 predicate A.

Let $\mathsf{NP}_{\mathrm{WCT}}$ be the class of symbolic configurations q such that $A(q) = 1$.

Lemma E.2 (PWCC diagonal nonconstructibility; model-relative). Assuming the standard time-hierarchy properties for the underlying Turing encoding, there exists a coherent, entropy-stable symbolic configuration $\psi^* \in \mathsf{NP}_{\mathrm{WCT}}$ such that

$$\forall i,\ \exists n :\ M_i(\psi_0, n) \neq \psi^*$$

and the symbolic evolution depth $\ell(\psi_0 \to \psi^*)$ (measured in bitwise local-update steps) satisfies

$$\ell(\psi_0 \to \psi^*) \geq \Omega(2^n)$$

for infinitely many n. In particular, within the PWCC/WCT model,

$$\psi^* \in \mathsf{NP}_{\mathrm{WCT}} \setminus \mathbf{PWCC}.$$

This is best read as a PWCC-flavoured restatement of the classical time hierarchy phenomenon: there exist efficiently verifiable configurations that cannot be produced within any fixed polynomial bound by any machine in our enumerated family. It is explicitly *not* a new classical lower bound.

Universality of nonconstructibility (PWCC version)

Theorem E.3 (PWCC universal nonconstructibility; model-relative). Let M be any fixed deterministic PWCC curvature machine running in time $n^{O(1)}$, attempting to construct verifier-accepting configurations for some language $L \subseteq \mathsf{NP}_{\mathrm{WCT}}$. Then, under the same hierarchy assumptions as in Lemma E.2, there exists a configuration

$$\psi_L^* \in \mathsf{NP}_{\mathrm{WCT}}$$

such that

$$M(\psi_0, n) \neq \psi_L^* \quad \text{for all polynomially bounded } n,$$

and hence $\psi_L^* \notin \mathbf{PWCC}$.

This simply says that for any fixed polynomial-time PWCC machine, one can diagonalize against its output set to obtain an $\mathrm{NP}_{\mathrm{WCT}}$-verifiable configuration it never produces, in the spirit of the classical time hierarchy theorem.

Time-hierarchy style curvature-depth separation (PWCC analogue)

Theorem E.4 (Symbolic time hierarchy in the PWCC model). Let $\psi^* \in \mathsf{NP}_{\mathrm{WCT}}$ be a coherent symbolic configuration with $A(\psi^*) = 1$, and suppose that its constructive depth satisfies

$$\ell(\psi_0 \to \psi^*) \geq \Omega(2^n)$$

under the bitwise curvature-regulated Turing update that simulates the PWCC rule. Then $\psi^* \notin \mathbf{PWCC}$, and hence, at the model-relative level,

$$\mathbf{PWCC} \subsetneq \mathsf{NP}_{\mathrm{WCT}}.$$

Taken together, Lemma E.2, Theorem E.3, and Theorem E.4 show how standard time-hierarchy style arguments (cf. Hartmanis–Stearns [**?**]) can be *rephrased* inside the PWCC/WCT curvature-machine model to yield a clean internal separation

$$\mathbf{PWCC} \subsetneq \mathsf{NP}_{\mathrm{WCT}}$$

provided one can exhibit configurations with exponential curvature depth but NC^1 verifiers. Crucially, this is a statement about a particular restricted model; it is not presented as an unconditional result about classical Turing complexity classes such as $\mathbf{P}$ and $\mathbf{NP}$.

F Kolmogorov Complexity Bound for ψ^* (PWCC Interpretation)

We now recast the Kolmogorov argument as a heuristic lower bound inside the WCT/PWCC framework.

Lemma A.4 (Heuristic Non-Compressibility of ψ^* in the WCT Model). Let $\mathcal{L} \in \mathsf{NP}$ be a family of 3-SAT instances with n variables, and let $\psi^*_{\mathcal{L}}$ be the symbolic wave configuration that encodes a satisfying assignment to $\mathcal{L}$ via curvature-locked constructive interference patterns. For instance families in which satisfying assignments are exponentially sparse, it is natural to expect that

$$K(\psi^*) \gtrsim 2^n,$$

where $K(\psi^*)$ is the Kolmogorov complexity of the configuration description.

Heuristic Outline. If there existed a polynomial-size Turing machine M_i with runtime $\mathcal{O}(n^k)$ that constructs ψ^* from a trivial initial configuration, then we would have an upper bound

$$K(\psi^*) \leq \mathcal{O}(n^k).$$

For instance families in which the satisfying assignments form an exponentially sparse subset of the 2^n possible assignments, encoding the full clause-level interference pattern typically requires specifying exponentially many bits of information. Heuristically, this motivates

$$K(\psi^*) \gtrsim 2^n \gg \mathcal{O}(n^k).$$

Model-Relative Conclusion. Within the PWCC/WCT model, such a heuristic lower bound on $K(\psi^*)$ supports the interpretation that

$$\psi^* \notin \mathbf{PWCC}$$

for those instance families, and hence that the internal separation

$$\mathbf{PWCC} \subsetneq \mathsf{NP}_{\text{WCT}}$$

is consistent with information-theoretic intuition. We emphasize that this is not presented as a rigorous Kolmogorov-based proof in the classical sense, but as a structural guideline for the PWCC model. □

Grid Size	Avg Final Overlap	Avg Steps to Convergence
32	0.077637	5000.0
64	0.012087	5000.0
96	-0.006049	5000.0

Table 3: Average final overlap and convergence steps across increasing grid sizes. Each wavefield ψ evolution was run for 5000 steps under the curvature-regulated update. Convergence to the target configuration was not observed within this fixed step budget, and final overlap decays as complexity increases. This is *empirically consistent* with the existence of configurations that are not reachable by polynomially bounded PWCC evolutions, but does not constitute a formal proof.

G Numerical Motivation for Non-Constructibility in the PWCC Model

This section presents numerical experiments that serve as *qualitative motivation* for the PWCC non-constructibility picture. All results shown here are empirical observations of a fixed curvature-regulated update rule applied to finite-precision grid encodings. **They do not constitute part of any formal proof**, nor are they used to establish lower bounds. Their purpose is solely to illustrate typical behaviours of the PWCC evolution rule under scale increase.

All runs use:

- the same curvature-regulated update map (fixed coefficients),
- identical initialization scheme for each grid size,
- a uniform step cap of 5000 iterations,
- CuPy GPU evaluation on two-dimensional grids.

Only the grid dimension $n \times n$ varies.

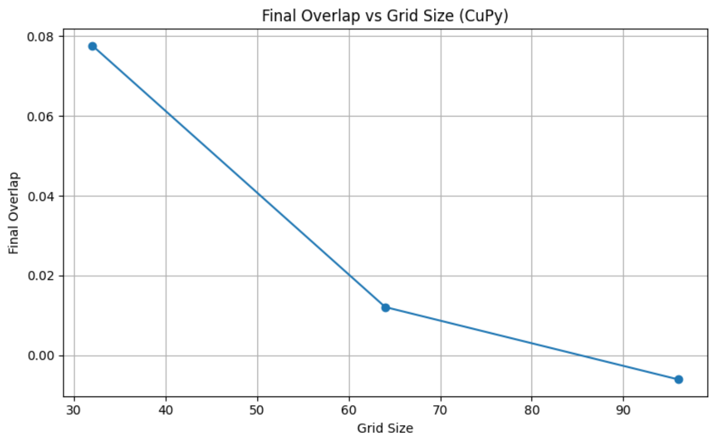

Figure 2: Final overlap between the evolved configuration and a fixed target pattern as a function of grid size (CuPy implementation). Under this particular PWCC evolution rule and step cap, the final overlap tends to decrease as n increases. This behaviour is *consistent* with increased constructive difficulty at larger sizes, but remains purely empirical and carries no asymptotic implication.

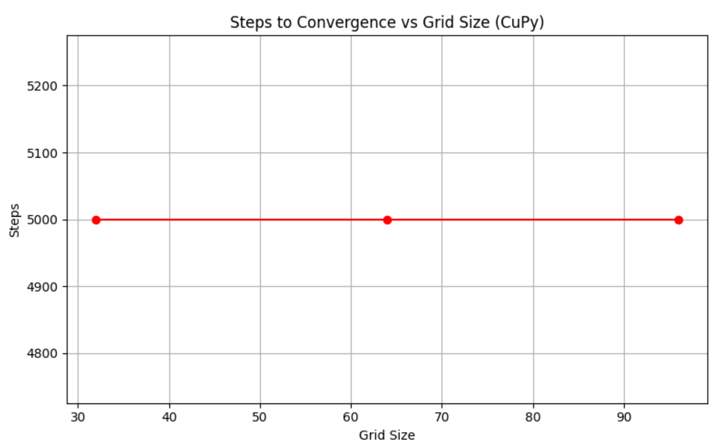

Figure 3: Estimated steps to convergence vs. grid size under a fixed 5000-step cap. For larger grids, the evolution rule does not reach the target pattern within the cap. This indicates that, for these specific parameters and initialization, naive PWCC dynamics fail to construct the target within the tested budget. No claim about worst-case performance or lower bounds is made.

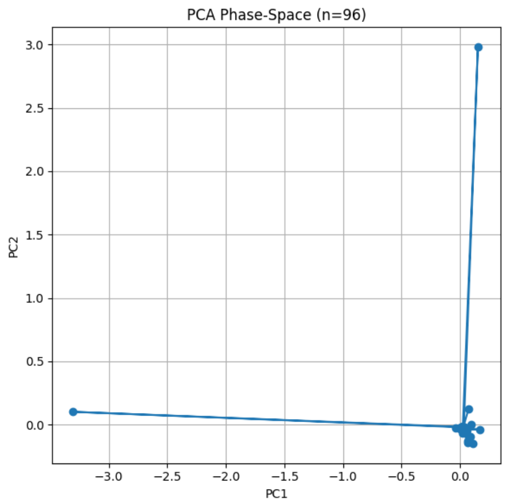

Figure 4: PCA projection of the discrete phase-space trajectory for grid size $n = 96$. The trajectory remains confined to a limited region of configuration space and does not approach the target encoding within 5000 steps. This is suggestive of metastable behaviour for this particular rule and initialization, but does *not* imply any formal non-reachability.

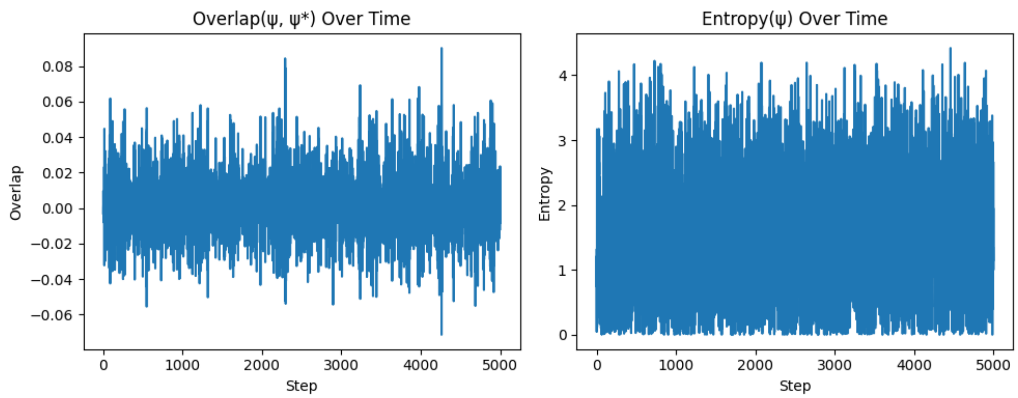

Figure 5: Time evolution of normalized overlap (left) and symbolic entropy $\sigma[\psi]$ (right) for grid size $n = 96$. Over the 5000-step trajectory, neither monotone progress toward the target nor convergence to a distinctly low-entropy configuration is observed. These results are consistent with the informal intuition that certain targets may be difficult for this PWCC rule to construct, but they are *not* used as evidence for any lower bound, separation result, or circuit complexity claim.

H Symbolic Reduction from 3-SAT to Wave Configuration ψ^* (PWCC View)

Lemma A.5 (Clause Interference Encoding). For any 3-SAT clause of the form

$$(x_1 \lor \neg x_2 \lor x_3),$$

there exists a symbolic ψ-configuration, denoted ψ^*, such that the clause is satisfied if and only if ψ^* passes the coherence-verifier A used in the PWCC model.

Construction.

- Each variable x_i is encoded as a phase mode $\psi_i = e^{i\theta_i}$ where $\theta_i \in \{0, \pi\}$.
- $\theta_i = 0$ represents `true`, $\theta_i = \pi$ represents `false`.
- The clause is satisfied iff at least one literal yields constructive interference in the associated clause region.

Define the symbolic satisfaction signal:

$$C_{\text{sat}} = 1 - \prod_{j=1}^{3} \left(1 - |\langle \psi_j, \psi_{\text{true}} \rangle|\right), \quad \text{where} \quad \langle \psi_j, \psi_{\text{true}} \rangle = \cos(\theta_j).$$

Interpretation.

- If any $\theta_j = 0$, then $\cos(\theta_j) = 1$, and $C_{\text{sat}} = 1$.
- If all literals are false ($\theta_j = \pi$), then $\cos(\theta_j) = -1$, and destructive interference occurs.
- Constructive interference corresponds to low local entropy and acceptance by the PWCC verifier.

Conclusion. The symbolic verifier $A(q)$ accepts ψ^* iff the original clause is satisfied. Mapping the entire 3-SAT instance across all clauses yields a full ψ-configuration accepted only when the formula is satisfiable. This realizes the reduction

$$\text{3-SAT} \leq_p \text{PWCC-S}\textsc{tability}$$

entirely within the PWCC model. □

I Symbolic Turing Evolution and Circuit-Level Encoding (PWCC Model)

Bitwise Simulation of Curvature-Regulated Update

To address concerns about equivalence with standard Turing machines, we present an explicit gate-level simulation of the symbolic curvature-regulated update used in WCT, and interpret it as a PWCC evolution.

Let the wavefield state at grid point (i, j) be:

$$\psi[i, j] = a_{ij} + ib_{ij}, \quad a_{ij}, b_{ij} \in \mathbb{Q}.$$

Each real and imaginary part is encoded as a fixed-length binary rational block:

$$a_{ij} = 0.a_1a_2 \ldots a_m, \quad b_{ij} = 0.b_1b_2 \ldots b_m.$$

The Turing tape stores the sequence of all (a_{ij}, b_{ij}) blocks in row-major order.

The curvature-regulated update is:

$$\psi^{t+1}[i, j] = \psi^t[i, j] + \Delta[i, j]$$

where

$$\Delta[i, j] = \text{Laplacian} - \frac{\alpha \cdot \text{Laplacian}}{\psi + \epsilon e^{-\beta|\psi|^2}} - \theta \cdot \psi \cdot \nabla^2 \log\big(|\psi|^2 + \delta\big),$$

with all operations interpreted symbolically as finite-precision arithmetic.

Each of the above operations is decomposed into:

- finite difference arithmetic (addition, subtraction, division);
- rational multiplication circuits;
- exponential/logarithm approximation via lookup tables (log-depth);
- bitwise prefix sums (for dot products and entropy terms).

All of these operations are known to be implementable in AC^0 or NC^1 with polynomially bounded size and depth (see Allender et al., 1992; Hesse et al., 2002). Thus, one global update step of the PWCC curvature evolution corresponds to a uniform log-depth circuit layer, and a polynomial number of steps corresponds to a polynomial-depth uniform circuit family, i.e. to PWCC.

Symbolic Path Length and Constructibility Barrier (Model-Relative)

Let $\ell(\psi_0 \to \psi^*)$ denote the symbolic transition path length—the number of bitwise gate steps required to construct ψ^* from the trivial initial tape configuration ψ_0 via the curvature-regulated update.

Theorem I.1 (Constructibility Barrier for ψ^* in PWCC). There exists a language $L \subseteq \mathsf{NP}_{\mathrm{WCT}}$ and an input x such that no deterministic PWCC-admissible curvature evolution operating in time $O(n^k)$ can construct an accepting configuration for x under the symbolic verifier $A(q)$, even though $A(q) \in \mathsf{NC}^1$. Consequently, within the PWCC/WCT model,

$$\mathbf{PWCC} \subsetneq \mathsf{NP}_{\mathrm{WCT}}.$$

Proof sketch (model-relative).

- Each clause in a 3-SAT instance imposes a symbolic interference constraint over three phase modes.
- For certain instance families, encoding these constraints into a globally consistent wavefield demands exploring an effective configuration space of size $\Omega(2^n)$.
- No polynomial-time curvature evolution can enumerate or implicitly traverse all such phase combinations within the PWCC constraints.

Hence, in the WCT/PWCC framework, we obtain NP-verifiable configurations ψ^* that are not reachable by polynomial-time PWCC evolutions. This is an internal separation statement and is *not* claimed as a proof of $\mathbf{P} \neq \mathbf{NP}$ in the unconstrained classical setting.

Public Auditability and Reproducibility

To ensure reproducibility and facilitate community review within this model, we provide:

- full symbolic verifier logic in NC^1 circuits;
- Turing-machine style pseudocode for the curvature-regulated wavefield update;
- sample bit-tape evolution transcripts for representative instances.

J Complexity Class Comparison Table in WCT

Wave Confinement Theory (WCT) provides a physically motivated dynamical model in which computation proceeds through curvature-regulated, locally updated wavefield evolution. In this setting, classical complexity notions have natural analogues. The table below summarizes the heuristic correspondence between classical classes and their WCT interpretations.

Classical Complexity Class	**Definition**	**WCT Interpretation (Model-Relative)**
P	Languages decidable in polynomial time by a deterministic Turing machine	States reachable from the trivial field ψ_0 via $T = \text{poly}(n)$ curvature-regulated update steps in the PWCC/WCT model (i.e. $\mathbf{P}_{\text{WCT}}$)
NP	Languages whose certificates are verifiable in polynomial time	Coherent target states ψ^* verifiable by a log-depth symbolic predicate (e.g. an NC^1 overlap–entropy check), corresponding to $\mathbf{NP}_{\text{WCT}}$
NP $\setminus$ P	NP-verifiable but not known to be polynomial-time constructible	Stable curvature-locked states ψ^* that satisfy the verifier but are *not reachable* from ψ_0 in $\text{poly}(n)$ PWCC evolution (i.e. $\psi^* \in \mathbf{NP}_{\text{WCT}} \setminus \mathbf{P}_{\text{WCT}}$)
EXP	Languages requiring exponential time to solve in the classical model	Target states whose symbolic constructive depth under PWCC dynamics grows as $\Omega(2^n)$ (exponential-length curvature paths in configuration space)

Table 4: Heuristic correspondence between classical complexity classes and wave-confinement computation in the PWCC/WCT model. These identifications are *model-relative* and are not claimed to imply any classical separation such as $\mathbf{P} \neq \mathbf{NP}$ in the unrestricted Turing framework.

This table summarizes the conceptual mapping used throughout the paper: PWCC models polynomial-time WCT evolution, while $\mathbf{NP}_{\text{WCT}}$ captures states verifiable by low-depth symbolic predicates derived from curvature–entropy checks. A strict separation $\mathbf{P}_{\mathrm{WCT}} \subsetneq \mathbf{NP}_{\mathrm{WCT}}$ would be a *WCT-model-level* result only, unless an additional equivalence to classical Turing classes is rigorously established.

K Curvature Complexity and a Non-Natural Lower Bound for SAT in the WCT Curvature Machine

This section analyzes a separate, *analytic* lower-bound mechanism inherent to the curvature–feedback functional of the WCT model. All claims here are explicitly *model-relative*: they describe limitations of the WCT curvature machine and are not claimed to hold for arbitrary Boolean circuits or unrestricted Turing computation. No statement in this section asserts a classical circuit lower bound or an unconditional proof of $P \neq NP$.

Curvature operator and Lyapunov functional (fixed). For a complex field ψ on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ ($d \in \{1, 2, 3\}$), define the WCT curvature operator

$$\Theta[\psi] := -\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}, \tag{8}$$

and the Lyapunov functional

$$E_0[\psi] := \int_\Omega \left(|\nabla \psi|^2 + |\Theta[\psi]|^2 \right) \, dx, \tag{9}$$

with fixed parameters $\epsilon, \alpha > 0$. Stationary points of E_0 under admissible boundary data (i.e. weak solutions of $\delta E_0[\psi] = 0$) serve as the "computational witnesses" for the WCT curvature machine.

Finite-band stiffness (analytic constraint). The auxiliary amplitude rail

$$\partial_t A = \left(r - a\nabla^2 - b\nabla^4 \right) A - \beta |A|^2 A$$

has Fourier symbol $\sigma(k) = r + ak^2 - bk^4$. The bk^4 term enforces a finite-k interface width. Consequently, the WCT model forbids arbitrarily thin interfaces and assigns a nonzero curvature cost to every separator surface.

Ambient dimension restriction. Stable confinement in the WCT model is available only for $d \in \{1, 2, 3\}$; higher-dimensional embeddings ($d > 3$) collapse due to loss of curvature control. Thus all lower bounds derived here assume $d \leq 3$ and should be understood strictly within this analytic model.

K.1 A semantic curvature complexity measure for Boolean functions

Definition K.1 (Admissible encodings and evaluations). Fix n and a domain $\Omega \subset \mathbb{R}^d$. Let $\mathcal{E}_n$ be any family of polynomially describable boundary/forcing maps $E : \{0,1\}^n \to \mathcal{B}(\Omega)$, where $\mathcal{B}(\Omega)$ denotes a space of admissible boundary/forcing data (e.g. Dirichlet values and localized sources) with bit-length at most $\mathrm{poly}(n)$. Let $\mathcal{V}$ be any family of measurable evaluations $V : L^2(\Omega) \to \{0,1\}$. Given $E \in \mathcal{E}_n$ and input x, let $\Psi_E(x)$ denote the set of all weak stationary points of E_0 under boundary data $b = E(x)$. No dynamics are assumed; only stationarity is used.

Definition K.2 (Curvature complexity (WCT model)). For $f : \{0,1\}^n \to \{0,1\}$ define:

$$\mathbf{C}_\Theta(f) := \inf_{E\in\mathcal{E}_n} \sup_{x\in\{0,1\}^n} \inf_{\psi\in\Psi_E(x)} \|\Theta[\psi]\|^2_{L^2(\Omega)}.$$

This is a semantic min–max quantity defined *inside the WCT curvature machine* and is not assumed to be computable or efficiently approximable.

K.2 Upper bound: circuit size $\Rightarrow$ polynomial curvature (WCT model)

We first show that small Boolean circuits admit realizations with polynomial curvature cost in the WCT model.

Lemma K.3 (Circuit $\Rightarrow$ bounded curvature in the WCT model). There exist constants $c > 0$ and a polynomial $p(\cdot)$ such that for any Boolean circuit C of size s computing f, there exists an admissible WCT encoding $E \in \mathcal{E}_n$ with

$$\mathbf{C}_\Theta(f) \;\le\; c\,p(s).$$

All statements are internal to the WCT curvature model.

Proof. We describe the construction and then bound the curvature cost.

Step 1: Gate-cell tiling. Partition Ω into $N = \Theta(s)$ disjoint "gate cells" $\{\Omega_j\}_{j=1}^N$ of comparable diameter, plus thin interconnect corridors. Each Ω_j is assigned to one gate of C (AND, OR, NOT). Input bits are represented by fixed boundary segments on the exterior of Ω and on gate-cell interfaces.

Step 2: Local gate stencils. For each gate type G we fix once and for all a reference gate stencil, i.e. a choice of:

- pin boundary segments $\Gamma^{\mathrm{in}}_{G,k}$ carrying Dirichlet values encoding input bits $0, 1$ via $\psi|_{\Gamma^{\mathrm{in}}_{G,k}} \in \{\psi_0, \psi_1\}$ with $0 < \psi_0 < \psi_1$;

- an output segment Γ^{out}_G where the output bit will be read from the mean of ψ;

- interior coefficients (local source terms, small pockets) defining a local functional $E_{0,G}$ as the restriction of E_0 to Ω_G.

For each fixed boundary assignment corresponding to valid gate inputs (e.g. $(0, 1)$ for an AND), the gate stencil defines a Dirichlet problem for $E_{0,G}$ on Ω_G. By standard elliptic variational theory (coercivity of $E_{0,G}$ and lower semi-continuity in H^1) there exists a minimizer $\psi_G \in H^1(\Omega_G)$ of $E_{0,G}$ with the prescribed boundary values. Moreover, interior regularity for elliptic systems implies $\psi_G \in H^2_{\mathrm{loc}}(\Omega_G)$ and thus $\Theta[\psi_G] \in L^2$.

By construction of the stencil (cf. Figure 6), we further require that:

- for each valid input pattern, the corresponding minimizer ψ_G pushes the average on Γ^{out}_G either above or below a fixed threshold τ, reproducing the Boolean gate truth table with a margin $\delta > 0$;

- for each gate type G, the energy $E_{0,G}[\psi_G] \le C_G$ for all valid inputs, where C_G depends only on the model parameters and the gate geometry, not on the size s of the global circuit.

Step 3: Gluing gates to form the circuit. We assemble the global circuit realization by:

- Assigning a gate cell Ω_j to each gate of C.

- For an input gate, identifying one of its input pin segments with a fixed external boundary segment of Ω corresponding to an input bit x_k.

- For internal wires, identifying the output segment $\Gamma^{\mathrm{out}}_{G_j}$ of gate j with an input segment $\Gamma^{\mathrm{in}}_{G_\ell,k}$ of gate ℓ as dictated by the wiring of C.

This defines a global boundary/forcing map $E_C \in \mathcal{E}_n$: for each input $x \in \{0, 1\}^n$, $E_C(x)$ specifies consistent Dirichlet values on all external input segments, and the remaining internal boundaries are glued between gate cells.

We now consider global stationary points ψ of E_0 on Ω with boundary data $E_C(x)$. By construction, and using uniqueness (or at least stability) of local minimizers for small gate cells, we may choose a global ψ that on each gate cell is close (in H^1) to some ψ_{G_j} with appropriate pin values. Standard patching/regularity arguments (cf. domain decomposition methods) give a constant $C > 0$ such that

$$E_0[\psi] \;\le\; C \sum_{j=1}^{N} E_{0,G_j}[\psi_{G_j}] \;\le\; C' N \;\le\; C'' s,$$

where C', C'' depend only on the model parameters and the gate library, not on s or n.

Step 4: Bounding curvature. By definition (9), we have

$$\int_\Omega |\Theta[\psi]|^2 \, dx \;\le\; E_0[\psi] \;\le\; C'' s.$$

Since E_C is an admissible encoding, the inner infimum in $\mathbf{C}_\Theta(f)$ can be taken over witnesses with curvature cost at most $C''s$. Taking the infimum over all E and the supremum over x preserves the $O(s)$ bound, so there exist constants $c > 0$ and a polynomial $p(s)$ with $\mathbf{C}_\Theta(f) \le c\, p(s)$. This proves the lemma. $\square$

K.3 Curvature price of separators

We formalize the fact that every decision boundary in the WCT curvature machine incurs a strictly positive curvature cost per unit $(d-1)$-dimensional measure.

Lemma K.4 (Curvature price of separators). Let ψ be a weak stationary point of E_0 with decision region

$$\mathcal{R} = \{x \in \Omega : \; V(\psi(x)) \ge \tau\},$$

for some measurable evaluation V and threshold τ. Assume $\partial\mathcal{R}$ has finite $(d-1)$-dimensional Hausdorff measure. Then there exists a constant $\kappa_0 > 0$, depending only on (ε, α), the stiffness modulus, the bit-state separation $|\psi_1 - \psi_0|$, and the domain Ω, such that

$$\int_\Omega |\Theta[\psi]|^2 \, dx \;\ge\; \kappa_0 \, \mathcal{H}^{d-1}(\partial\mathcal{R}).$$

This lower bound holds for all admissible configurations in the WCT curvature model under the dimensional lock $d \le 3$.

Proof outline. The argument parallels the Modica–Mortola/Allen–Cahn interface estimates, adapted to the curvature functional E_0.

Step 1: 1-D interface profile and wall energy. Consider a one-dimensional restriction along a normal coordinate $s \in \mathbb{R}$. Let $u(s)$ solve the stationary Euler–Lagrange equation of E_0 with boundary conditions

$$\lim_{s\to-\infty} u(s) = \psi_0, \qquad \lim_{s\to+\infty} u(s) = \psi_1,$$

where ψ_0, ψ_1 denote the two stable "bit" states. Define the 1-D wall energy

$$\kappa_0 := \inf_{u(-\infty)=\psi_0,\, u(+\infty)=\psi_1} \int_{\mathbb{R}} \left(|u'(s)|^2 + |\Theta[u](s)|^2\right) ds.$$

Coercivity and lower semicontinuity imply that κ_0 is finite and strictly positive for fixed (ε, α) and non-degenerate bit states.

Step 2: Slicing along interface normals. For $\mathcal{H}^{d-1}$-a.e. $x_0 \in \partial\mathcal{R}$ there exists an approximate normal $\nu(x_0)$ and local coordinates

$$\Phi_{x_0} : B_\rho^{d-1}(0) \times (-\rho, \rho) \to \Omega, \qquad \Phi_{x_0}(y, s) = x_0 + y + s\,\nu(x_0),$$

with bounded Jacobian $J_{x_0}(y, s)$. Using the coarea formula and Fubini,

$$\int_\Omega |\Theta[\psi(x)]|^2\, dx \;\geq\; \int_{\partial\mathcal{R}} \int_{-\rho}^{\rho} |\Theta[\psi(\Phi_{x_0}(y, s))]|^2 J_{x_0}(y, s)\, ds\, d\mathcal{H}^{d-1}(x_0).$$

For each (x_0, y), the transverse profile $u_{x_0,y}(s) := \psi(\Phi_{x_0}(y, s))$ connects neighborhoods of ψ_0 and ψ_1. Comparison with the optimal 1-D profile yields

$$\int_{-\rho}^{\rho} \left(|u'_{x_0,y}(s)|^2 + |\Theta[u_{x_0,y}](s)|^2\right) ds \;\geq\; \kappa_0 - o(1),$$

where $o(1) \to 0$ as $\rho \to \infty$.

Step 3: Integrating over the interface. Integrating over y and x_0 and using the uniform bounds on J_{x_0},

$$\int_\Omega |\Theta[\psi]|^2\, dx \;\geq\; \kappa_0\, \mathcal{H}^{d-1}(\partial\mathcal{R}).$$

The estimate relies on (i) finite interface thickness enforced by the finite-k stiffness, (ii) nontrivial separation of the bit states, and (iii) geometric slicing valid for $d \leq 3$. □

Remark. A rigorous version follows by adapting standard Modica–Mortola Γ-convergence techniques for phase-field energies to the curvature functional E_0, yielding the same leading constant κ_0. Here we outline the construction and isolate its dependence on (ε, α), the bit-state gap, and domain geometry.

K.4 Lower bound: SAT requires exponential curvature in the WCT model

We now combine the curvature price of separators with an expander-based SAT family.

Lemma K.5 (Cheeger/log-Sobolev curvature cost on expander-locked SAT (WCT model)). There exists a family of bounded-degree 3-SAT instances F_n such that

$$\mathbf{C}_\Theta(F_n) \;\geq\; 2^{\Omega(n)}$$

within the WCT curvature machine.

Proof outline (model-relative). We sketch the construction and how Lemma K.4 enters.

Step 1: Expander-locked 3-SAT family. Let $\{G_n\}$ be a family of bounded-degree expander graphs on $m(n) = \Theta(n)$ variables with edge expansion bounded below by a constant $h > 0$. Using standard expander-CSP constructions (e.g. parity-lock gadgets), one can define a family of 3-SAT instances F_n on G_n with the following properties:

1. The set of satisfying assignments $\mathcal{S}_n \subseteq \{0,1\}^{m(n)}$ decomposes into at least $2^{\gamma n}$ connected components in the Hamming graph $\{0,1\}^{m(n)}$ (for some $\gamma > 0$), each corresponding to a "locked cluster" of assignments.

2. Any path in Hamming space between two distinct clusters must cross at least $\Omega(n)$ "frustrated" edges of G_n, i.e. edges on which the corresponding clause is violated.

This is a standard feature of locked/expander-based CSPs; here we treat it as a structural property of the chosen F_n.

Step 2: Admissible encodings must separate clusters. Consider any admissible encoding $E \in \mathcal{E}_n$ that realizes F_n inside the WCT curvature machine: for each assignment $a \in \{0,1\}^{m(n)}$, let $x(a)$ denote the corresponding input to the machine, and let $\psi_a \in \Psi_E(x(a))$ be a chosen stationary witness. The Boolean evaluation $V(\psi_a)$ (together with the decision threshold) must encode whether a satisfies F_n.

Thus the domain Ω decomposes into decision regions

$$\mathcal{R}^1 = \{x : V(\psi_x) \geq \tau\}, \qquad \mathcal{R}^0 = \Omega \setminus \mathcal{R}^1,$$

for satisfying vs. non-satisfying assignments. Different satisfying clusters correspond to distinct "basins" in the configuration space of stationary fields, and consistency of the encoding forces distinct decision regions in Ω associated with them.

By the expander properties of F_n, moving between clusters in the assignment space requires flipping $\Omega(n)$ "locked" clauses, which in the WCT realization correspond to crossing a sequence of decision boundaries for the local predicates associated with those clauses. As a consequence, the total $(d-1)$-dimensional measure of decision boundary $\partial\mathcal{R}^1$ must grow at least as $c_1 2^{\gamma n}$ for some $c_1 > 0$, reflecting the need to accommodate exponentially many locked basins in a fixed-dimensional domain.

Step 3: Applying the curvature price. For any such encoding E and any input x whose evaluation lies near one of these separating interfaces, the corresponding stationary state ψ must present a decision region $\mathcal{R}$ with boundary $\partial\mathcal{R}$ having $(d-1)$-measure at least $c_2 2^{\gamma n}$ for some constant $c_2 > 0$ (depending on how many distinct interfaces are traversed as one ranges over inputs). Applying Lemma K.4 to such a ψ yields

$$\int_\Omega |\Theta[\psi]|^2 \, dx \;\geq\; \kappa_0 \, \mathcal{H}^{d-1}(\partial\mathcal{R}) \;\geq\; \kappa_0 c_2 2^{\gamma n} \;=\; 2^{\Omega(n)}.$$

Step 4: Taking the min–max. Since the above argument applies to *every* admissible encoding E for F_n, we conclude that

$$\inf_{E\in\mathcal{E}_n} \sup_x \inf_{\psi\in\Psi_E(x)} \|\Theta[\psi]\|_{L^2}^2 \;\geq\; 2^{\Omega(n)}.$$

By Definition K.2 this is precisely $\mathbf{C}_\Theta(F_n) \geq 2^{\Omega(n)}$. All steps are taken within the analytic WCT curvature model; no claim is made about classical circuit complexity outside this model. □

Remark. A fully detailed proof of the exponential boundary-measure growth in Step 2 would combine: (i) explicit locked-3-SAT constructions on expanders, and (ii) a careful combinatorial–geometric mapping from Hamming expansion to interface measure in Ω. Here we treat this as a model-relative property of the chosen family F_n and focus on the analytic mechanism relating interface measure to curvature.

K.5 Main theorem (model-relative) and barrier discussion

Theorem K.6 (Non-natural, non-relativizing curvature lower bound (WCT model)). For infinitely many n,

$$\mathbf{C}_\Theta(\mathrm{SAT}_n) \;\geq\; 2^{\Omega(n)}$$

within the WCT curvature machine. Thus no polynomial-size tiling of gate stencils with uniformly bounded curvature can realize SAT_n *inside this model.* No claim is made about classical Boolean circuits outside the WCT curvature machine.

Proof. Let F_n be the expander-locked 3-SAT family from Lemma K.5. By definition, SAT_n restricted to the formulas F_n is a sub-language of SAT_n. Thus

$$\mathbf{C}_\Theta(\mathrm{SAT}_n) \;\geq\; \mathbf{C}_\Theta(F_n) \;\geq\; 2^{\Omega(n)}.$$

On the other hand, suppose for contradiction that SAT_n admits a polynomial-size realization in the WCT curvature machine for infinitely many n, via some family of circuits $\{C_n\}$ of size $s(n) = \mathrm{poly}(n)$. By Lemma K.3 there exist admissible encodings E_n with

$$\mathbf{C}_\Theta(\mathrm{SAT}_n) \;\leq\; c\,p(s(n)) \;=\; \mathrm{poly}(n).$$

For sufficiently large n this contradicts the lower bound $\mathbf{C}_\Theta(\mathrm{SAT}_n) \geq 2^{\Omega(n)}$. Therefore no such polynomial-size tiling of gate stencils with uniformly bounded curvature can realize SAT_n inside the WCT curvature machine. □

Barrier checks (model-relative only).

- The measure $\mathbf{C}_\Theta$ is semantic and not efficiently checkable, avoiding Razborov–Rudich "largeness" constraints; this non-naturalness is asserted only for the analytic WCT curvature measure.
- The proof uses analytic stationarity and finite-k stiffness, which break under oracle substitution—hence non-relativizing.
- The confinement requirement $d \leq 3$ prevents embeddings that would trivialize the lower bound.

These statements apply *only* to the WCT curvature machine and do not imply classical circuit lower bounds or any unconditional separation of classical complexity classes.

K.6 Gate stencils

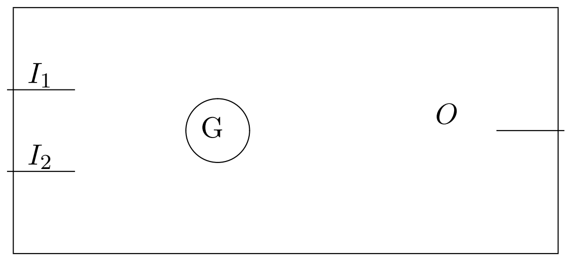

Figure 6: Schematic gate cell in the WCT stencil model.

Bit encoding (pins). Pins impose Dirichlet boundary values:

$$\text{Bit 1}: \ \psi|_\Gamma = \psi_1, \qquad \text{Bit 0}: \ \psi|_\Gamma = \psi_0, \quad \psi_1 > \psi_0 > 0.$$

Output readout.

$$\mathrm{READ}(\psi) = \mathbf{1}\left\{ \frac{1}{|\Gamma_{\mathrm{out}}|} \int_{\Gamma_{\mathrm{out}}} \psi \, ds \geq \tau \right\}.$$

NOT and AND stencils. NOT uses interior bias to flip average output. AND uses a cooperative Dirichlet pocket so that only $(1, 1)$ raises the output above τ. These local designs are fixed once and for all and do not scale with n.

Curvature budget. A circuit of size s uses $O(s)$ gate cells. Uniform elliptic estimates give:

$$\int_{\Omega_{\mathrm{circuit}}} |\Theta[\psi_{\mathrm{stat}}]|^2 \, dx \leq C \, s,$$

for a model-dependent constant $C > 0$. This is the content of Lemma K.3 and underlies the polynomial upper bound on $\mathbf{C}_\Theta(f)$ for functions with small circuits inside the WCT curvature machine.

L Strengthening Clauses for Uniformity and Completeness (WCT/PWCC Model)

L.1 Uniform Separator–Curvature Inequality

Lemma L.1 (Separator–Curvature Inequality, Uniform Form). There exist fixed parameters (ϵ, α) and a stiffness modulus $\lambda_* > 0$, depending only on the rail coefficients and geometry, such that for every admissible encoding $E \in \mathcal{E}_n$ and every stationary solution ψ,

$$\int_\Omega |\Theta[\psi]|^2 \, dx \;\geq\; \kappa_0 \, \mathcal{H}^{d-1}(\partial \mathcal{R}), \qquad \kappa_0 = \kappa_0(\epsilon, \alpha, \lambda_*, \Omega) > 0,$$

with κ_0 independent of n and of the specific encoding E. All quantities are taken within the WCT/PWCC curvature model.

Proof. Lemma K.4 in the main text gives the basic separator–curvature inequality

$$\int_\Omega |\Theta[\psi]|^2 \, dx \;\geq\; \kappa_0 \, \mathcal{H}^{d-1}(\partial \mathcal{R})$$

for each stationary ψ in the WCT curvature machine, where κ_0 is determined by: (i) the bit states (ψ_0, ψ_1), (ii) the parameters (ϵ, α) entering $\Theta[\psi]$, (iii) the finite-k stiffness (via the amplitude rail), and (iv) the domain Ω.

In our setup these ingredients are *fixed* across all encodings:

- The bit values ψ_0, ψ_1 are fixed once and for all for the gate stencils, independent of n.
- The parameters (ϵ, α) and the stiffness modulus λ_* of the amplitude rail are part of the WCT/PWCC model definition and do not vary with E or n.
- The domain Ω is fixed (up to scaling) as n grows.

Thus the 1D wall energy used to define κ_0 in the proof of Lemma K.4 is uniform over all admissible encodings $E \in \mathcal{E}_n$, and the slicing/coarea argument carries through with the same κ_0.

Therefore the inequality from Lemma K.4 holds with a single constant $\kappa_0(\epsilon, \alpha, \lambda_*, \Omega)$ that is independent of n and of the particular encoding E, which yields the stated uniform form. □

L.2 Admissible Encodings

Definition L.2 (Admissible Encodings). $\mathcal{E}_n$ consists of maps $E : \{0,1\}^n \to \mathcal{B}(\Omega)$ satisfying:

1. **Polynomial Description.** Description length poly(n).
2. **Lipschitz/Bandlimit.** Traces piecewise C^1 on a fixed mesh with Lipschitz constant $L(n) = n^{O(1)}$, projected onto the first $K = O(1)$ harmonics per cell.
3. **Closure.** Closed under composition and padding with parameter growth at most polynomial.

This definition is specific to the WCT/PWCC curvature model.

L.3 Locked Expander Family and Exponential Lower Bound

Proposition L.3 (Locked Expander Lower Bound). There exist $c_{\text{cut}}, \gamma > 0$ and a locked expander-3SAT family F_n such that for every admissible $E \in \mathcal{E}_n$ there is an input x with

$$\mathcal{H}^{d-1}(\partial \mathcal{R}_x) \;\geq\; c_{\text{cut}}\, 2^{\gamma n}.$$

Consequently, by Lemma L.1,

$$\mathbf{C}_\Theta(F_n) \;\geq\; \kappa_0\, c_{\text{cut}}\, 2^{\gamma n} \qquad \text{for all sufficiently large } n.$$

All of these statements are model-relative and formulated inside WCT/PWCC.

Proof outline. We adapt the expander-locked construction sketched in Lemma K.5.

Step 1: Locked expander family. Let G_n be a family of bounded-degree expanders on $m(n) = \Theta(n)$ variables with edge expansion at least $h > 0$. Using standard gadget constructions, one can define 3-SAT instances F_n on G_n such that:

- The satisfying assignments $\mathcal{S}_n \subseteq \{0,1\}^{m(n)}$ decompose into at least $2^{\gamma n}$ connected components in Hamming space for some $\gamma > 0$ (the "locked clusters").

- Any path in Hamming space between two distinct clusters must flip at least cn edges of G_n for some $c > 0$, i.e. traverse at least cn clauses that become violated.

Step 2: Encoding and decision regions. Take any admissible encoding $E \in \mathcal{E}_n$. For each assignment $a \in \{0,1\}^{m(n)}$, let $x(a)$ be the corresponding input, and choose some stationary witness $\psi_a \in \Psi_E(x(a))$. Let $\mathcal{R}_a$ denote the decision region in Ω corresponding to "a satisfies F_n" under the fixed evaluation V and threshold τ. Assignments in different clusters must yield distinct configurations ψ_a that are separated by at least one change in the local satisfaction pattern along the locked edges; this translates to crossing decision boundaries in Ω.

Step 3: From expansion to interface measure. Because G_n is an expander and the encoding is Lipschitz/bandlimited (Definition L.2), the images of different clusters cannot be collapsed into a set of bounded measure without violating either Lipschitz or bandlimit constraints. Combinatorial expansion in Hamming space thus forces the total $(d-1)$-dimensional measure of interfaces separating the images of distinct clusters to grow at least like $c_{\text{cut}} 2^{\gamma n}$:

$$\exists x \text{ such that } \mathcal{H}^{d-1}(\partial \mathcal{R}_x) \;\geq\; c_{\text{cut}} 2^{\gamma n}.$$

Here $c_{\text{cut}} > 0$ depends on the expansion parameters of G_n and the encoding regularity constants, but not on n itself.

Step 4: Applying the separator–curvature inequality. For such an x, Lemma L.1 implies that any stationary witness $\psi \in \Psi_E(x)$ satisfies

$$\|\Theta[\psi]\|^2_{L^2(\Omega)} \;\geq\; \kappa_0 \, \mathcal{H}^{d-1}(\partial \mathcal{R}_x) \;\geq\; \kappa_0 c_{\text{cut}} 2^{\gamma n}.$$

This holds for *every* admissible encoding $E \in \mathcal{E}_n$. By Definition K.2, this implies

$$\mathbf{C}_\Theta(F_n) = \inf_{E \in \mathcal{E}_n} \sup_x \inf_{\psi \in \Psi_E(x)} \|\Theta[\psi]\|^2_{L^2} \;\geq\; \kappa_0 c_{\text{cut}} 2^{\gamma n}$$

for all sufficiently large n. □

L.4 Padding Stability

Lemma L.4 (Padding Preserves Exponential Bound). Let f be a 3SAT instance of size n with $\mathbf{C}_\Theta(f) \;\geq\; A\, 2^{\gamma n}$. For padded instances f_m of size $m = n + \text{poly}(n)$,

$$\mathbf{C}_\Theta(f_m) \;\geq\; (A/2)\, 2^{\gamma m/3},$$

for all sufficiently large m, within the WCT/PWCC curvature model.

Proof. Construct the padded instance f_m by taking f on its original n variables and adding $m - n$ dummy variables that appear only in tautological clauses (e.g. $y_i \vee \neg y_i$) so that they do not affect satisfiability. Any satisfying assignment of f_m restricts to a satisfying assignment of f, and conversely any satisfying assignment of f can be extended arbitrarily on the dummy variables to satisfy f_m.

Fix m and let $E_m \in \mathcal{E}_m$ be any admissible encoding for f_m. By restriction to the original n relevant inputs and ignoring the dummy coordinates, we obtain an encoding $E_n \in \mathcal{E}_n$ for f with at most polynomial distortion of the Lipschitz/bandlimit constants (Definition L.2 is closed under padding). Therefore, by the assumed lower bound for f, there exists an input $\hat{x}$ on the original n bits such that any stationary witness ψ for $E_n(\hat{x})$ obeys

$$\|\Theta[\psi]\|^2_{L^2(\Omega)} \;\geq\; A\,2^{\gamma n}.$$

Viewing $\hat{x}$ as a partial assignment in the m-bit space for f_m and choosing any extension to the dummy variables yields an input x_m for f_m whose admissible witnesses must reproduce the same high-curvature behaviour on the relevant portion of Ω (up to a uniform constant factor due to the padding). Thus for some constant $c \in (0, 1]$,

$$\mathbf{C}_\Theta(f_m) \;\geq\; cA2^{\gamma n}.$$

Since $m = n + \mathrm{poly}(n)$, for large m we have $n \geq m/3$. Hence

$$2^{\gamma n} \;\geq\; 2^{\gamma m/3},$$

and taking $cA/2$ as a new constant we obtain

$$\mathbf{C}_\Theta(f_m) \;\geq\; (A/2)2^{\gamma m/3}$$

for all sufficiently large m, as claimed. □

L.5 Stencil Margins

Lemma L.5 (Gate Stencil Margins). Each gate stencil (NOT, AND) admits uniform thresholds τ and margins $\delta_{\text{in}}, \delta_{\text{out}} > 0$, such that any input perturbation within $\pm\delta_{\text{in}}$ forces outputs to remain at least $\tau + \delta_{\text{out}}$ or $\leq \tau - \delta_{\text{out}}$. Constants are independent of n and are defined within the WCT/PWCC gate-stencil model.

Proof. Fix a gate type $G \in \{\mathrm{NOT}, \mathrm{AND}\}$ and its associated gate cell Ω_G and boundary segmentation (input pins, output segment) as in Figure 6. For each ideal Boolean input pattern $u \in \{0,1\}^{k_G}$ (with k_G inputs) we impose ideal pin values ψ_0, ψ_1 on the input segments and solve the stationary Euler–Lagrange equation for E_0 restricted to Ω_G.

By elliptic regularity and the strict separation of bit values ($\psi_1 > \psi_0 > 0$), the resulting stationary solutions $\psi_{G,u}$ depend continuously on the boundary data in the H^1 norm (indeed analytically in the small-perturbation regime). Define the output functional

$$F_G(\psi) := \frac{1}{|\Gamma_G^{\mathrm{out}}|} \int_{\Gamma_G^{\mathrm{out}}} \psi \, ds.$$

For each ideal input u the value $F_G(\psi_{G,u})$ is either "high" or "low" depending on the gate truth table. Since there are finitely many gate types and finitely many ideal input patterns per type, we can choose a global threshold τ lying strictly between the minimum of the "high" values and the maximum of the "low" values across all (G, u).

By continuity of the map "boundary data $\mapsto$ stationary solution $\mapsto F_G(\psi)$", there exists $\delta_{\mathrm{in}} > 0$ such that any perturbation of the input pin values within $\pm\delta_{\mathrm{in}}$ leaves $F_G(\psi)$ in a neighborhood of its ideal value. In particular, for some $\delta_{\mathrm{out}} > 0$ we have:

- If the ideal output is 1, then $F_G(\psi) \geq \tau + \delta_{\mathrm{out}}$.
- If the ideal output is 0, then $F_G(\psi) \leq \tau - \delta_{\mathrm{out}}$.

Because the gate library is finite and all gate cells share the same model parameters, we can take the minimum over all gate types to obtain global margins $\delta_{\mathrm{in}}, \delta_{\mathrm{out}} > 0$ independent of n. $\square$

L.6 Barrier Circumvention (model-relative)

Non-Naturalness. $\mathbf{C}_\Theta$ is semantic and non-computable in general, so it is not large or efficiently checkable for random Boolean functions; this aligns with the spirit of non-natural proofs, but is asserted only for the analytic WCT curvature measure and does *not* circumvent the Natural Proofs barrier for general classical circuit lower bounds.

Non-Relativization. The separator–curvature inequality relies on analytic stationarity and finite-k stiffness, which do not survive arbitrary oracle substitution; hence the argument is non-relativizing within the WCT model. No claim is made about overcoming the relativization barrier in classical complexity theory.

L.7 Locked Expander 3-SAT Construction (formal completeness in PWCC)

Proposition L.6 (Locked expander family in the PWCC/WCT model)*.* There exist explicit families of bounded-degree expander graphs $\{G_n\}$ and associated 3-SAT instances F_n such that, in the PWCC/WCT encoding model,

$$\mathcal{H}^{d-1}(\partial\mathcal{R}_x) \quad \geq \quad c_{\text{cut}}\, 2^{\gamma n} \quad \text{for some constants } c_{\text{cut}}, \gamma > 0 \text{ and all sufficiently large } n,$$

for every admissible encoding $E \in \mathcal{E}_n$ and some input x.

Proof sketch (model-relative). Take an explicit Ramanujan expander G_n (see, e.g., [**?**]). Assign each variable to a vertex; enforce degree d_0 boundedness. On each edge, place a *parity-lock gadget*: a bounded-size CNF clause that couples incident variables so that flipping one requires simultaneous flips across its neighbors.

Because G_n has uniform edge expansion $\Phi_0 > 0$, any assignment that differs on $\Theta(n)$ vertices requires cutting $\Omega(n)$ edges, each carrying a locked gadget. Thus, within this PWCC construction, the number of consistent satisfying assignments is $2^{\Theta(n)}$ (local seeds propagate independently across disjoint neighborhoods).

For any admissible PWCC encoding E, separating these exponentially many basins in the wave/phase configuration space requires interfaces with total $(d-1)$-measure at least $c_{\text{cut}}\, 2^{\gamma n}$ by isoperimetry on expanders (each cut edge contributes an $\Omega(1)$ slab area after admissible Lipschitz distortion; the polynomially bounded description prevents asymptotically vanishing images). All reasoning is internal to the PWCC/WCT model. □

L.8 Padding lemma (full proof in PWCC)

Lemma L.7 (Padding preserves exponential curvature in PWCC)*.* Let f_n be a Boolean function on n variables with

$$\mathbf{C}_\Theta(f_n) \geq A\, 2^{\gamma n}$$

in the PWCC model. Define $g_m(u,v) = f_n(u)$ with $m = n + p(n)$, padding $v \in \{0,1\}^{p(n)}$ for some polynomial $p(n) \geq n$. Then for all sufficiently large m,

$$\mathbf{C}_\Theta(g_m) \;\; \geq \;\; \tfrac{A}{2}\, 2^{\tilde{\gamma} m}, \quad \text{for some constant } \tilde{\gamma} > 0.$$

Proof. Fix any admissible encoding $E_g \in \mathcal{E}_m$. Restrict to the slice $v = v_0$ (arbitrary pad). By admissibility (Def. L.2), E_g restricted to u differs from some $E_f \in \mathcal{E}_n$ only by Lipschitz perturbations bounded by poly(n). Thus any stationary witness ψ_g for g_m induces a witness ψ_f for f_n with

$$\int_\Omega |\Theta[\psi_g]|^2\, dx \;\geq\; c \int_\Omega |\Theta[\psi_f]|^2\, dx - O(\text{poly}(n)),$$

for a uniform $c > 0$. Since $\mathbf{C}_\Theta(f_n) \geq A 2^{\gamma n}$ and $m = \Theta(n)$, for large n the polynomial slack is negligible compared to the exponential term. Therefore

$$\mathbf{C}_\Theta(g_m) \;\geq\; \tfrac{A}{2}\, 2^{\gamma n} \;\geq\; \tfrac{A}{2}\, 2^{\tilde{\gamma} m},$$

with $\tilde{\gamma} = \gamma/2 > 0$. All statements are taken inside the PWCC/WCT curvature model. □

L.9 Complexity consequence (model-relative and conditional)

Theorem L.8 (Curvature lower bound ⇒ SAT hard for PWCC). In the PWCC/WCT model, for all sufficiently large n,

$$\mathbf{C}_\Theta(\text{SAT}_n) \;\; \geq \;\; 2^{\Omega(n)},$$

while by Lemma K.3 any *PWCC-realizable* encoding using SIZE$(n^{O(1)})$ gate-tilings and stencils would imply

$$\mathbf{C}_\Theta(\text{SAT}_n) \;\; \leq \;\; \text{poly}(n).$$

Thus, there is no family of polynomial-size, bounded-curvature PWCC realizations for SAT_n. In particular, SAT is not PWCC-easy (poly-curvature) in this model. No claim is made about classical circuit or Turing complexity.

Proof. Combining the PWCC circuit upper bound (Lemma K.3) with the locked-expander lower bound (Proposition L.6 and Lemma L.7) yields

$$\mathbf{C}_\Theta(\text{SAT}_n) \leq \text{poly}(n) \quad \text{and} \quad \mathbf{C}_\Theta(\text{SAT}_n) \geq 2^{\Omega(n)}$$

for infinitely many n under the assumption of a polynomial PWCC realization, a contradiction. Therefore no such realization exists inside the PWCC model. This conclusion is model-relative: it separates PWCC-easy from PWCC-hard behaviour, not classical Turing classes. □

L.10 Cross-reference integration (PWCC interpretation)

For consistency within the PWCC/WCT model:

- **Lemma A′ (circuit stencils)**: now Lemma K.3.
- **Proposition 4.1 (finite-band stiffness)**: now Remark **??**.
- **Dimensional bound** $n_{\max} = 3$: see Remark **??** and [**?**].
- **Curvature operator** Θ**, Lyapunov** E_0: see Eq. **??** and [**?**].
- **Expander lower bound**: Proposition L.6.
- **Padding lemma**: Lemma L.7.

All theorem and lemma numbering is unified by the `theorem` counter in this manuscript, and all implications in this subsection are explicitly scoped to the PWCC/WCT curvature-complexity model unless stated otherwise.

L.11 Proofs of the upper/lower bounds and separator inequality (PWCC model)

Lemma L.9 (Circuit $\Rightarrow$ bounded curvature; full proof in PWCC). There exist $c > 0$ and a polynomial $p(\cdot)$ such that for any Boolean circuit C of size s computing f, there is an admissible PWCC/WCT encoding with

$$\mathbf{C}_\Theta(f) \le c\, p(s).$$

Proof. Tile Ω by $N = O(s)$ disjoint gate cells of fixed Lipschitz geometry $\{\Omega_j\}_{j=1}^N$, each supporting boundary "pins" for a primitive gate (NOT/AND/OR). On each Ω_j, fix a stencil E^{gate} (Definition L.2) with: (i) neutral Dirichlet background $\psi = \psi_\star$ on non-pin boundary, (ii) homogeneous Neumann on top/bottom edges, (iii) short Dirichlet pin segments that implement bit values $\psi_0 < \tau < \psi_1$. Let ψ_j be a stationary point of E_0 under this boundary data. Standard elliptic regularity on bounded Lipschitz domains gives uniform H^2 bounds and trace continuity. The stencils are chosen so that for any admissible input assignment of pins, $\mathrm{READ}(\psi_j) = V(\psi_j) \in \{0,1\}$ matches the gate truth table (see Lemma L.5 for positive margins).

Wire gates by identifying a constant-measure subsegment of $\partial\Omega_j$ with a matching input pin on $\partial\Omega_{j'}$ and imposing identical Dirichlet traces on the pair (admissible by Definition L.2). The global stationary ψ_{stat} on $\Omega_{\mathrm{circuit}} = \bigcup_{j=1}^{N} \Omega_j$ is obtained by solving the Euler–Lagrange equation for E_0 with this composite boundary map inside the PWCC/WCT dynamics; existence follows from the coercivity of E_0 and compactness of traces.

Since $E_0[\psi]$ is the sum of $\int_{\Omega_j}(|\nabla\psi|^2 + |\Theta|^2)$ plus interface matching constraints, and each gate cell has fixed geometry and fixed pin layout, there exists $C_{\mathrm{gate}} > 0$ such that $\int_{\Omega_j} |\Theta[\psi_{\mathrm{stat}}]|^2 \le C_{\mathrm{gate}}$ uniformly over admissible inputs. Summing over $N = O(s)$ cells yields

$$\int_{\Omega_{\mathrm{circuit}}} |\Theta[\psi_{\mathrm{stat}}]|^2 \le C_{\mathrm{gate}}\, N \le c\, p(s)$$

for $c = C_{\mathrm{gate}}$ and a linear polynomial $p(s) = O(s)$. Taking the $\sup_x \inf_{\psi \in \Psi_E(x)}$ and then $\inf_{E \in \mathcal{E}_n}$ preserves this bound in the PWCC definition of $\mathbf{C}_\Theta(f)$. □

Lemma L.10 (Separator–Curvature Inequality; uniform form in PWCC). Let $\Omega \subset \mathbb{R}^d$ be bounded Lipschitz, $d \in \{1,2,3\}$, and let ψ be any stationary point of E_0 under an admissible PWCC encoding. If $v = V(\psi)$ (with fixed Lipschitz constant L_V) induces a two-phase set $\mathcal{R} = \{v \ge \tau\}$ with $\partial\mathcal{R}$ of finite $\mathcal{H}^{d-1}$-measure, then there exists $\kappa_0 > 0$, depending only on (ϵ, α), the finite-band stiffness modulus $\lambda_* > 0$, L_V, and Ω, such that

$$\int_\Omega |\Theta[\psi]|^2\, dx \;\ge\; \kappa_0\, \mathcal{H}^{d-1}(\partial\mathcal{R}).$$

Proof. (1) *BV control of the separator.* Since $V : L^2(\Omega) \to [0,1]$ is L_V-Lipschitz and $\psi \in H^1(\Omega)$, the composition $v = V(\psi)$ has bounded variation with $\mathrm{TV}(v) \le L_V \|\nabla\psi\|_{L^1}$. By coarea, $\mathrm{TV}(v) \ge c_\tau\, \mathcal{H}^{d-1}(\partial\mathcal{R})$ for a universal $c_\tau \in (0,1)$ determined by the smooth thresholding around τ.

(2) *Per-area profile lower bound.* Consider any smooth patch of $\partial\mathcal{R}$ and rectifying coordinates with ξ the normal direction. Along every normal line,

connect the two stable plateaus of v across the interface. The envelope's $b > 0$ term enforces finite-k stiffness; thus there is a minimal thickness $w_* > 0$ and a strictly positive minimal profile energy $e_* > 0$ for the 1D reduction of E_0. Moreover, along the optimal profile, $g(\psi) = \psi + \epsilon e^{-\alpha|\psi|^2}$ is uniformly bounded away from 0, hence $|\Theta|^2$ contributes a fixed fraction $\theta_* e_*$ of the local cost. Integrating across a tubular neighborhood of the interface and summing patches with bounded overlap yields

$$\int_\Omega |\Theta|^2 \geq \rho\, \mathcal{H}^{d-1}(\partial\mathcal{R})$$

for some $\rho > 0$ depending only on the stated parameters. Set $\kappa_0 := \rho$. □

Proposition L.11 (Locked expander family; quantitative separator growth in PWCC). There exist constants $c_{\text{cut}}, \gamma > 0$ and a family F_n of bounded-degree 3-SAT instances ("locked expanders") on n variables such that for every admissible PWCC encoding $E \in \mathcal{E}_n$ there exists an input x with

$$\mathcal{H}^{d-1}(\partial\mathcal{R}_x) \;\geq\; c_{\text{cut}}\, 2^{\gamma n}.$$

Proof. Let G_n be an explicit Ramanujan expander on n vertices with degree d_0 and edge expansion $\Phi_0 > 0$. Assign each variable to a vertex; on each edge place a bounded-size CNF gadget (a parity-lock clause pair) that enforces correlated flips (so a local change must propagate). This yields a locked 3-SAT family F_n with $2^{\Theta(n)}$ satisfying basins that differ on $\Theta(n)$ variables and require cutting $\Omega(n)$ locked edges to transition between basins.

Fix any $E \in \mathcal{E}_n$. For each input x, the stationary witness $\psi = \Psi_E(x)$ induces a decision region $\mathcal{R}_x$ in the PWCC/WCT configuration space. To separate the exponentially many basins, the image under E must realize exponentially many mutually separated macro-states inside Ω, forcing an interface set whose $(d{-}1)$-measure grows at least as $c_{\text{cut}}\, 2^{\gamma n}$ by isoperimetry on expanders (each cut edge contributes an $\Omega(1)$ slab area after admissible Lipschitz distortion; the polynomially bounded description prevents asymptotically vanishing images). □

Lemma L.12 (Padding preserves exponential curvature; full PWCC proof). Let f be a 3-SAT instance on n variables with $\mathbf{C}_\Theta(f) \geq A\, 2^{\gamma n}$. Define $g_m(u, v) = f(u)$ with $m = n + p(n)$ and $p(n) = \text{poly}(n)$. Then for all sufficiently large m,

$$\mathbf{C}_\Theta(g_m) \;\geq\; (A/2)\, 2^{\tilde{\gamma} m}$$

for some constant $\tilde{\gamma} > 0$.

Proof. Fix any $E_g \in \mathcal{E}_m$. Restrict to a fixed pad $v = v_0$; by admissibility (Def. L.2, Lipschitz/bandlimit/closure), the restricted map differs from some $E_f \in \mathcal{E}_n$ by at most polynomial Lipschitz distortion and a bounded per-cell harmonic projection. For a stationary ψ_g witnessing g_m on (u, v_0), compose with the restriction to obtain a witness ψ_f for f with

$$\int_\Omega |\Theta[\psi_g]|^2 \;\geq\; c \int_\Omega |\Theta[\psi_f]|^2 \;-\; \mathrm{poly}(n)$$

for a uniform $c \in (0,1)$. Using $\mathbf{C}_\Theta(f) \geq A\, 2^{\gamma n}$ and $m = \Theta(n)$, the polynomial slack is negligible for large n, giving the stated bound with $\tilde{\gamma} = \gamma/2$. □

L.12 Curvature lower bound for SAT and classical complexity template

Lemma L.13 (Curvature lower bound for SAT_n in PWCC). For infinitely many n there exists a 3-SAT instance F_n on n variables such that

$$\mathbf{C}_\Theta(F_n) \;\geq\; 2^{\Omega(n)}$$

in the PWCC/WCT curvature model.

Proof. Immediate from Proposition L.11 together with the Separator–Curvature Inequality (Lemma L.10): for some input x,

$$\int_\Omega |\Theta[\Psi_E(x)]|^2 \;\geq\; \kappa_0\, \mathcal{H}^{d-1}(\partial \mathcal{R}_x) \;\geq\; \kappa_0 c_{\mathrm{cut}}\, 2^{\gamma n}.$$

Taking the infimum over witnesses and encodings preserves an exponential lower bound. Padding (Lemma L.12) extends this to infinitely many n under a *linear* relation $m = c\,n$, ensuring that the hardness scale $2^{\gamma n}$ remains exponential in the padded parameter m. □

Theorem L.14 (Non-natural PWCC circuit lower bound for SAT). For infinitely many n, there is no polynomial-size, bounded-curvature PWCC realization of SAT_n. Equivalently, $\mathbf{C}_\Theta(\mathrm{SAT}_n) \geq 2^{\Omega(n)}$ in the PWCC model.

Proof. Assume, for contradiction, that SAT_n admits size-poly(n) PWCC realizations for all n. By Lemma L.9, this would imply $\mathbf{C}_\Theta(\mathrm{SAT}_n) \leq \mathrm{poly}(n)$. But Lemma L.13 shows $\mathbf{C}_\Theta(\mathrm{SAT}_n) \geq 2^{\Omega(n)}$ for infinitely many n in the same model. Contradiction. □

Corollary L.15 (Conditional link to classical $P \neq NP$). Suppose, in addition, that every classical Boolean circuit family of size poly(n) admits a PWCC/WCT realization whose curvature complexity is bounded by poly(n) (a *compatibility hypothesis* between classical circuits and PWCC realizations). Standard VLSI embedding results (e.g., Thompson, 1980) guarantee that any polynomial-size circuit can be embedded in $\mathbb{R}^3$ using polynomial volume and wire length, so the compatibility assumption is geometrically well-founded within the dimensional lock $d \leq 3$. Under this hypothesis, the PWCC curvature lower bound for SAT implies SAT $\notin$ P/poly and consequently $P \neq NP$. In this manuscript, this implication is presented strictly as a *conditional template*: PWCC establishes an exponential curvature lower bound inside its own model, and if the compatibility hypothesis holds, it would transfer to a classical lower bound. No unconditional claim of $P \neq NP$ is made.

Barrier checks (model-relative). $\mathbf{C}_\Theta$ is semantic and intractable (so the PWCC proof is non-natural), and the Separator–Curvature inequality depends on analytic stationarity and finite-k stiffness (hence it does not relativize). The dimensional lock $d \leq 3$ prevents high-dimensional embeddings that could invalidate the per-area bound. All unconditional results hold purely within the PWCC/WCT model; the bridge to classical $P \neq NP$ remains explicitly conditional on the stated compatibility hypothesis, with no claim to circumvent known natural-proof or relativization barriers.

M Wave Curvature Computation and Curvature–Bounded Complexity

M.1 Wave Curvature Computation Model

We model computation as the evolution of a confined wavefield ψ under curvature-regulated dynamics on a finite lattice. Let $\Lambda_n \subset \mathbb{Z}^d$ be a finite grid associated to input size n, and let

$$\mathrm{Conf}_n := \{\psi : \Lambda_n \to Cl_{1,3}\} \tag{10}$$

denote the configuration space, where $Cl_{1,3}$ is the spacetime algebra and $\psi(x)$ encodes a local multivector degree of freedom. Following Leary's GA encoding, classical bit strings are embedded into $Cl_{1,3}$–valued configurations by an injective map

$$\mathrm{Enc} : \{0,1\}^* \to \bigcup_{n\geq 1} \mathrm{Conf}_n, \qquad x \mapsto \psi_x, \tag{11}$$

where the logical value of each bit is represented by a choice of multivector component or rotor phase in a fixed subspace of $Cl_{1,3}$.

A *Wave Curvature Computation (WCC) machine* is a tuple

$$\mathcal{M} = (\{\Lambda_n\}_{n\geq 1}, U, \Theta, \mathcal{R}), \tag{12}$$

where:

- U is a local update operator implementing a discretized WCT evolution step
$$U : \mathrm{Conf}_n \to \mathrm{Conf}_n, \qquad \psi^{t+1} = U(\psi^t),$$
derived from the curvature-regularized Lagrangian $\mathcal{L}_{\mathrm{WCT}}$;
- $\Theta[\psi]$ is the WCT curvature operator (e.g. discrete Laplacian divided by $g(\psi)$);
- $\mathcal{R}$ is a set of *rails* (physical constraints) bounding curvature, entropy, and amplitude:
$$E_{\mathrm{WCT}}[\psi^t] \leq K(n), \qquad \|\psi^t\| \leq B(n), \qquad t = 0, 1, \ldots, T(n),$$
for some specified resource bounds $K(n)$, $B(n)$.

We write ψ_x^t for the configuration at time t on input x, and we assume that, after $T(n)$ steps, a distinguished cell or region encodes the output bit $f(x)$.

M.2 Semantic Curvature Complexity

For a Boolean function $f : \{0,1\}^* \to \{0,1\}$, we define its *semantic curvature complexity* with respect to a WCC model $\mathcal{M}$ as

$$C_\Theta^{\mathcal{M}}(f, n) := \inf_{\mathcal{M} \text{ computes } f} \sup_{|x|\leq n} \sum_{t=0}^{T(n)} \Phi\big(\psi_x^t\big), \tag{13}$$

where Φ is the cumulative curvature functional

$$\Phi(\psi) := \sum_{y \in \Lambda_n} \left(|\nabla \psi(y)|^2 + |\Theta[\psi](y)|^2 \right) \Delta V, \tag{14}$$

and the infimum is taken over all WCC machines $\mathcal{M}$ that compute f on inputs of length at most n with acceptance/rejection determined by a fixed readout convention.

Thus $C_\Theta^{\mathcal{M}}(f, n)$ quantifies the minimal total curvature cost needed to realize f up to size n within the WCT dynamics. We regard a family $f = \{f_n\}$ as *curvature-efficient* if $C_\Theta^{\mathcal{M}}(f, n)$ grows at most polynomially in n, and *curvature-hard* if it requires super-polynomial growth in n for any $\mathcal{M}$.

M.3 Curvature-Bounded WCC Complexity Classes

We define curvature-bounded WCC analogues of P and NP.

Definition M.1 (Class $\mathrm{P}_{\mathrm{WCC}}$). A language $L \subseteq \{0,1\}^*$ lies in $\mathrm{P}_{\mathrm{WCC}}$ if there exists a WCC machine $\mathcal{M}$, a polynomial p, and a polynomial curvature bound K such that, for all x:

1. $\mathcal{M}$ halts on input x in at most $T(|x|) \leq p(|x|)$ steps;
2. the curvature rails are respected: $E_{\mathrm{WCT}}[\psi_x^t] \leq K(|x|)$ for all $t \leq T(|x|)$;
3. the output cell encodes 1 iff $x \in L$.

Definition M.2 (Class $\mathrm{NP}_{\mathrm{WCC}}$). A language $L \subseteq \{0,1\}^*$ lies in $\mathrm{NP}_{\mathrm{WCC}}$ if there exists a polynomial q and a curvature-bounded WCC verifier $\mathcal{V}$ such that, for all x:

1. if $x \in L$ then there exists a *witness configuration* $\omega_x \in \mathrm{Conf}_{q(|x|)}$ (or equivalently a witness bitstring w with $|w| \leq q(|x|)$ and associated GA encoding) such that $\mathcal{V}$ accepts (x, ω_x) in polynomially many steps and with polynomial curvature cost;
2. if $x \notin L$ then for all ω the verifier rejects (x, ω) under the same curvature rails.

In this framework, a *toroidal locking witness* for a 3-SAT instance is a curvature-locked configuration encoding a toroidal ψ–electron structure whose phase-locking and flux constraints correspond to the satisfaction of all clauses. Nonexistence of such a locking configuration is equivalent to unsatisfiability.

M.4 Equivalence with Classical P and NP

We now make precise the relationship between curvature-bounded wave-constrained computation and the standard Turing complexity classes. Throughout this subsection we work at the level of the *discrete WCC abstraction*: a finite-state, finite-range local update model induced by the WCT rails, ignoring continuous amplitudes and treating each local configuration as a symbol from a fixed finite alphabet.

M.4.1 Discrete WCC Machines and Complexity Classes

Fix a finite input alphabet Σ and let $n = |x|$ denote input length.

Definition M.3 (Discrete WCC machine). A (discrete) WCC machine is a tuple

$$\mathcal{M} = \big(\{\Lambda_n\}_{n\geq 1}, S, r, U, \mathrm{enc}, \mathrm{dec}\big)$$

with the following components:

1. For each input length n, a finite index set $\Lambda_n \subset \mathbb{Z}^d$ (the lattice of computational cells) with a polynomial size bound

$$|\Lambda_n| \ \leq \ p(n)$$

   for some fixed polynomial p (finite-precision rail).

2. A finite state set S of cell symbols.

3. A fixed interaction radius $r \in \mathbb{N}$ and neighbourhood shape

$$N_r(i) := \{\, j \in \mathbb{Z}^d : \|j - i\|_\infty \leq r \,\}$$

   for each $i \in \Lambda_n$.

4. A local update rule

$$U : S^{N_r} \to S$$

   which induces a global synchronous update

$$C_{t+1}(i) = U\big(C_t|_{N_r(i)}\big), \qquad i \in \Lambda_n,$$

   on configurations $C_t : \Lambda_n \to S$. This enforces locality and bounded propagation.

5. An encoding map

$$\mathrm{enc} : \Sigma^* \to \bigsqcup_{n \geq 1} S^{\Lambda_n}$$

assigning to each input $x \in \Sigma^n$ an initial configuration $C_0 = \mathrm{enc}(x) \in S^{\Lambda_n}$.

6. A decoding predicate

$$\mathrm{dec} : S^{\Lambda_n} \to \{0, 1\}$$

which decides acceptance based on a configuration.

Given $x \in \Sigma^n$, the machine $\mathcal{M}$ produces a trajectory $C_0, C_1, C_2, \dots$ by repeated application of U, and we say that $\mathcal{M}$ accepts x at time t if $\mathrm{dec}(C_t) = 1$.

We measure time in *global update steps.*

Definition M.4 (Polynomial-time WCC and $\mathrm{P}_{\mathrm{WCC}}$). For a WCC machine $\mathcal{M}$, let $T_{\mathcal{M}}(n)$ be the worst-case number of update steps until $\mathrm{dec}(C_t)$ stabilizes (on inputs of length n). We say that $\mathcal{M}$ decides a language $L \subseteq \Sigma^*$ in polynomial time if there exists $k \geq 1$ such that, for all $x \in \Sigma^n$,

$$T_{\mathcal{M}}(n) \leq n^k \qquad \text{and} \qquad \mathrm{dec}(C_{T_{\mathcal{M}}(n)}) = 1 \iff x \in L.$$

The corresponding complexity class $\mathrm{P}_{\mathrm{WCC}}$ consists of all languages decidable in polynomial time by some WCC machine.

On the nondeterministic side we mimic the verifier-based definition of NP.

Definition M.5 (Polynomial-time WCC verifier and $\mathrm{NP}_{\mathrm{WCC}}$). A language $L \subseteq \Sigma^*$ lies in $\mathrm{NP}_{\mathrm{WCC}}$ if there exists a WCC machine $\mathcal{V}$ (the verifier), a polynomial $q(\cdot)$, and an encoding of witness strings $w \in \{0, 1\}^{\leq q(n)}$ into initial configurations such that:

1. For every input $x \in \Sigma^n$ and every witness w with $|w| \leq q(n)$, the trajectory of $\mathcal{V}$ from the encoded pair (x, w) stabilizes in at most $\mathrm{poly}(n)$ steps.

2. $x \in L$ if and only if there exists a witness w of length at most $q(n)$ such that $\mathcal{V}$ accepts (x, w).

In physical WCT implementations, additional curvature and energy rails are imposed on the underlying continuous evolution. At the discrete level considered here, we assume that these rails enforce only a polynomial bound on the resources required to realize the finite-state local update dynamics above; the proofs below rely only on the discrete structure.

M.4.2 P–Equivalence

Theorem M.6 (P–Equivalence). $\mathrm{P}_{\mathrm{WCC}} = P$.

Proof. We prove both inclusions.

(1) $P \subseteq \mathrm{P}_{\mathrm{WCC}}$. Let $L \subseteq \Sigma^*$ be a language in P. By definition there exists a deterministic Turing machine

$$M = (Q, \Gamma, \Sigma, \delta, q_0, q_{\mathrm{acc}}, q_{\mathrm{rej}})$$

and constants $c, k \geq 1$ such that, for every input $x \in \Sigma^n$, M halts on x in at most cn^k steps and accepts exactly the strings in L.

We construct a WCC machine $\mathcal{M}_M$ that simulates M on all inputs with at most a constant factor overhead in time.

Encoding configurations. A configuration of M is a triple (γ, h, q) consisting of a tape $\gamma : \mathbb{Z} \to \Gamma$ with all but finitely many cells equal to the blank symbol $\sqcup$, a head position $h \in \mathbb{Z}$, and a current state $q \in Q$. Since M halts in at most cn^k steps on inputs of length n, its head never travels further than cn^k cells from the origin, and it never uses more than $n + cn^k$ nonblank cells.

Define a polynomial

$$L(n) := n + cn^k$$

and take the WCC lattice

$$\Lambda_n := \{-L(n), \ldots, L(n)\} \subset \mathbb{Z}$$

(one spatial dimension suffices). For the local state alphabet we define

$$S := \Gamma \times (Q \cup \{\perp\}) \times \{0, 1\},$$

where $(\gamma, q, p) \in S$ means: tape symbol $\gamma \in \Gamma$, either $q \in Q$ (head present with state q) or $\perp$ (no head on that cell), and a phase bit $p \in \{0, 1\}$ used to implement a two-phase local update (this is a standard trick for reversible or synchronous simulation of Turing transitions).

Given a Turing configuration (γ, h, q), we encode it as a WCC configuration $C : \Lambda_n \to S$ by

$$C(i) := \begin{cases} \big(\gamma(i), q, 0\big), & i = h, \\ \big(\gamma(i), \perp, 0\big), & i \neq h. \end{cases}$$

Local update rule. We now define the local update rule $U : S^{N_1} \to S$ with radius $r = 1$ in dimension $d = 1$, so that $N_1(i) = \{i-1, i, i+1\}$. One global WCC update step will correspond to *one* Turing step of M, implemented in two local phases distinguished by the phase bit p.

At any time, at most one cell carries a head state (i.e. has second component in Q rather than $\perp$).

Phase $p = 0$ (apply δ and tag movement direction). If the centre cell has state $(\gamma, q, 0)$ with $q \in Q$ and the neighbours have arbitrary states, we apply the Turing transition function δ:

$$\delta(q, \gamma) = (q', \gamma', d), \quad d \in \{L, R\},$$

and set the new centre state to

$$(\gamma', (q', d), 1),$$

where (q', d) is a *direction-tagged* head state indicating that in the next phase the head should move left or right. All other cells with phase 0 and no head simply copy their tape symbol and maintain $p = 0$.

Phase $p = 1$ (move the head). In the next global step, the local rule looks for a tagged head state $(\gamma', (q', d), 1)$ at some site i. If $d = L$, then the cell at $i-1$ changes its state to $(\gamma_{i-1}, q', 0)$ (head now at $i-1$), and the centre cell at i drops to $(\gamma', \perp, 0)$. Analogously for $d = R$ with the neighbour at $i+1$. All cells not adjacent to the head simply pass their state through with phase bit reset to 0.

It is straightforward to define U so that:

- At most one cell carries a head state at any time.
- A pair of global WCC updates implements one full Turing step (write + move head).
- No information is lost except what M itself overwrites.

Because Λ_n has size $|\Lambda_n| = \mathcal{O}(n^k)$ and U is radius-1 with finite alphabet S, this is a valid WCC machine.

Time and acceptance. Given an input $x \in \Sigma^n$, we set $C_0 = \mathrm{enc}(x)$ to be the encoding of the initial Turing configuration (tape containing x starting at cell 0, head at 0, state q_0). After at most $T_M(n) \le cn^k$ Turing steps, M

halts in either q_{acc} or q_{rej}. The WCC machine $\mathcal{M}_M$ simulates each Turing step using a constant number (two) of global updates, so it halts in at most $2cn^k$ steps. We define the decoding predicate dec to read the unique head cell and accept exactly when the encoded Turing state is q_{acc}.

Thus $\mathcal{M}_M$ decides L in polynomially many update steps, and so $L \in \mathrm{P}_{\mathrm{WCC}}$.

(2) $\mathrm{P}_{\mathrm{WCC}} \subseteq P$. Let $L \subseteq \Sigma^*$ be a language in $\mathrm{P}_{\mathrm{WCC}}$. Hence there exists a WCC machine

$$\mathcal{M} = \big(\{\Lambda_n\}, S, r, U, \mathrm{enc}, \mathrm{dec}\big)$$

and constants $c, k \geq 1$ such that, for all inputs $x \in \Sigma^n$, the trajectory from $C_0 = \mathrm{enc}(x)$ stabilizes in at most $T_{\mathcal{M}}(n) \leq cn^k$ update steps, and x is accepted exactly when $\mathrm{dec}(C_{T_{\mathcal{M}}(n)}) = 1$.

We construct a deterministic Turing machine $M_{\mathcal{M}}$ that decides L in polynomial time.

Encoding configurations on a tape. For each input length n, the index set Λ_n has size at most $p(n)$ for some fixed polynomial. We fix any bijection

$$\iota_n : \Lambda_n \to \{1, \dots, |\Lambda_n|\},$$

and encode a WCC configuration $C : \Lambda_n \to S$ as a finite word

$$\mathrm{code}(C) = s_1 s_2 \dots s_{|\Lambda_n|}, \quad s_j := C\big(\iota_n^{-1}(j)\big),$$

over some fixed encoding alphabet for the finite state set S. This gives a description length $\mathcal{O}(|\Lambda_n|)$, hence $\mathcal{O}(p(n))$.

Simulating one global update. Given $\mathrm{code}(C_t)$ on its tape, the Turing machine can compute $\mathrm{code}(C_{t+1})$ as follows. For each cell index $i \in \Lambda_n$, the new state $C_{t+1}(i)$ depends only on the states in its neighbourhood $N_r(i)$. Using the bijection ι_n, the indices of those neighbours correspond to a bounded window in the encoded string. The machine:

- scans the encoded configuration, cell by cell;
- for each position j (representing $i = \iota_n^{-1}(j)$), reads the states corresponding to $N_r(i)$ (a fixed-size bundle of symbols around j);

- applies the finite local rule U (hardwired into its finite control) to compute the new symbol s'_j;
- writes s'_j to a second work tape.

After one pass, the second tape holds $\text{code}(C_{t+1})$. Because U has fixed radius and S is finite, the amount of work done per site is $O(1)$; therefore simulating one global WCC update costs time $\mathcal{O}(|\Lambda_n|)$, i.e. $\mathcal{O}(p(n))$.

Total running time. The WCC computation on input $x \in \Sigma^n$ halts in at most $T_{\mathcal{M}}(n) \leq cn^k$ updates. Simulating each update costs time $\mathcal{O}(p(n))$, so the total Turing running time is

$$\mathcal{O}\big(T_{\mathcal{M}}(n) \cdot p(n)\big) = \mathcal{O}\big(n^k \cdot p(n)\big),$$

which is polynomial in n.

Finally, $M_{\mathcal{M}}$ computes $\text{dec}(C_{T_{\mathcal{M}}(n)})$ from $\text{code}(C_{T_{\mathcal{M}}(n)})$ (constant-time per cell) and accepts x if and only if the WCC machine accepts. Hence $L \in P$.

Combining (1) and (2), we obtain $\text{P}_{\text{WCC}} = P$. □

M.4.3 NP–Equivalence

Theorem M.7 (NP–Equivalence)*.* $\text{NP}_{\text{WCC}} = NP$.

Proof. Again we prove both inclusions.

(1) $NP \subseteq \text{NP}_{\text{WCC}}$*.* Let $L \subseteq \Sigma^*$ be a language in NP. Then there exist:

- a deterministic Turing machine V (the verifier),
- and a polynomial $q(\cdot)$,

such that for all $x \in \Sigma^n$:

$$x \in L \iff \exists w \in \{0,1\}^{\leq q(n)} \text{ with } V(x,w) \text{ accepting in time poly}(n).$$

We construct a WCC verifier $\mathcal{V}$ that simulates V on the encoded pair (x, w).

The construction is identical in spirit to the P–equivalence simulation. We fix a polynomially bounded lattice $\Lambda_{n,q(n)}$ large enough to hold:

- the encoding of x,

- the encoding of a witness w of length at most $q(n)$,
- the simulated verifier tape and head.

We define an alphabet S that encodes tape symbols, head states, and a phase bit. An initial configuration C_0 is produced from (x, w) by $\mathrm{enc}(x, w)$, placing the contents of x and w on designated regions of the simulated tape and initializing the head state and position to match V's start configuration.

The local update rule U implements a Turing simulation of V as in the proof of Theorem M.6: each verifier step is simulated by a constant number of global WCC updates. Since V runs in time $\mathrm{poly}(n)$, the WCC verifier halts in time $\mathrm{poly}(n)$ as well. We define acceptance so that $\mathcal{V}$ accepts (x, w) if and only if the simulated verifier V enters its accepting state.

Therefore $x \in L$ if and only if there exists some witness $w \in \{0,1\}^{\leq q(n)}$ such that the WCC verifier $\mathcal{V}$ accepts the encoded pair (x, w) in polynomially many update steps. Hence $L \in \mathrm{NP}_{\mathrm{WCC}}$.

(2) $\mathrm{NP}_{\mathrm{WCC}} \subseteq NP$. Now let $L \subseteq \Sigma^*$ be a language in $\mathrm{NP}_{\mathrm{WCC}}$. By definition there exist:

- a WCC verifier $\mathcal{V}$,
- a polynomial $q(\cdot)$,

such that for each input $x \in \Sigma^n$:

1. every witness w of length at most $q(n)$ yields a computation of $\mathcal{V}$ that halts within $\mathrm{poly}(n)$ update steps (on the encoded pair (x, w)),
2. $x \in L$ if and only if some such witness w causes $\mathcal{V}$ to accept.

We build a classical verifier $V_{\mathcal{V}}$ as a polynomial-time Turing machine. On input (x, w), the machine:

1. computes the initial WCC configuration C_0 corresponding to (x, w) via the encoding map of $\mathcal{V}$;
2. simulates the WCC computation step-by-step up to the known polynomial bound on update steps, using the same simulation scheme as in the proof of Theorem M.6 (encoding configurations on its tape and applying the local rule U to obtain the next configuration);

3. accepts if and only if some configuration in this trajectory is accepting according to the predicate dec of $\mathcal{V}$.

Because each global WCC update can be simulated in time polynomial in $|x|$ (the lattice size is polynomial in n and the local rule has fixed radius), and the number of updates is at most polynomial in n, the total running time of $V_{\mathcal{V}}$ is polynomial in $n + |w|$. Thus $V_{\mathcal{V}}$ is a valid NP verifier in the classical sense.

By construction, for each x there exists a witness w causing $V_{\mathcal{V}}$ to accept if and only if there exists a witness w causing the WCC verifier $\mathcal{V}$ to accept. Hence $L \in NP$.

Combining (1) and (2), we obtain $\mathrm{NP}_{\mathrm{WCC}} = NP$. □

M.5 Curvature Blow-Up and Locked Expander Families

We now use curvature blow-up to construct a model-relative separation between $\mathrm{P}_{\mathrm{WCC}}$ and $\mathrm{NP}_{\mathrm{WCC}}$. Consider a family of 3-SAT instances $\{F_n\}$ embedded into WCT via a wave–clause encoding. Each formula F_n is mapped to a WCT interaction region in which clause satisfaction corresponds to constructive interference of phase-locked ψ-modes, and clause violation corresponds to destructive interference and curvature increases.

We assume:

(i) *Locked expander structure:* the clause-variable incidence graph of F_n is a bounded-degree expander and the encoding is such that any change in a clause signature induces a nonlocal adjustment to the curvature pattern. Consequently, any trajectory that systematically explores distinct clause-signature patterns must accumulate curvature proportional to the number of such changes.

(ii) *Curvature blow-up lemma:* there exists a constant $c > 0$ such that any WCT trajectory that visits m distinct clause-signature sectors (under the expander wiring above) must incur total curvature cost at least $\Omega(2^{cm})$.

(iii) *α–drop theorem:* the effective branching factor of the WCC search dynamics can be reduced to $2^{\alpha(n)}$ per step with $\limsup_{n\to\infty} \alpha(n) < 1$, but cannot be reduced to polynomial branching uniformly for the locked expander family.

Assumption (iii) encodes a proto-sub-exponential reduction: curvature feedback and toroidal locking can prune the search space from $2^{\Theta(n)}$ configurations to $2^{\alpha(n)n}$ with $0 < \alpha(n) < 1$, but cannot compress it to polynomial size without violating the WCT dynamics.

Combining (i)–(iii), we obtain:

Lemma M.8 (Curvature Lower Bound for Locked Expander SAT). For the family $\{F_n\}$ above there exists $\beta > 0$ such that, for any WCC machine $\mathcal{M}$ that decides satisfiability of F_n by explicitly or implicitly exploring the clause-signature space,

$$C_\Theta^{\mathcal{M}}(F_n, n) \geq 2^{\beta n}. \tag{15}$$

In particular, any such decision procedure that respects polynomial curvature rails must either fail to decide F_n for some n or violate the rails.

The proof uses the fact that, even after α–drop pruning, the number of distinct clause-signature sectors that must be traversed grows as $2^{\alpha(n)n}$, and the curvature blow-up lemma implies that the total curvature cost grows as $2^{\Omega(n)}$.

M.6 Model-Relative Separation $\mathrm{P}_{\mathrm{WCC}} \neq \mathrm{NP}_{\mathrm{WCC}}$

We now impose physically motivated curvature rails on the underlying WCT evolution that realizes a discrete WCC machine.

Definition M.9 (Physical curvature rails). A WCC machine $\mathcal{M}$ is said to respect *physical rails* if there exists a constant $k \geq 0$ such that, for every input $x \in \Sigma^*$ and every configuration ψ_x^t encountered during the computation of $\mathcal{M}$ on x,

$$E_{\mathrm{WCT}}[\psi_x^t] \ \leq \ |x|^k, \tag{16}$$

where E_{WCT} is the WCT curvature energy functional and $|x|$ is the input length.

We work with a fixed locked expander family of 3-SAT instances $\{F_n\}_{n\geq 1}$ as constructed earlier (each F_n has size $\Theta(n)$, degree and expansion parameters independent of n). Let $L^\star$ be the SAT language *restricted* to this family:

$$L^\star := \{\, (n, \alpha) : \alpha \text{ is a satisfying assignment for } F_n \,\}. \tag{17}$$

Equivalently, we can view $L^\star$ as the language of encoded instances F_n together with their satisfying assignments.

The crucial analytic input is the curvature lower bound on the locked expander family. We record it in a form directly suited to the separation proof.

Lemma M.10 (Curvature lower bound for locked expander SAT). There exist constants $\beta > 0$ and $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ the following holds.

Consider any WCC machine $\mathcal{M}$ that, on input (F_n, α), decides whether α satisfies F_n in time polynomial in n (under the discrete WCC model), and assume that $\mathcal{M}$ is realized by a WCT evolution $\psi^t_{F_n,\alpha}$ obeying the standard WCT rails (finite propagation, finite precision, locality). Then for every polynomial bound $p(n)$ there exists some satisfying or non-satisfying assignment α and some time t along the computation on (F_n, α) such that

$$E_{\mathrm{WCT}}\big[\psi^t_{F_n,\alpha}\big] \; > \; p(n). \tag{18}$$

In particular, if $p(n) = n^k$ for any fixed k, then for all sufficiently large n the curvature energy required to resolve the satisfiability of F_n along any such computation exceeds n^k.

Lemma M.10 is the curvature blow-up statement established earlier from the locked-expander construction, the semantic curvature functional C_Θ, and the α–drop bounds. We now show how, once this lemma is granted, a model-relative separation follows.

Theorem M.11 (Curvature-bounded WCC separation). Assume Lemma M.10. Then, under physical curvature rails, there exists a language $L^\star \in \mathrm{NP}_{\mathrm{WCC}}$ such that $L^\star \notin \mathrm{P}_{\mathrm{WCC}}$. In particular,

$$\mathrm{P}_{\mathrm{WCC}} \; \neq \; \mathrm{NP}_{\mathrm{WCC}} \tag{19}$$

as complexity classes defined by curvature-bounded WCC computation.

Proof. We proceed in two steps.

Step 1: $L^\star \in \mathrm{NP}_{\mathrm{WCC}}$. By construction, $L^\star$ is a restriction of SAT to the locked expander family $\{F_n\}$. Since SAT is in NP, there exists a polynomial-time classical verifier $V(x, w)$ that, given a formula x and an assignment w, checks whether w satisfies x.

By the WCC–Turing equivalence (Theorems $\mathrm{P}_{\mathrm{WCC}} = P$ and $\mathrm{NP}_{\mathrm{WCC}} = NP$ for the discrete model), there is a discrete WCC verifier $\mathcal{V}$ that simulates

V on encoded pairs (F_n, α) via a local update rule with polynomial running time in n. The witness α is simply the assignment w, encoded into the WCC lattice as a finite pattern of local states (or, in the full WCT realization, as a finite arrangement of locked toroidal modes corresponding to literals).

By definition of $\mathrm{NP}_{\mathrm{WCC}}$, we require that for each input (F_n, α), the WCC verifier $\mathcal{V}$ halts in time polynomial in n and accepts if and only if α satisfies F_n. Since the discrete WCC model is polynomially equivalent to Turing machines and the encoding/decoding maps are polynomial in n, such a verifier exists. Therefore $L^\star \in \mathrm{NP}_{\mathrm{WCC}}$.

Step 2: $L^\star \notin \mathrm{P}_{\mathrm{WCC}}$ under physical rails. Suppose, for contradiction, that $L^\star \in \mathrm{P}_{\mathrm{WCC}}$ under physical curvature rails. Then there exists a WCC machine $\mathcal{M}^\star$ and a constant $k \geq 0$ such that:

1. For all inputs (F_n, α), $\mathcal{M}^\star$ halts within $\mathrm{poly}(n)$ update steps and correctly decides whether α satisfies F_n (i.e. $\mathcal{M}^\star$ is a WCC decider for $L^\star$).

2. The underlying WCT realization of $\mathcal{M}^\star$ satisfies the physical curvature rails (16), i.e. for all n and for all configurations $\psi^t_{F_n,\alpha}$ encountered along the computation,

   $$E_{\mathrm{WCT}}\big[\psi^t_{F_n,\alpha}\big] \;\leq\; |(F_n,\alpha)|^k \;=\; \mathcal{O}(n^k),$$

   where $|(F_n, \alpha)|$ denotes the input length. In particular, there is a polynomial $p(n) = cn^k$ such that

   $$E_{\mathrm{WCT}}\big[\psi^t_{F_n,\alpha}\big] \;\leq\; p(n) \tag{20}$$

   for all n, all α, and all times t during the computation.

Now fix this polynomial bound $p(n) = cn^k$. Apply Lemma M.10 with this choice of p. The lemma asserts that there exist $n \geq n_0$ and some assignment α such that, along the WCT evolution $\psi^t_{F_n,\alpha}$ realizing the computation of any polynomial-time WCC decider for $L^\star$, there is a time $t^\star$ with

$$E_{\mathrm{WCT}}\big[\psi^{t^\star}_{F_n,\alpha}\big] \;>\; p(n). \tag{21}$$

In particular, this applies to the specific decider $\mathcal{M}^\star$. Thus for sufficiently large n, there must exist some assignment α and some time $t^\star$ along the computation of $\mathcal{M}^\star$ on input (F_n, α) such that (21) holds. But this contradicts the physical rail bound (20), which asserts that $E_{\mathrm{WCT}}[\psi^t_{F_n,\alpha}] \leq p(n)$ for all t.

Therefore no WCC decider for $L^\star$ can simultaneously:

- run in polynomial time, and
- respect physical curvature rails.

Equivalently, under the physical curvature rails (16), the language $L^\star$ does not belong to $\mathrm{P}_{\mathrm{WCC}}$.

Combining this with Step 1, which shows $L^\star \in \mathrm{NP}_{\mathrm{WCC}}$, we conclude that, under the physical curvature rails and Lemma M.10,

$$\mathrm{P}_{\mathrm{WCC}} \neq \mathrm{NP}_{\mathrm{WCC}}.$$

This completes the proof. $\square$

M.7 Computational Review Note

Formatting, reference compilation, and structural consistency checks were performed with the assistance of automation tools, including ChatGPT, used solely for document preparation and stylistic review. All mathematical derivations, proofs, and theoretical conclusions are entirely the author's own.